part1 : gradient descent

Summary:

1. **Backtracking Line Search**: Adjusts the step size (\alpha) to ensure sufficient decrease in (f), based on the condition:

$$f(x - \alpha \nabla f(x)) \leq f(x) - c\alpha ||\nabla f(x)||^2$$

   The step size is reduced iteratively until this condition is satisfied.

2. **Gradient Descent (GD)**: Iteratively updates the solution using:

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

   Stops when the gradient is small or the change in (x_k) is negligible, or after a maximum number of iterations.

3. **Outputs**:

   - Sequence of solutions (x_k), number of iterations, function values, gradients, and error norms.

In [ ]:
```
import numpy as np
import matplotlib.pyplot as plt
import random
from random import randint
```

In [ ]:
```
def backtracking(f, grad_f, x, alpha = 1, c = 0.8, tau = 0.25):
    while f(x - alpha*grad_f(x)) > f(x) - c * alpha * np.linalg.norm(grad_f(
        alpha = tau * alpha
        if alpha < 1e-3:
            break
    return alpha
```

In [ ]:
```
def GD(f, grad_f, x0, tolf, tolx, kmax, alpha= 1, bt= False):
    """
    The function returns the following:
        •       The sequence of solutions  x_k  at each iteration.
        •       The total number of iterations.
        •       The function values  f(x_k)  at each iteration.
        •       The gradient values  \nabla f(x_k)  at each iteration.
        •       The gradient norms (errors) at each iteration.
```

```
"""
    xk = x0
    f_vals = [f(xk)]
    grad_vals = [grad_f(xk)]
    err_vals = [np.linalg.norm(grad_f(xk))]
    x_vals = [xk]
    iteration = 0

    while iteration < kmax:
        x_prec = xk

        xk = xk - alpha * grad_f(xk)

        if bt:
            alpha = backtracking(f, grad_f, xk)

        x_vals.append(xk)
        f_vals.append(f(xk))
        grad_vals.append(grad_f(xk))
        err_vals.append(np.linalg.norm(grad_f(xk)))

        iteration+=1

        if np.linalg.norm(grad_f(xk)) < tolf * np.linalg.norm(grad_f(x0)):
            break

        if np.linalg.norm(xk - x_prec) < tolx * np.linalg.norm(x0):
            break

    return (x_vals, iteration, f_vals, grad_vals, err_vals)
```

In [ ]:

```
def my_plot(xk_vals, k, f_vals, grad_vals, err_vals, f, title, with_contour=
    # Create a grid of x1 and x2 values for the contour plot
    xv = np.linspace(-10, 10, 100).T
    yv = np.linspace(-10, 10, 100).T
    xx, yy = np.meshgrid(xv, yv)

    # Calculate the objective function values on the grid
    zz = f([xx, yy])

    # Convert xk_vals to numpy array
    xk_vals = np.array(xk_vals)

    # If with_contour is True, plot the contours
    if with_contour:
        plt.contour(xx, yy, zz)

    # Plot the optimization path (xk_vals) with label
    plt.plot(xk_vals[:,0], xk_vals[:,1], '--ro', label="Optimization path")
```

```
    # Scatter the true minimum point with label
    plt.scatter(xtrue[0], xtrue[1], color='green', label="True minimum", mar

    # Add the legend
    plt.legend()

    # Set the title, labels, and grid
    plt.title(title)
    plt.xlabel("x1")
    plt.ylabel("x2")
    plt.grid()

    # Display the plot
    plt.show()

def my_plot_2D(xk_vals, k, f_vals, grad_valks, err_vals, f, title):
    x_vals = np.linspace(-3, 3, 100)
    y_vals = []
    for x in x_vals:
        y_vals.append(f([x]))
    plt.plot(x_vals, y_vals)
    plt.scatter(xk_vals, f_vals, c='green')
    plt.title(title)
    plt.show()

def plot_error(iters, errs, labels, title = "Error (2-norms of gradient)"):
    colors = []

    for i in range (len(iters)):
        colors.append('#%06X' % randint(0, 0xFFFFFF))

    colors = plt.get_cmap("tab20c")
    i= 0

    for item in zip(iters, errs, labels):
        plt.plot(item[0], item[1], c=colors(i/(len(iters)-1)), label = item[
        i+=1
    plt.title(title)

    plt.legend(loc="upper left")
    plt.show()
```

In [ ]:

```
def function_testing(f, grad_f, x0=0, title="", alpha = 1e-1, bt = False, or
    # Set tolerance values for stopping criteria
    tolf = 1e-8
    tolx = 1e-8
    kmax = 300

    # Perform gradient descent (GD), with or without backtracking
    if not bt:
```

```
        xk_vals, k, f_vals, grad_vals, err_vals = GD(f, grad_f, x0, tolf, tc
    else:
        # Use backtracking line search to adjust alpha
        alpha_bt = backtracking(f, grad_f, x0)
        xk_vals, k, f_vals, grad_vals, err_vals = GD(f, grad_f, x0, tolf, tc

    # Plot results: either with or without contours
    if not oneD and not isMatr:
        if not bt:
            my_plot(xk_vals, k, f_vals, grad_vals, err_vals, f, title + " wi
            my_plot(xk_vals, k, f_vals, grad_vals, err_vals, f, title + " wi
        else:
            my_plot(xk_vals, k, f_vals, grad_vals, err_vals, f, title + " wi
            my_plot(xk_vals, k, f_vals, grad_vals, err_vals, f, title + " wi

    elif not isMatr:
        if not bt:
            my_plot_2D(xk_vals, k, f_vals, grad_vals, err_vals, f, title + "
        else:
            my_plot_2D(xk_vals, k, f_vals, grad_vals, err_vals, f, title + "

    # Print the found minimum and the number of iterations
    print("Minimum Found =", xk_vals[k-1], "with", k, "iterations")

    to_ret = []

    # If check_err is True, calculate error at each iteration
    if check_err:
        if not isMatr:
            for xk in xk_vals:
                to_ret.append(np.linalg.norm((xk - xtrue)))  # Error as the
        else:
            xtrue = np.array(xtrue)
            for xk in xk_vals:
                xk = np.array(xk)
                to_ret.append(np.linalg.norm((xk - xtrue)))  # Error as the

    # Return the results
    if check_err:
        return np.arange(k+1), err_vals, to_ret  # Iterations, errors, and e
    return np.arange(k+1), err_vals  # Just iterations and errors
```

functions:

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- Test the algorithm above on the following functions:
  1. $f : \mathbb{R}^2 \to \mathbb{R}$ such that:

     $$f(x_1, x_2) = (x_1 - 3)^2 + (x_2 - 1)^2,$$

     for which the true solution is $x^* = (3, 1)^T$.

  2. $f : \mathbb{R}^2 \to \mathbb{R}$ such that:

     $$f(x_1, x_2) = 10(x_1 - 1)^2 + (x_2 - 2)^2,$$

     for which the true solution is $x^* = (1, 2)^T$.

  3. $f : \mathbb{R}^n \to \mathbb{R}$ such that:

     $$f(x) = \frac{1}{2}||Ax - b||_2^2,$$

     where $A \in \mathbb{R}^{ntimesn}$ is the Vandermonde matrix associated with the vector $v \in \mathbb{R}^n$ that contains n equispaced values in the interval $[0, 1]$, and $b \in \mathbb{R}^n$ is computed by first setting $x^* = (1, 1, \ldots, 1)^T$, and then $b = Ax^*$. Try for different values of n (e.g. $n = 5, 10, 15, \ldots$).

  4. $f : \mathbb{R}^n \to \mathbb{R}$ such that:

     $$f(x) = \frac{1}{2}||Ax - b||_2^2 + \frac{\lambda}{2}||x||_2^2,$$

     where $A \in \mathbb{R}^{ntimesn}$ and $b \in \mathbb{R}^n$ are the same of the exercise above, while λ is a fixed value in the interval $[0, 1]$. Try different values of λ and comment the result.

  5. $f : \mathbb{R} \to \mathbb{R}$ such that:

     $$f(x) = x^4 + x^3 - 2x^2 - 2x.$$

In [ ]:
```
random.seed(42)
lam = random.random()

def f1(x):
    return ((x[0] - 3)**2 + (x[1] - 1)**2)
def f2(x):
    return (10*(x[0] - 1)**2 + (x[1] - 2)**2)
def f3(x):
    x = np.array(x).T
    n = len(x)
    v = np.linspace(0, 1, n)
    A = np.vander(v)
    x_true = np.ones(n).T
    b= A @ x_true

    return ((np.linalg.norm((A @ x) - b)**2)/2)
def f4(x):
    n = len(x)
```

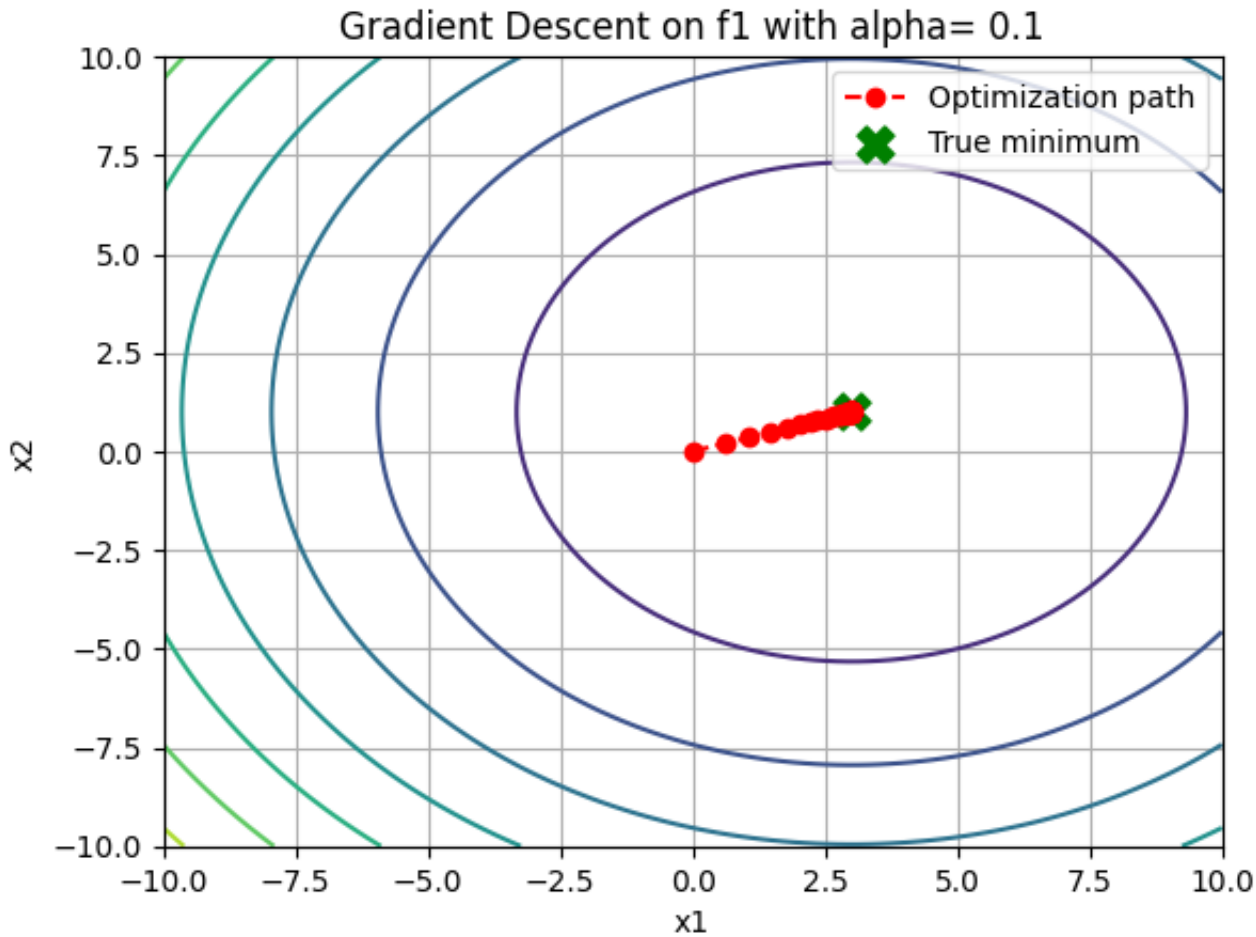
```
    v = np.linspace(0, 1, n)
    A = np.vander(v)
    x_true = np.ones(n).T
    b= A @ x_true

    return (((np.linalg.norm((A @ x) - b)**2)/2) + ((np.linalg.norm(x))**2)*
def f5(x):
    return x[0]**4 + x[0]**3 - 2*(x[0]**2) - 2*x[0]
```

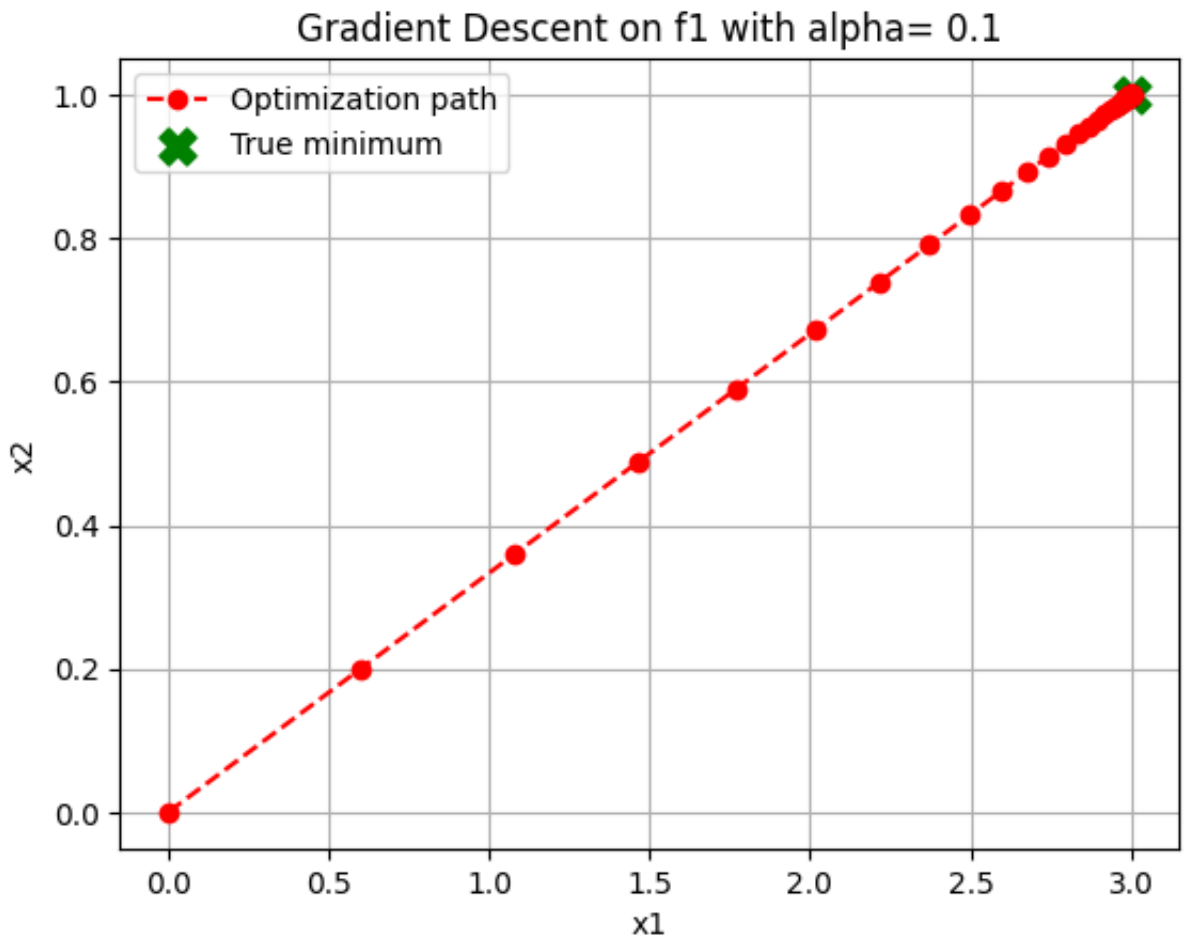
In [ ]:
```
def grad_f1(x):
    return np.array((2*x[0] - 6, 2*x[1] - 2))
def grad_f2(x):
    return np.array([20*(x[0] - 1), 2*(x[1] - 2)])
def grad_f3(x):
    n = len(x)
    v = np.linspace(0,1,n)
    A = np.vander(v)
    x_true = np.ones(n).T
    b = A @ x_true
    return A.T@(A@x-b)
def grad_f4(x):
    return grad_f3(x) + lam*np.array(x)
def grad_f5(x):
    return np.array(4*x[0]**3 + 3*x[0]**2 - 4*x[0] - 2)
```

- Use a contour plot for visualizing the optimization path in the function's input space.
- Use an error plot to analyze and compare convergence rates numerically over iterations.

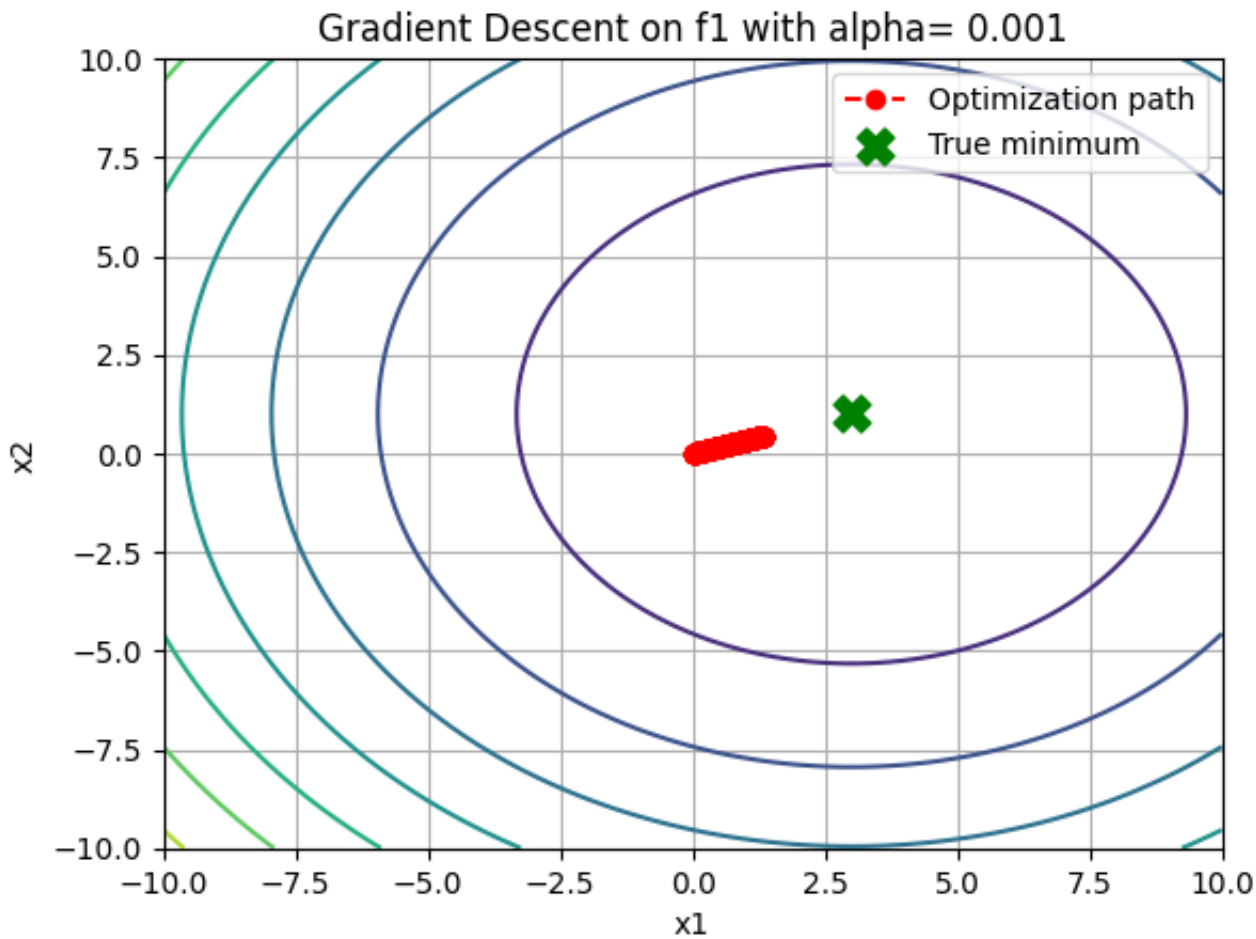
In [ ]:
```
iters = []
err_vals = []
labels = []
err_xtrue = []
xtrue = np.array([3,1]).T
el1, el2, el3 = function_testing(f1, grad_f1, x0 = np.array((0, 0)), title="
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-1")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f1, grad_f1, x0 = np.array((0, 0)), title="
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-3")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f1, grad_f1, x0 = np.array((0, 0)), title="
```

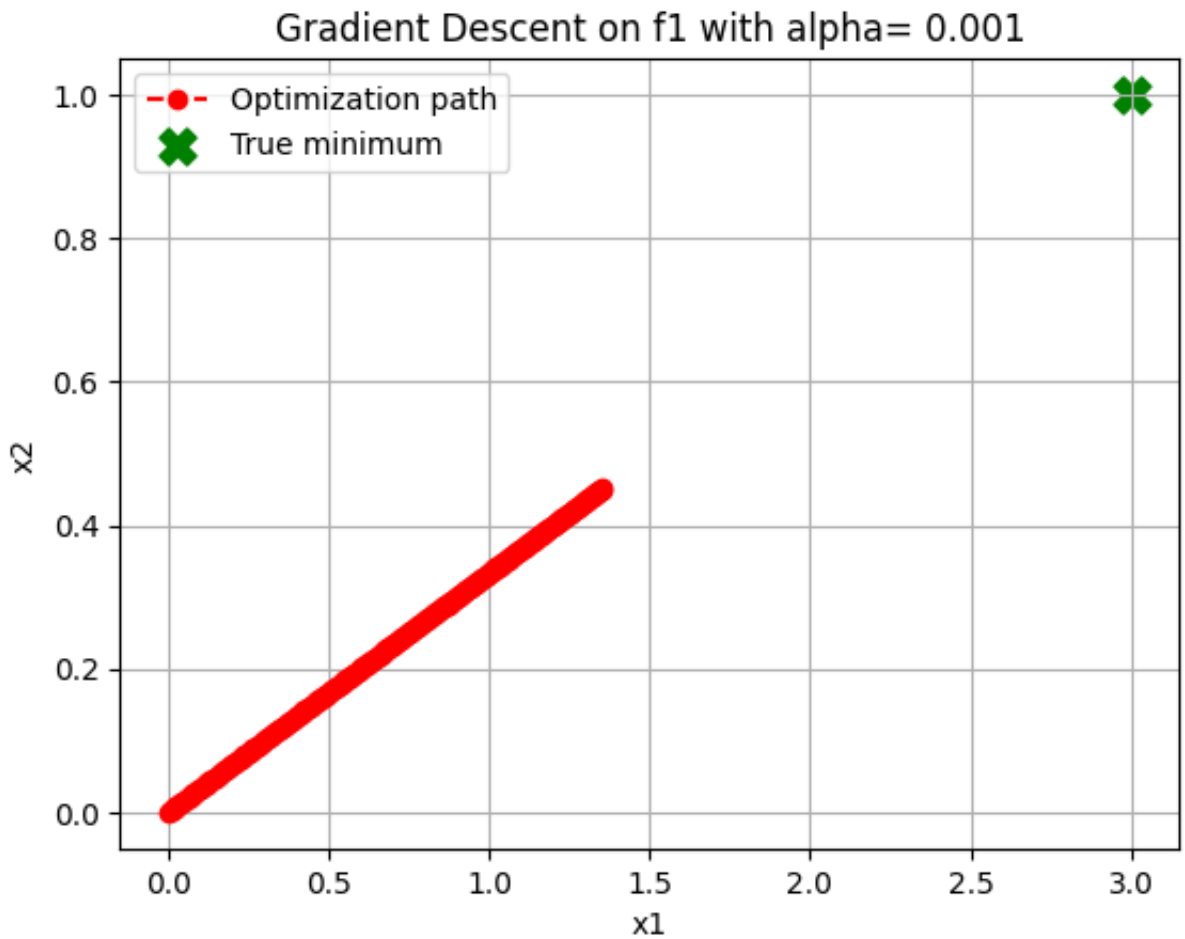
```
iters.append(el1)
err_vals.append(el2)
labels.append("Backtracking")
err_xtrue.append(el3)

plot_error(iters, err_vals, labels)
plot_error(iters, err_xtrue, labels, title="Error (distance from x_true)")
```



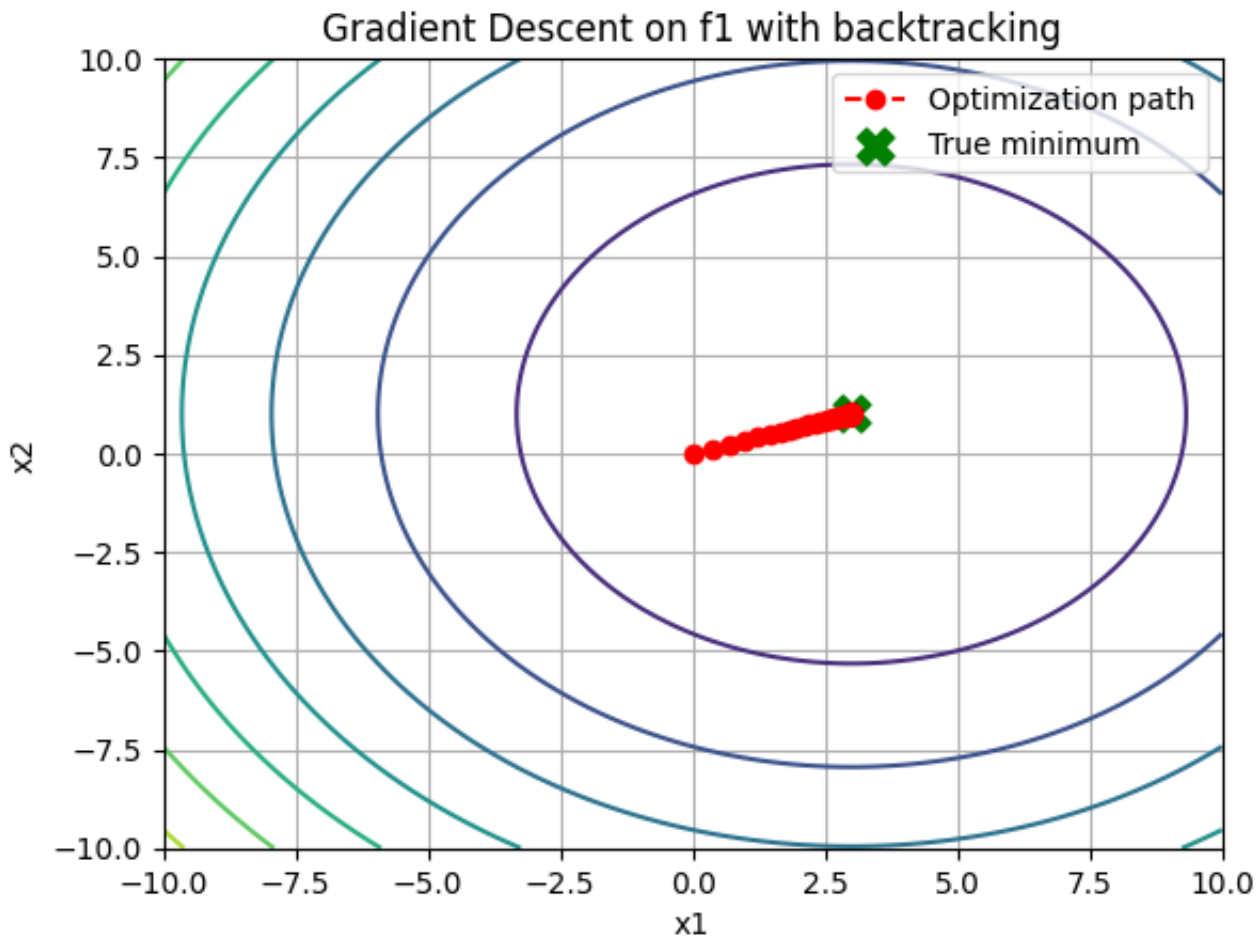


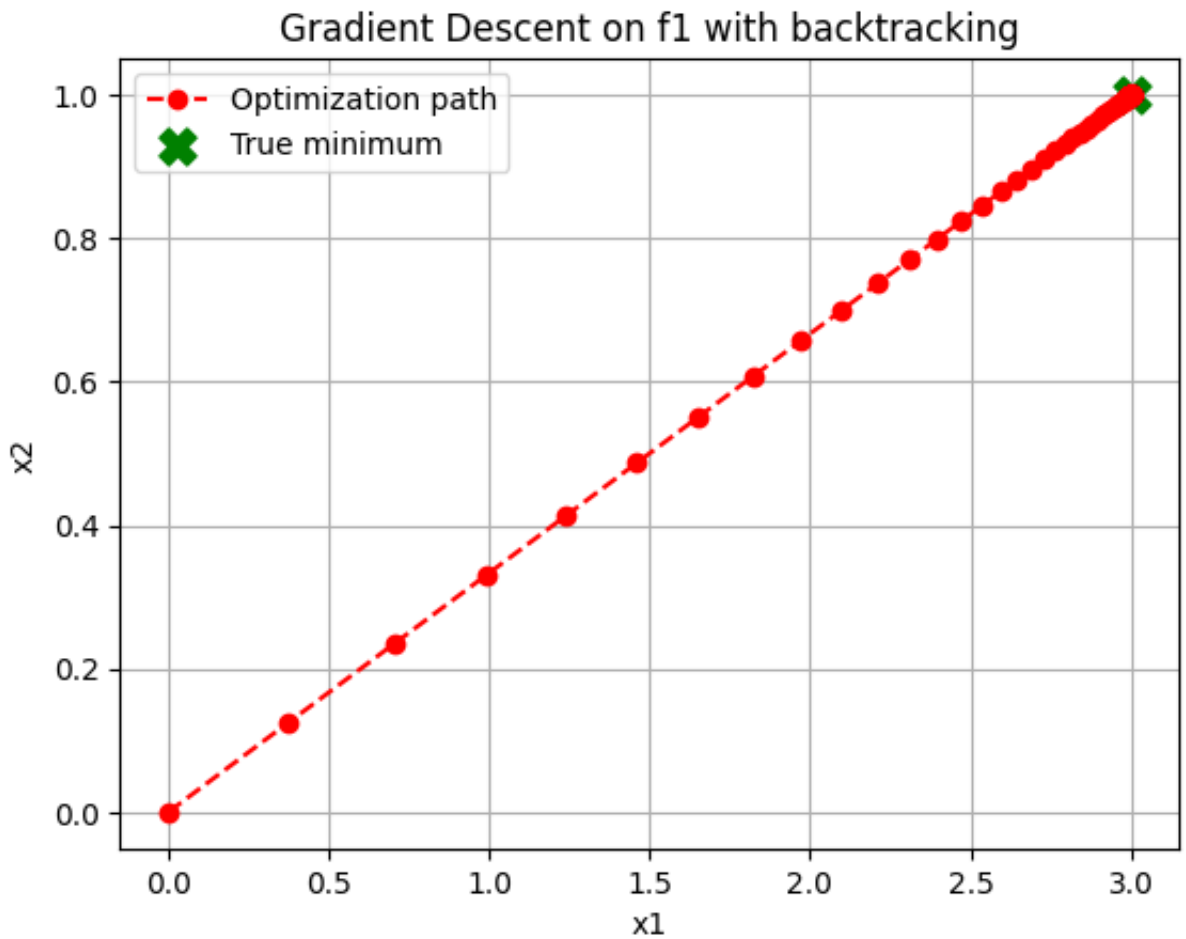

```
Minimum Found = [2.99999997 0.99999999] with 83 iterations
```


Gradient Descent on f1 with alpha= 0.001
Optimization path
True minimum
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
x2
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x1

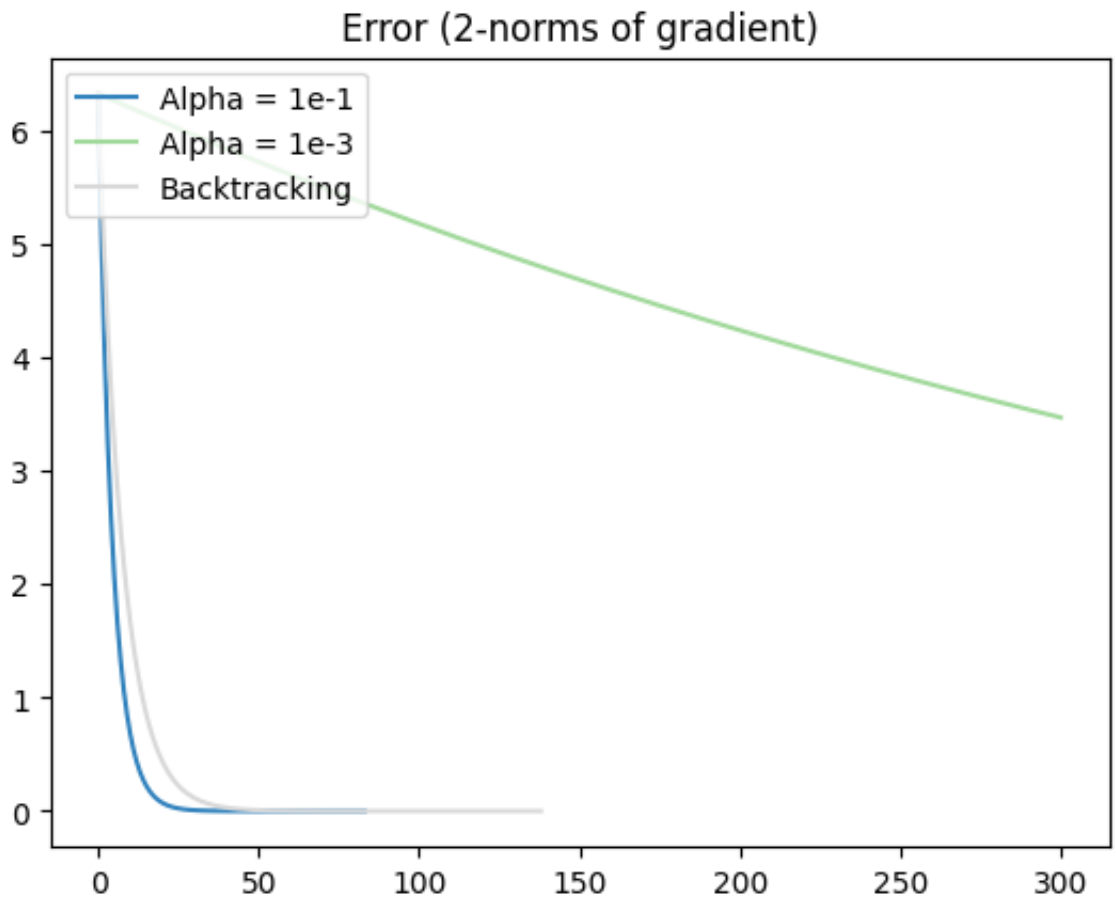


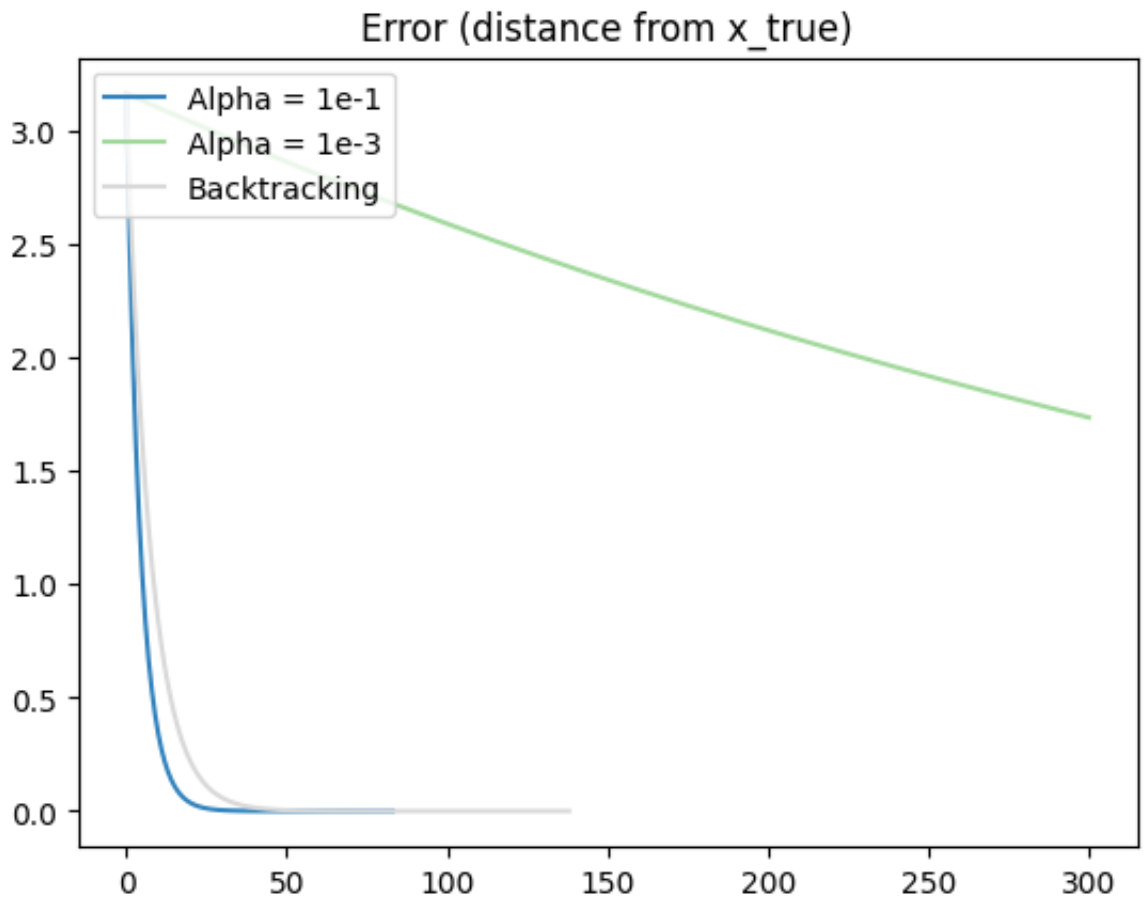

```
Minimum Found = [1.35125649 0.45041883] with 300 iterations
```


Gradient Descent on f1 with backtracking
Optimization path
True minimum
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
x2
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x1



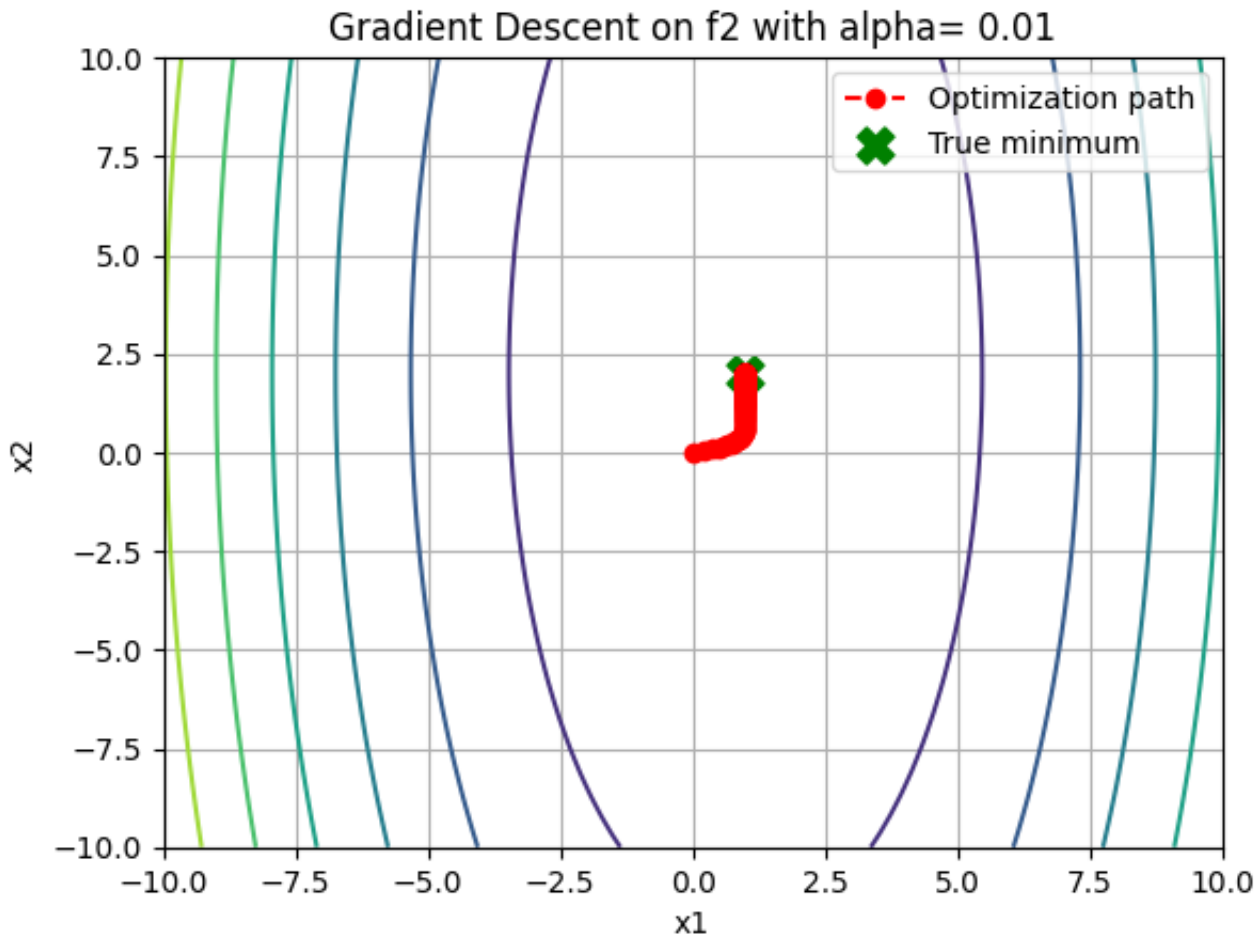

```
Minimum Found = [2.99999997 0.99999999] with 138 iterations
```

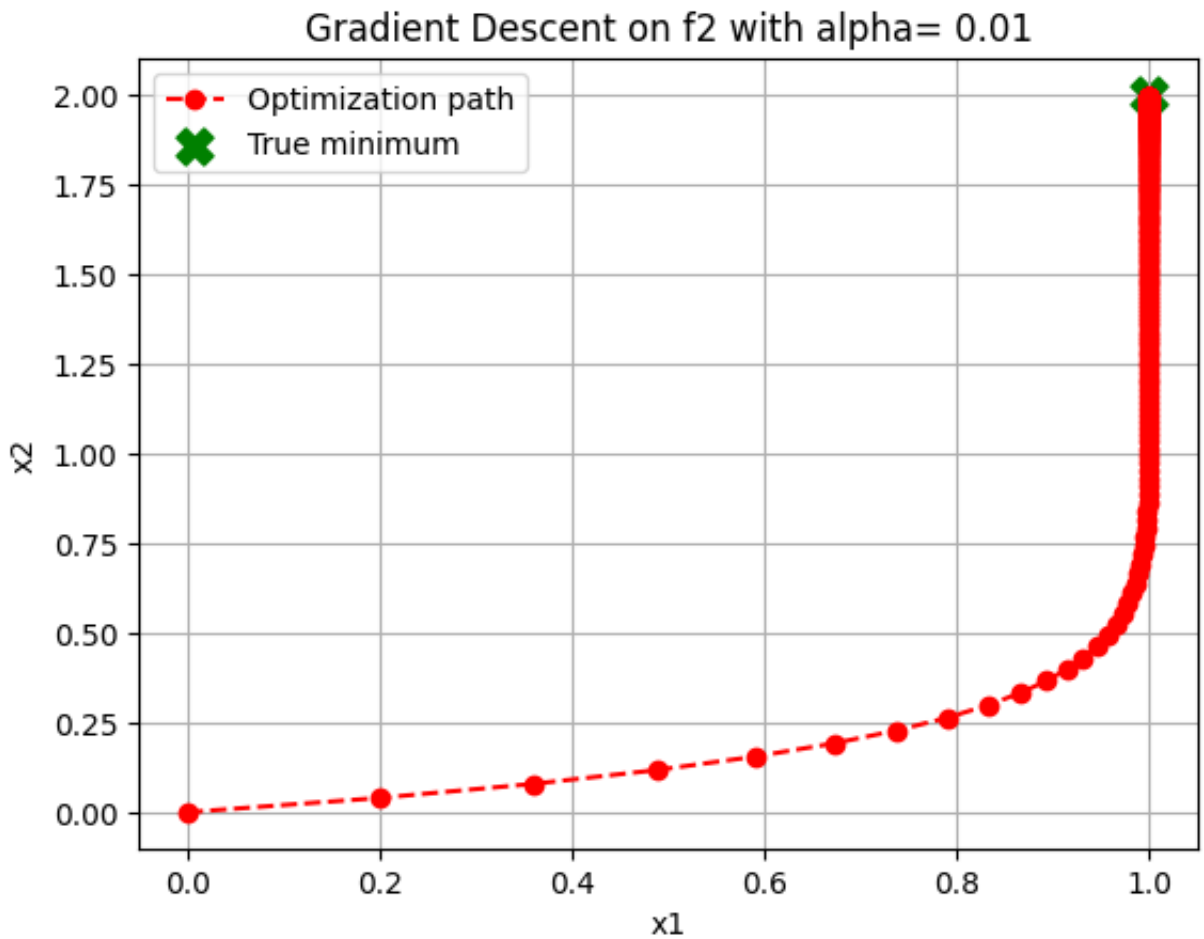

Error (2-norms of gradient)
Alpha = 1e-1
Alpha = 1e-3
Backtracking
0
1
2
3
4
5
6
0
50
100
150
200
250
300



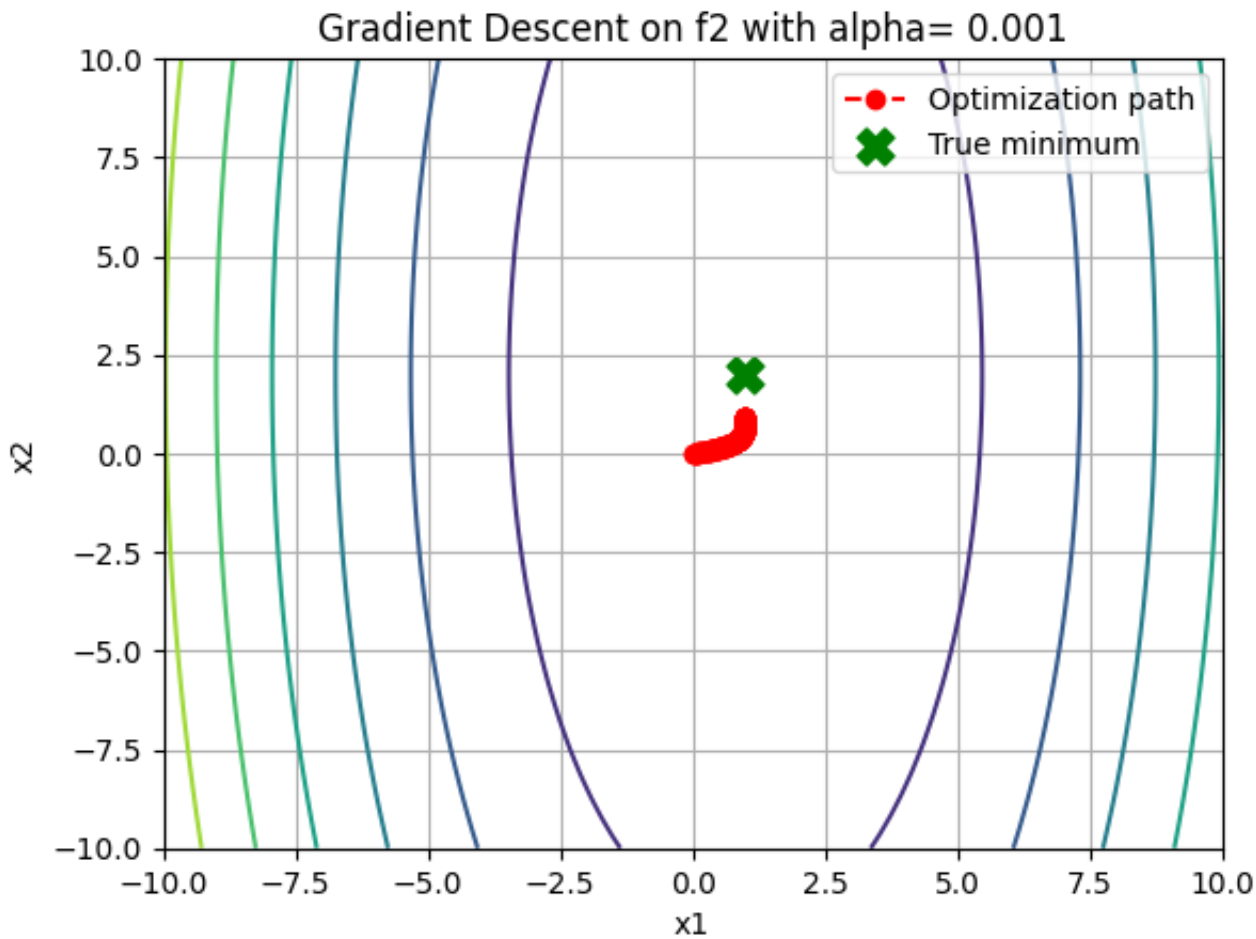
In [ ]:
```
iters = []
err_vals = []
labels = []
err_xtrue = []
xtrue = np.array([1,2]).T

el1, el2, el3 = function_testing(f2, grad_f2, x0 = np.array((0, 0)), title="
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-1")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f2, grad_f2, x0 = np.array((0, 0)), title="
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-3")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f2, grad_f2, x0 = np.array((0, 0)), title="
iters.append(el1)
err_vals.append(el2)
labels.append("Backtracking")
err_xtrue.append(el3)
```

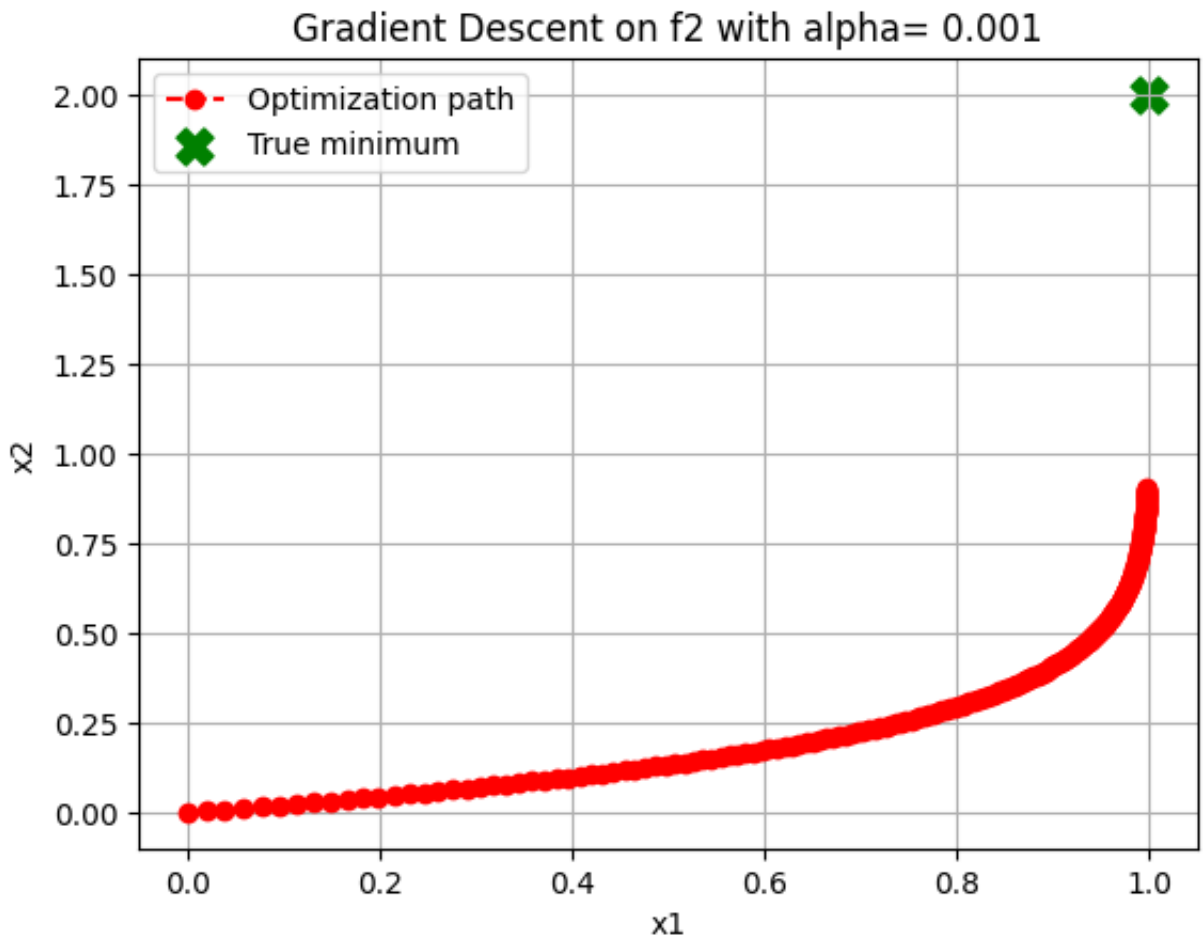
```
plot_error(iters, err_vals, labels)
plot_error(iters, err_xtrue, labels, title="Error (distance from x_true)")
```



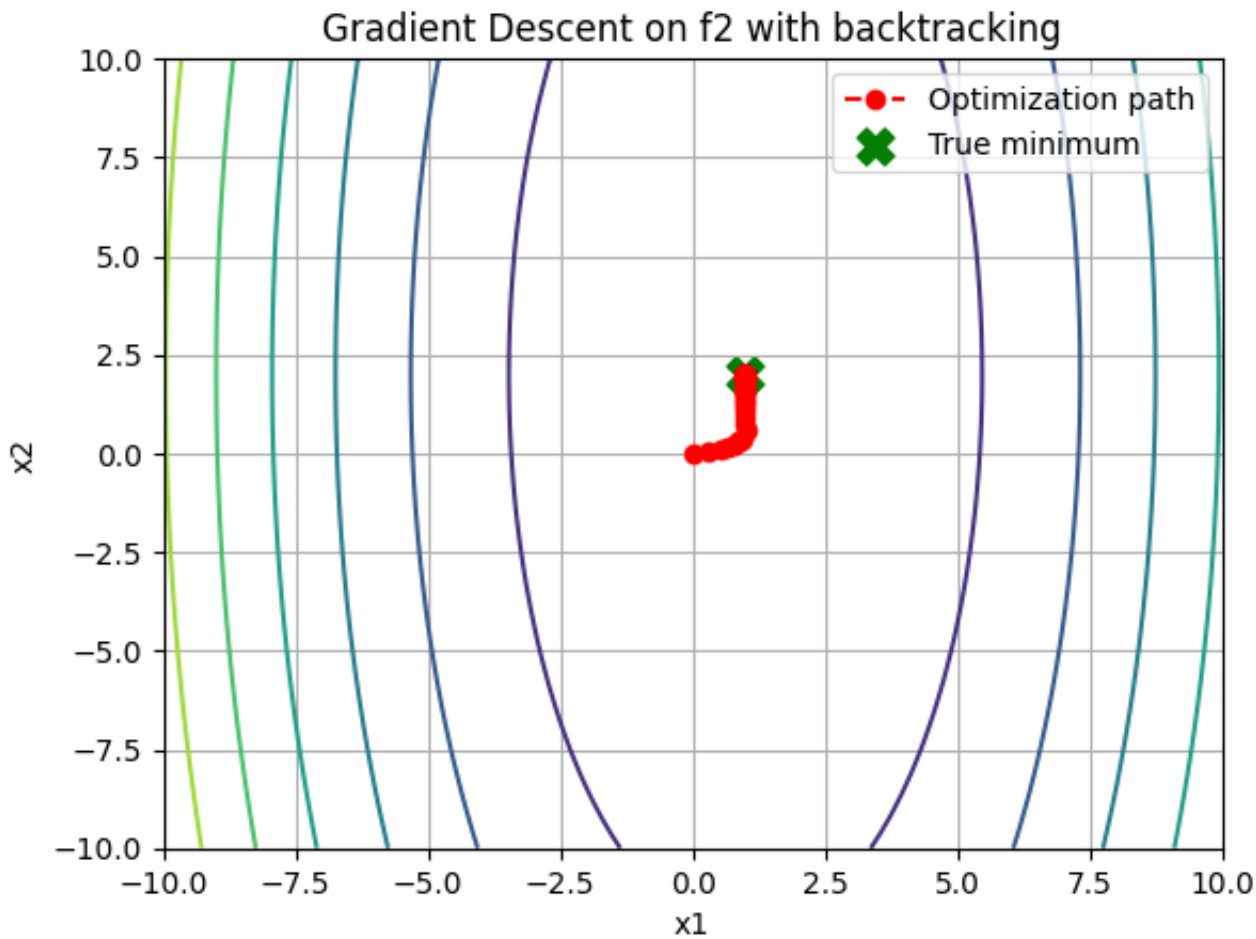


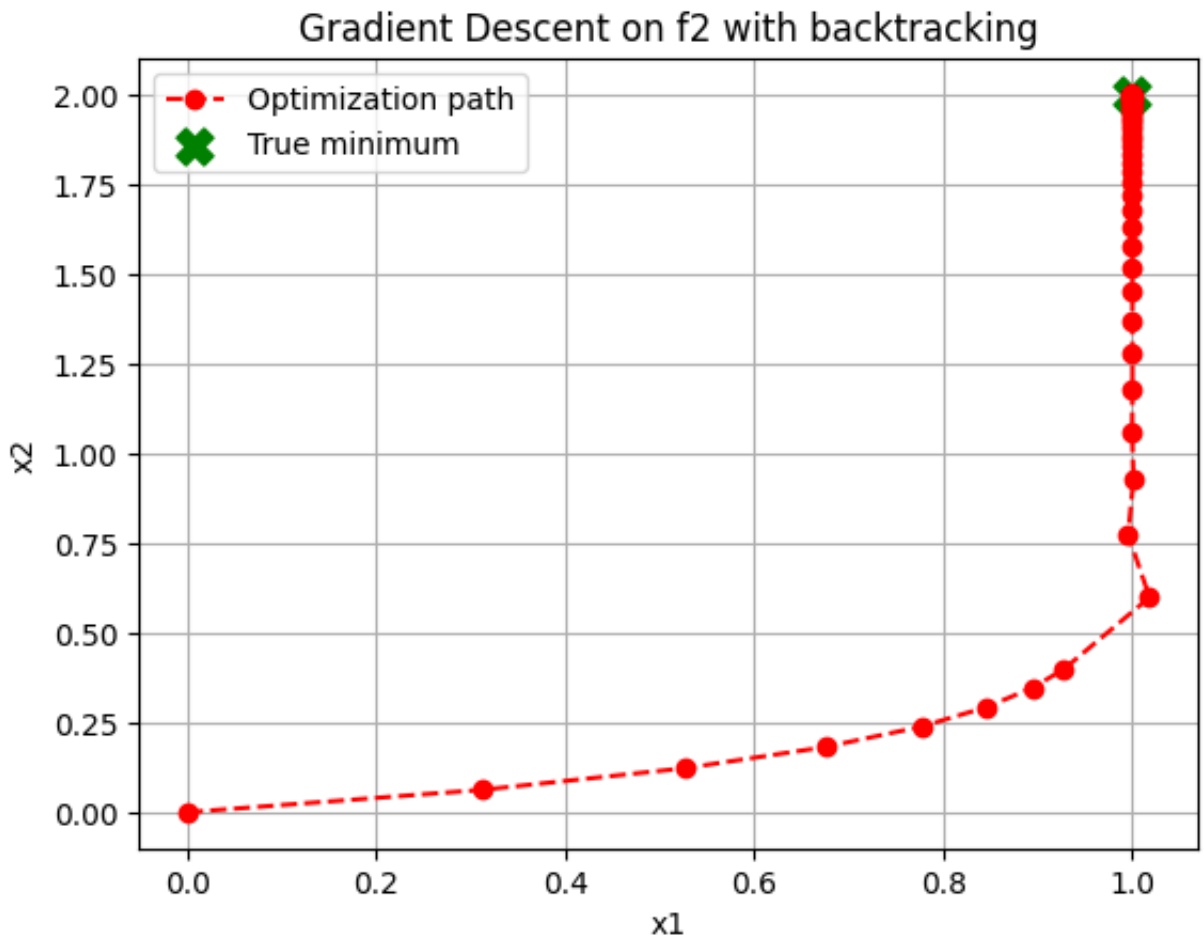

```
Minimum Found = [1.         1.99523978] with 300 iterations
```


Gradient Descent on f2 with alpha= 0.001
Optimization path
True minimum
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
x2
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x1

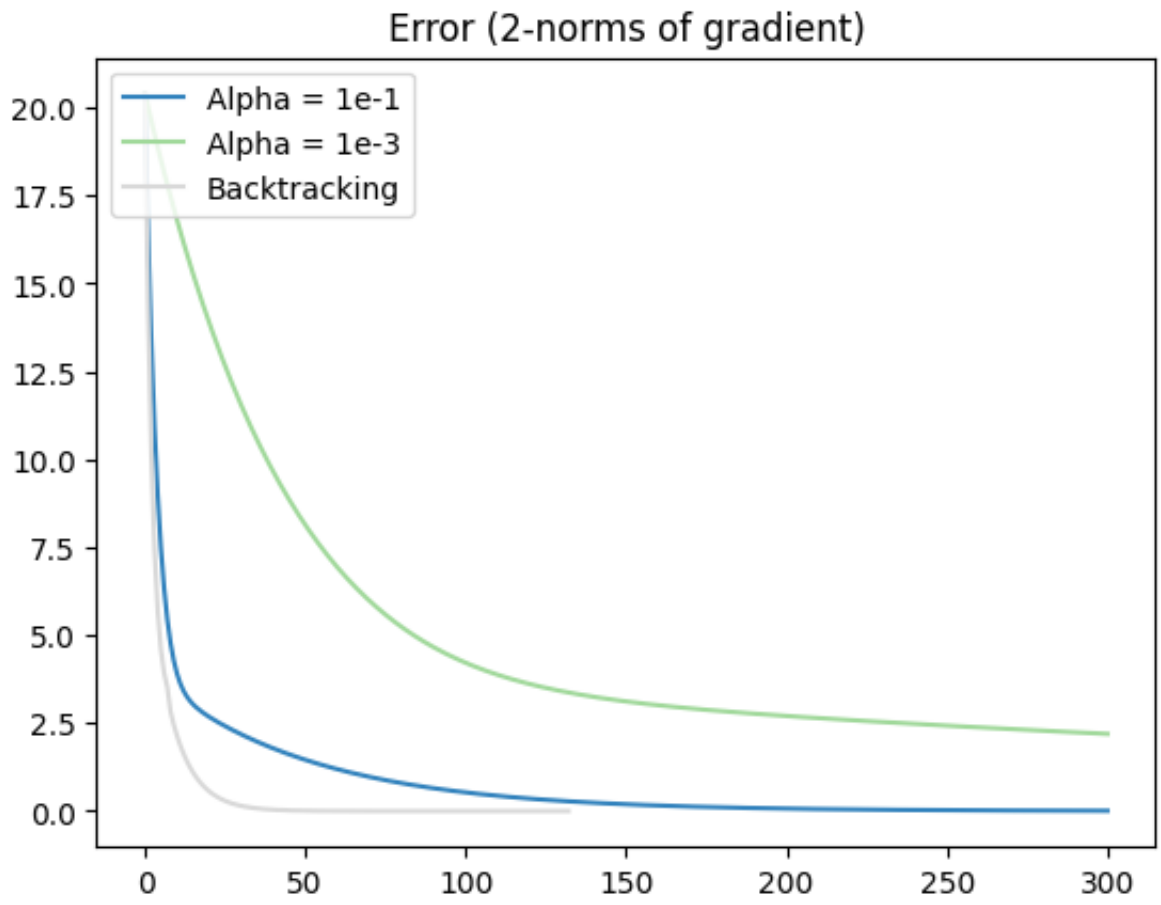


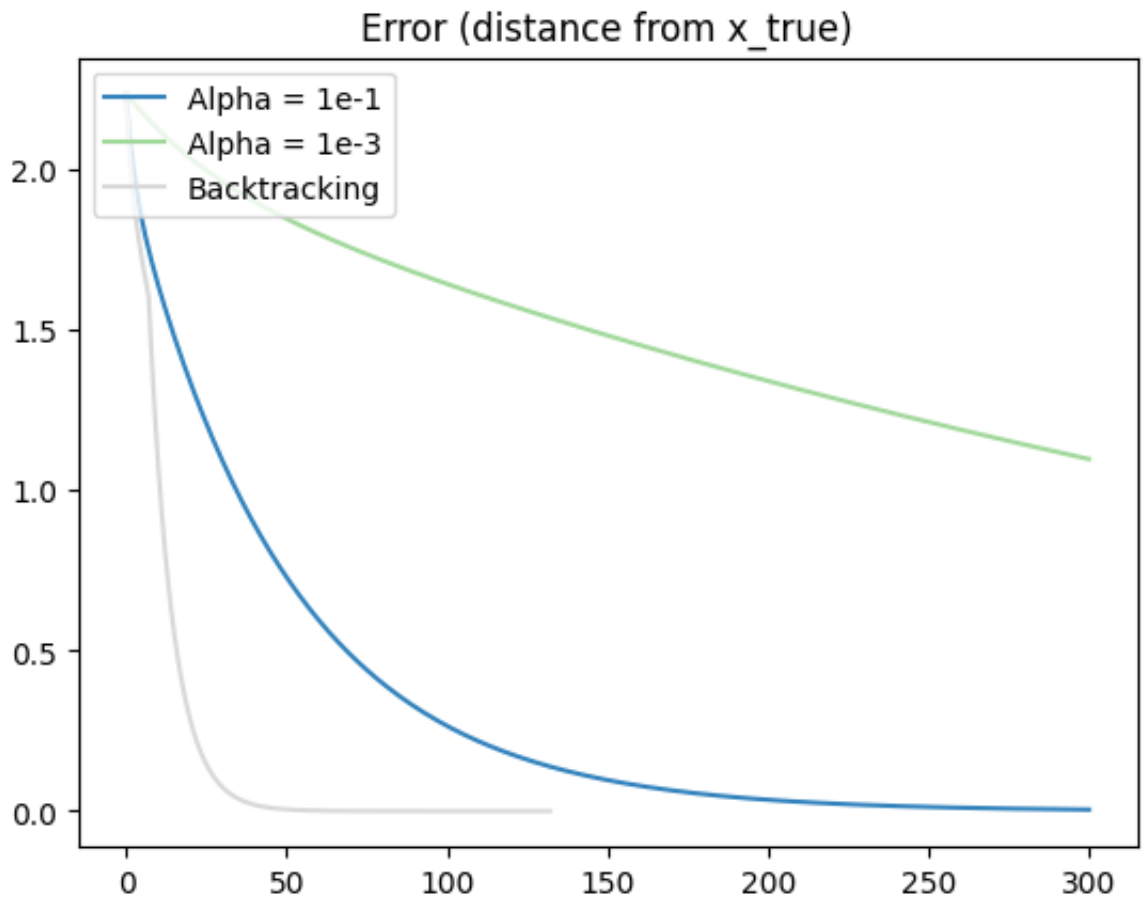

```
Minimum Found = [0.99761989 0.90083766] with 300 iterations
```


Gradient Descent on f2 with backtracking
Optimization path
True minimum
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
x2
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x1




```
Minimum Found = [1.        1.9999999] with 132 iterations
```

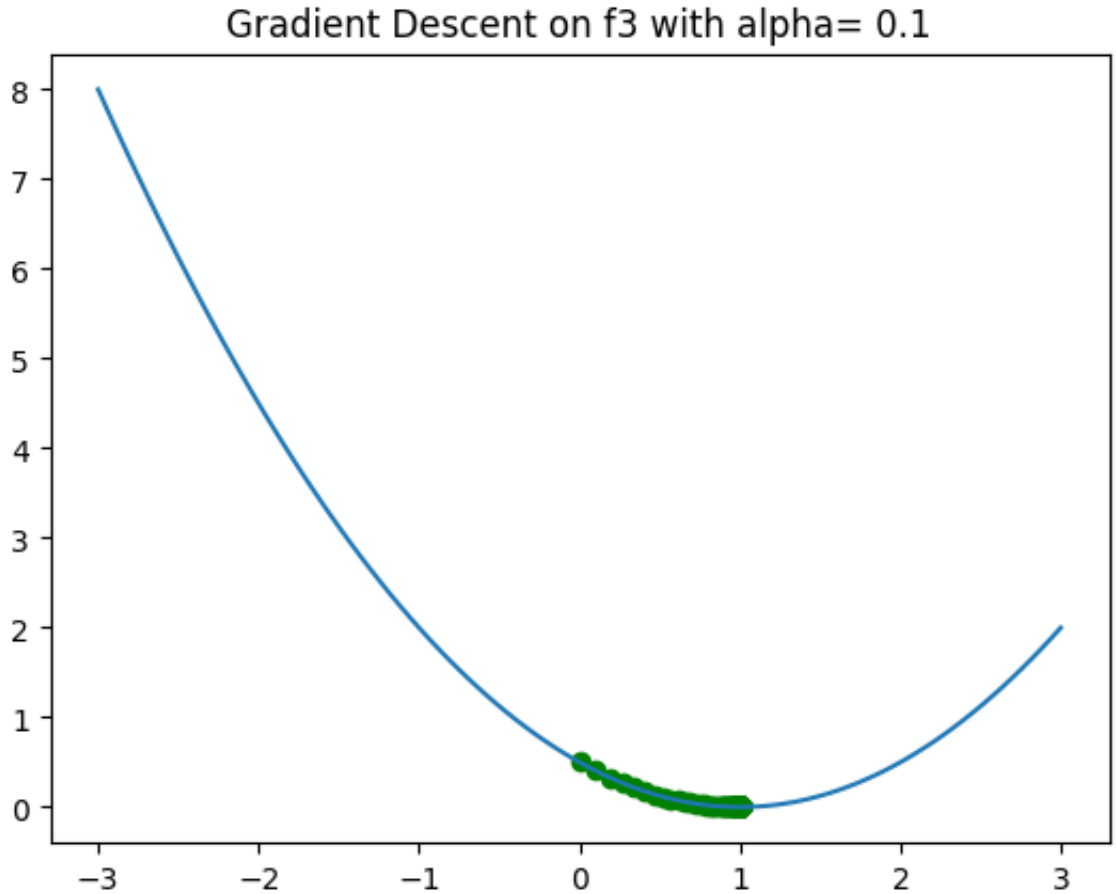

Error (2-norms of gradient)
Alpha = 1e-1
Alpha = 1e-3
Backtracking
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0
50
100
150
200
250
300



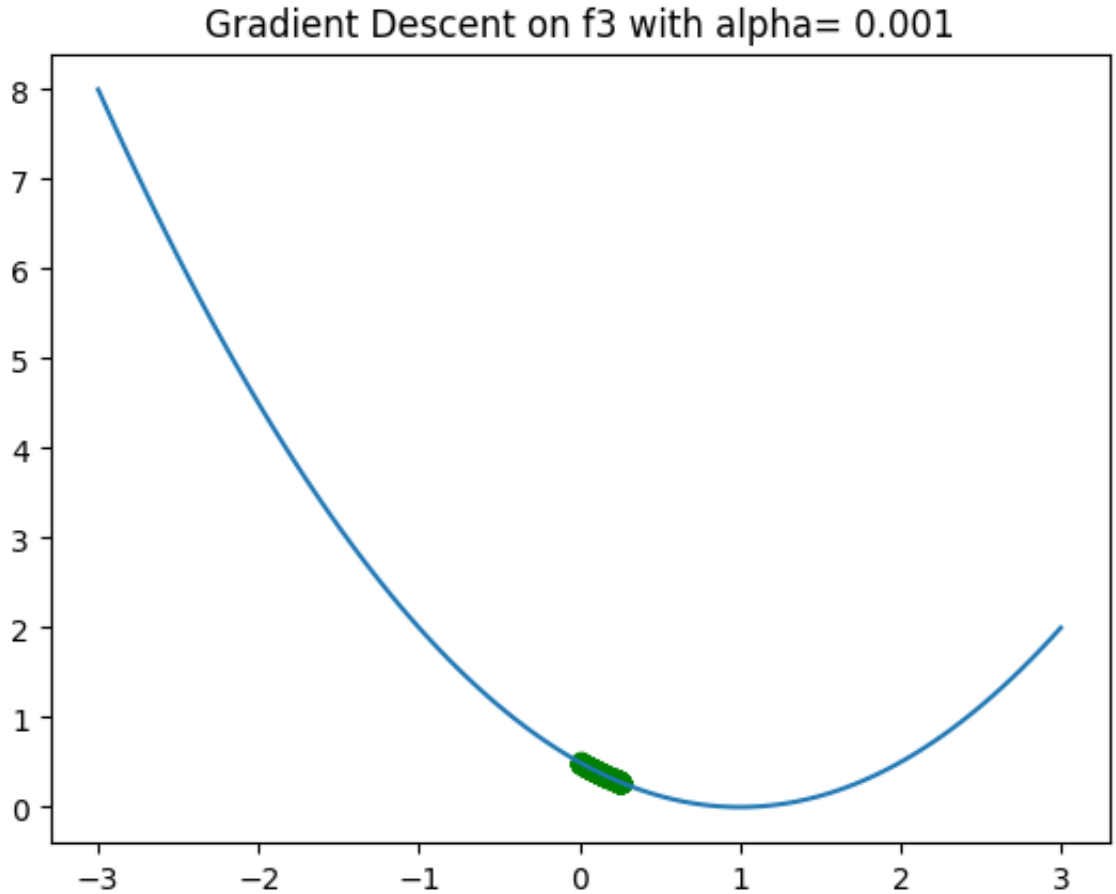
In [ ]:
```
iters = []
err_vals = []
labels = []
err_xtrue = []

print("n = 1")
xtrue = np.ones(1).T

el1, el2, el3 = function_testing(f3, grad_f3, x0 = [0], title="Gradient Desc
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-1")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f3, grad_f3, x0 = [0], title="Gradient Desc
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-3")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f3, grad_f3, x0 = [0], title="Gradient Desc
iters.append(el1)
err_vals.append(el2)
```

```
labels.append("Backtracking")
err_xtrue.append(el3)

plot_error(iters, err_vals, labels)
plot_error(iters, err_xtrue, labels, title="Error (distance from x_true)")
```

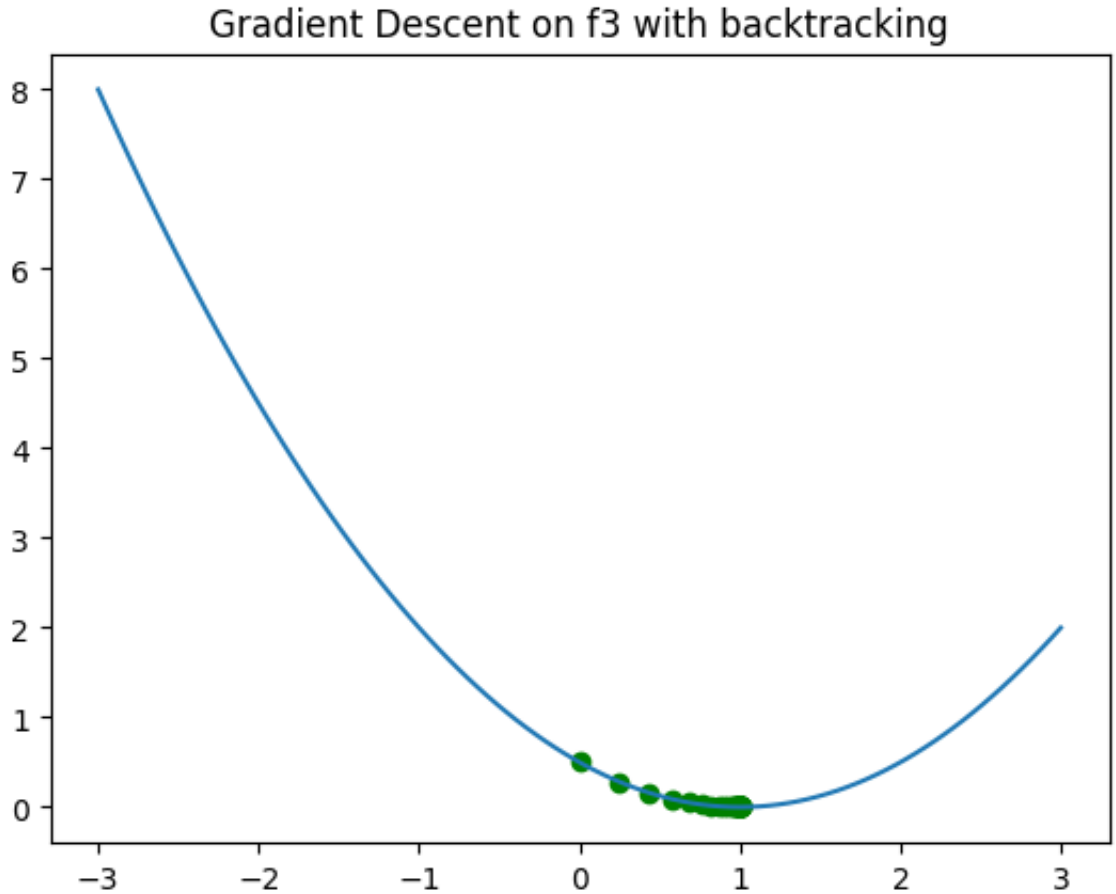
```
n = 1
```



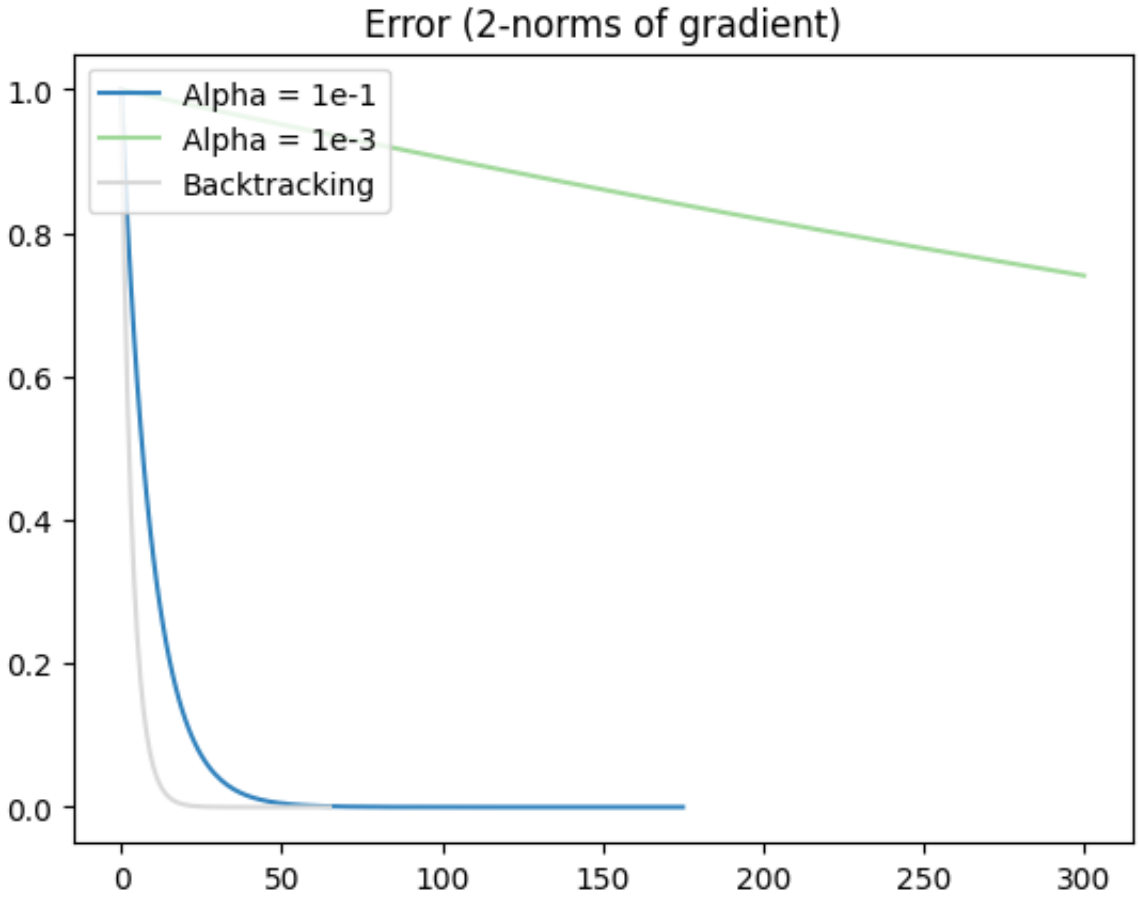

```
Minimum Found = [0.99999999] with 175 iterations
```

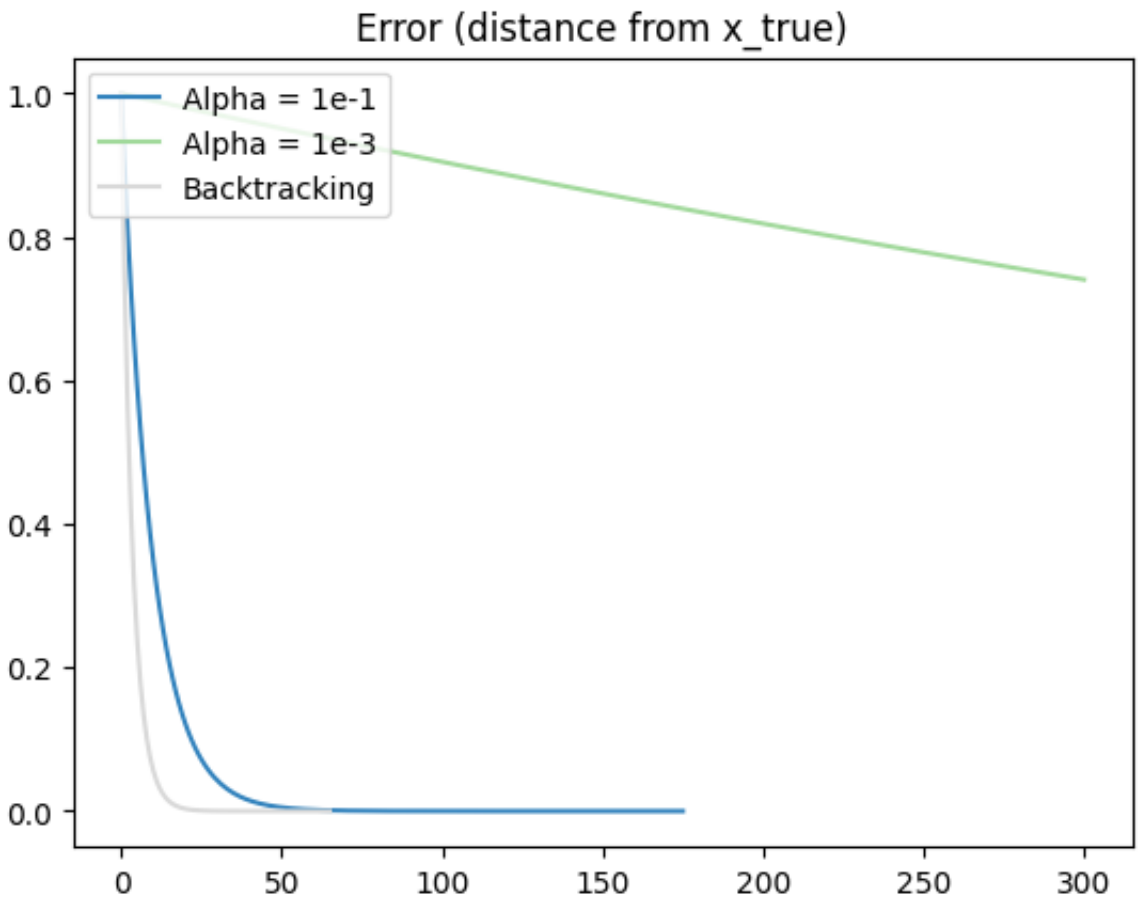



```
Minimum Found = [0.25855152] with 300 iterations
```




```
Minimum Found = [0.99999999] with 65 iterations
```

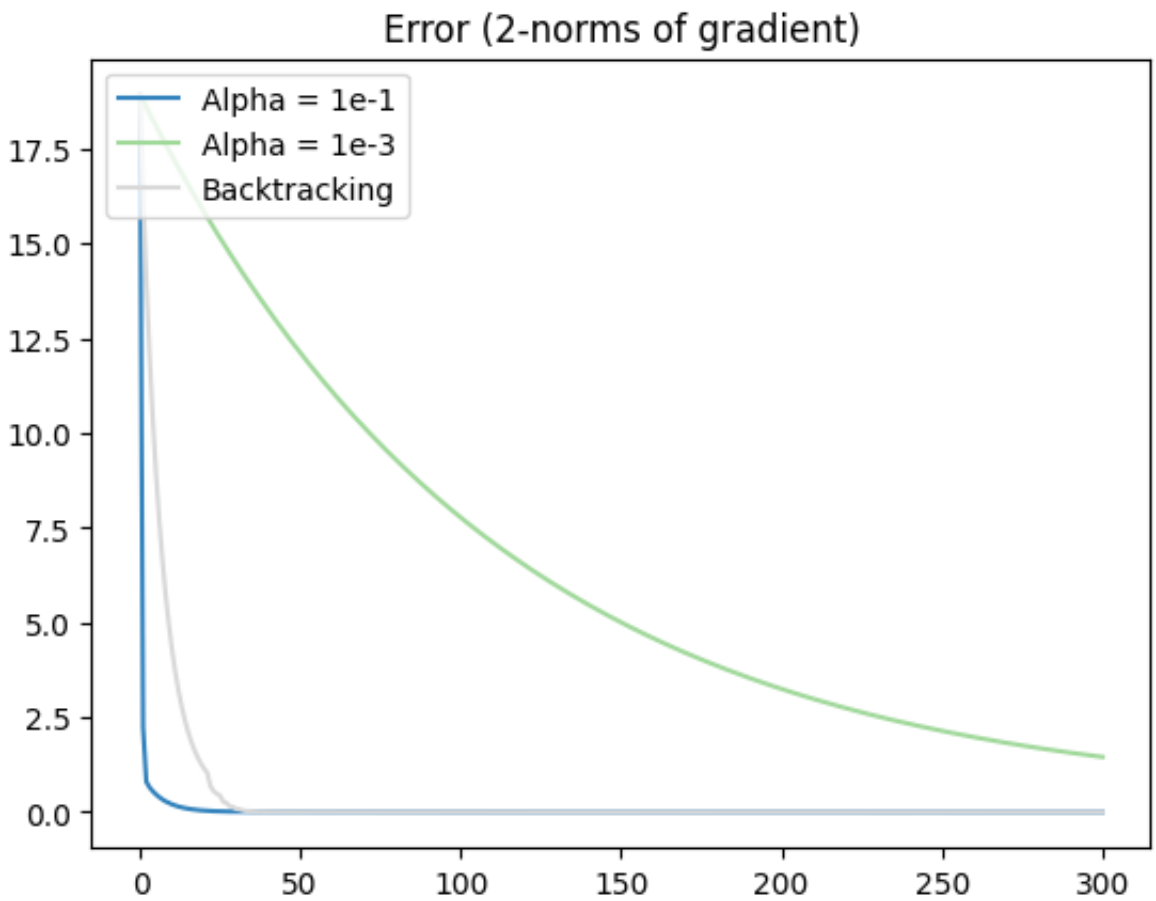

Error (2-norms of gradient)
Alpha = 1e-1
Alpha = 1e-3
Backtracking
1.0
0.8
0.6
0.4
0.2
0.0
0
50
100
150
200
250
300

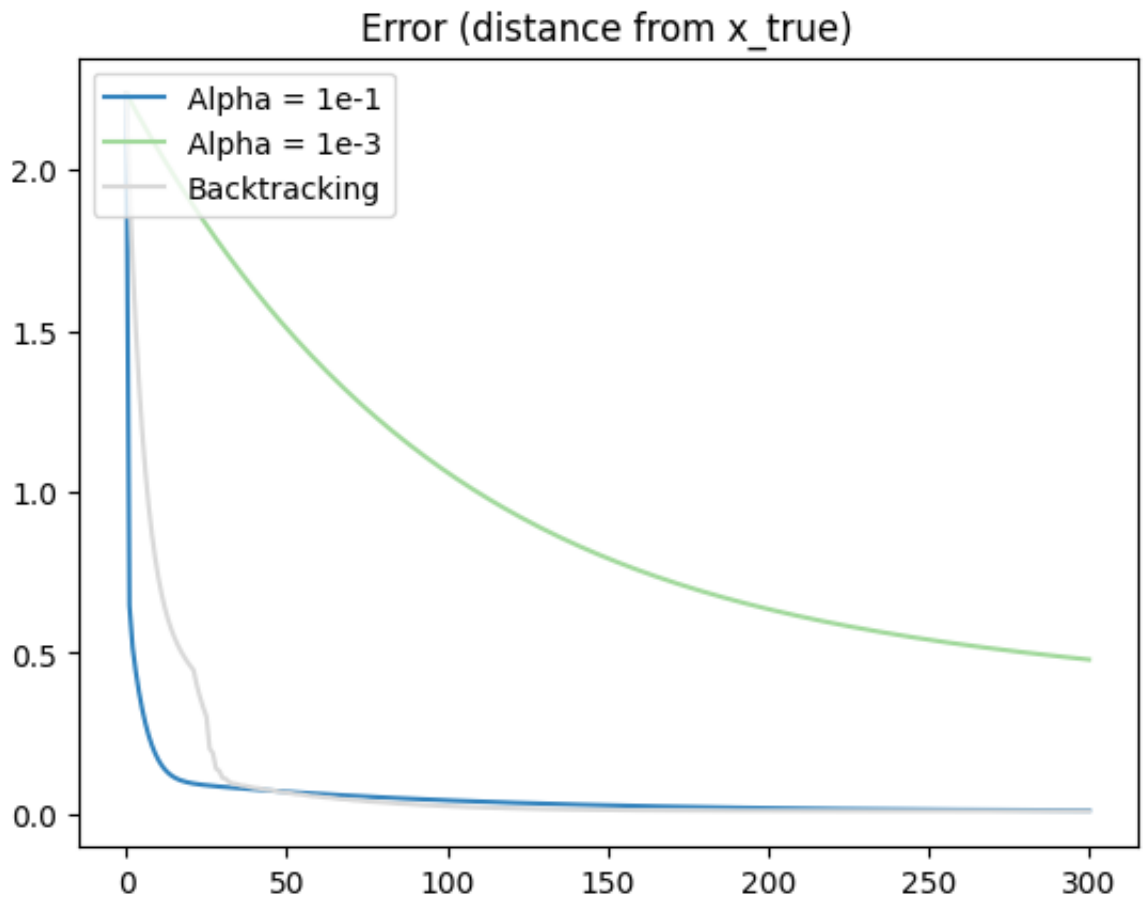


In [ ]:
```
iters = []
err_vals = []
labels = []
err_xtrue = []

print("n = 5")
xtrue = np.ones(5).T
x0 = np.zeros(5)

el1, el2, el3 = function_testing(f3, grad_f3, x0 = x0, title="Gradient Desce
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-1")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f3, grad_f3, x0 = x0, title="Gradient Desce
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-3")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f3, grad_f3, x0 = x0, title="Gradient Desce
iters.append(el1)
```

```
err_vals.append(el2)
labels.append("Backtracking")
err_xtrue.append(el3)

plot_error(iters, err_vals, labels)
plot_error(iters, err_xtrue, labels, title="Error (distance from x_true)")
```

```
n = 5
Minimum Found = [0.99163637 1.00089691 1.00736081 1.00026101 0.99911969] wit
h 300 iterations
Minimum Found = [0.70041381 0.74416544 0.81164195 0.92495403 1.18560809] wit
h 300 iterations
Minimum Found = [0.99477693 1.00214716 1.0061874  0.99660827 1.00014983] wit
h 300 iterations
```

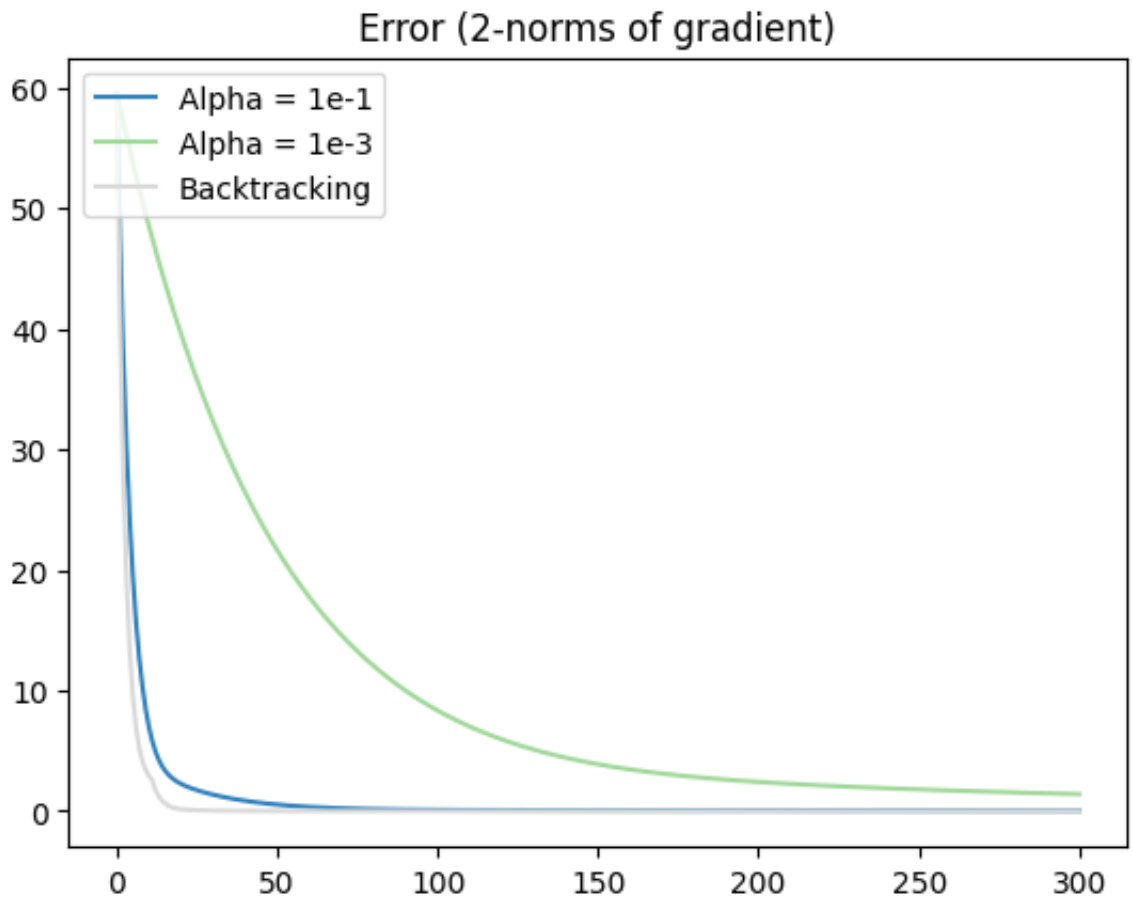


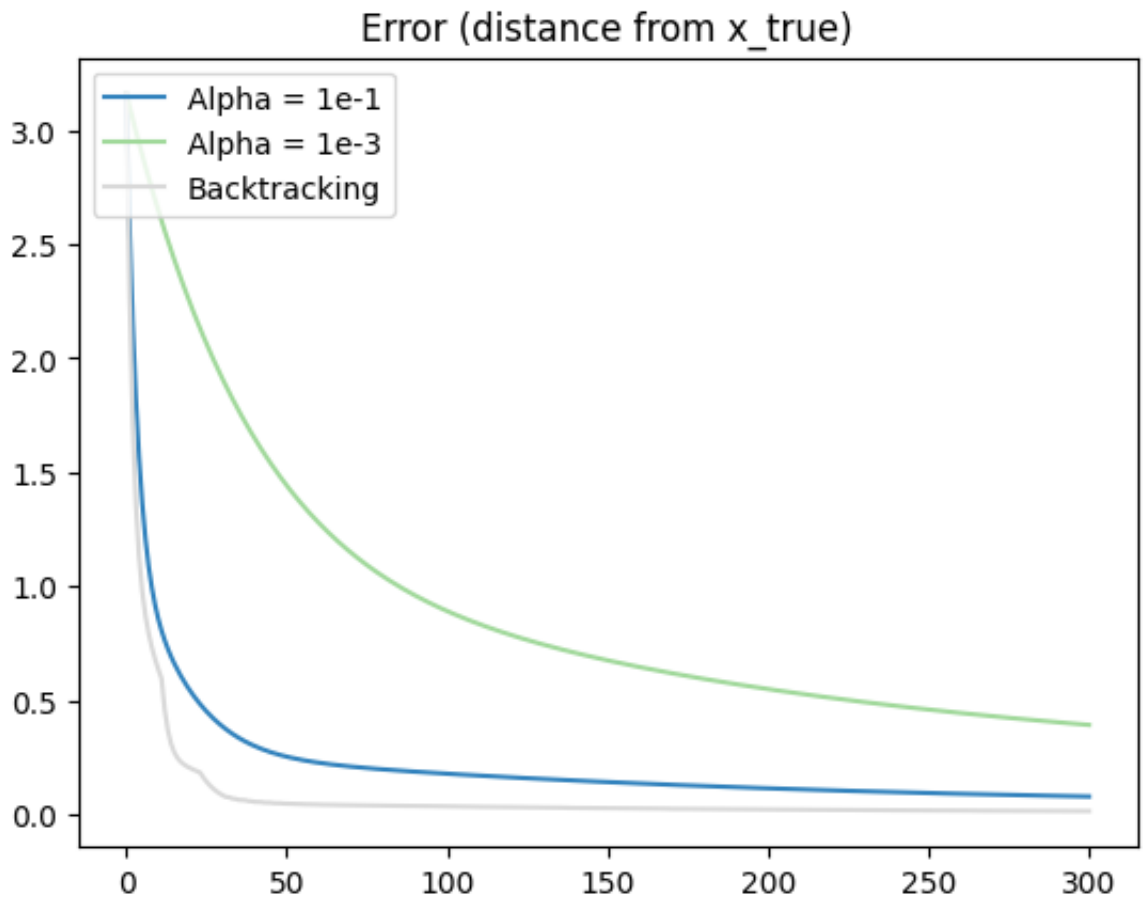


In [ ]:
```
iters = []
err_vals = []
labels = []
err_xtrue = []

print("n = 10")
xtrue = np.ones(10).T
x0 = np.zeros(10)

el1, el2, el3 = function_testing(f3, grad_f3, x0 = x0, title="Gradient Desce
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-1")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f3, grad_f3, x0 = x0, title="Gradient Desce
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-3")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f3, grad_f3, x0 = x0, title="Gradient Desce
iters.append(el1)
```

```
err_vals.append(el2)
labels.append("Backtracking")
err_xtrue.append(el3)

plot_error(iters, err_vals, labels)
plot_error(iters, err_xtrue, labels, title="Error (distance from x_true)")
```

```
n = 10
Minimum Found = [0.95648348 0.96778171 0.98009988 0.99324474 1.00672423 1.01
944161
 1.02904177 1.03046867 1.01291279 0.98395632] with 300 iterations
Minimum Found = [0.80664727 0.82637928 0.84966776 0.87735209 0.91052437 0.95
059288
 0.99926579 1.05802231 1.124548   1.13978212] with 300 iterations
Minimum Found = [0.99222119 0.99611011 0.99977037 1.00287691 1.00494811 1.00
532155
 1.0032568  0.9986253  0.99524933 1.00096048] with 300 iterations
```

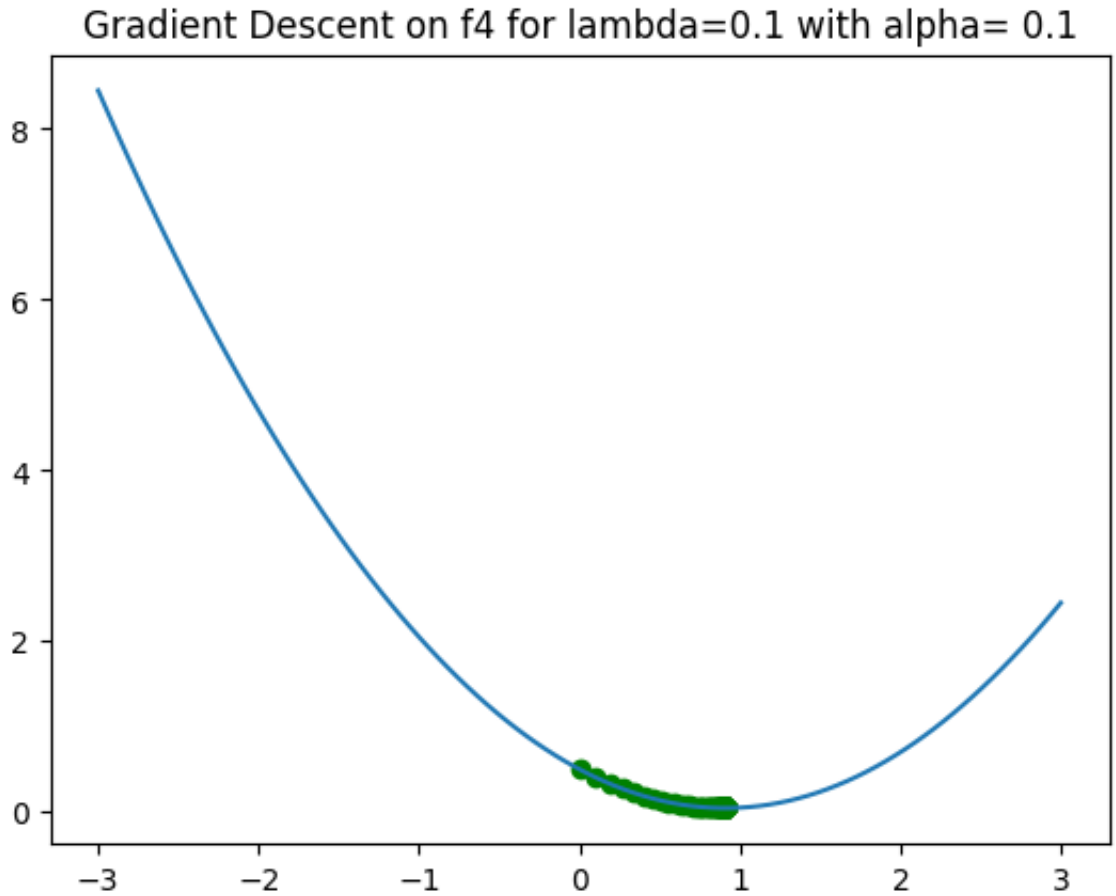




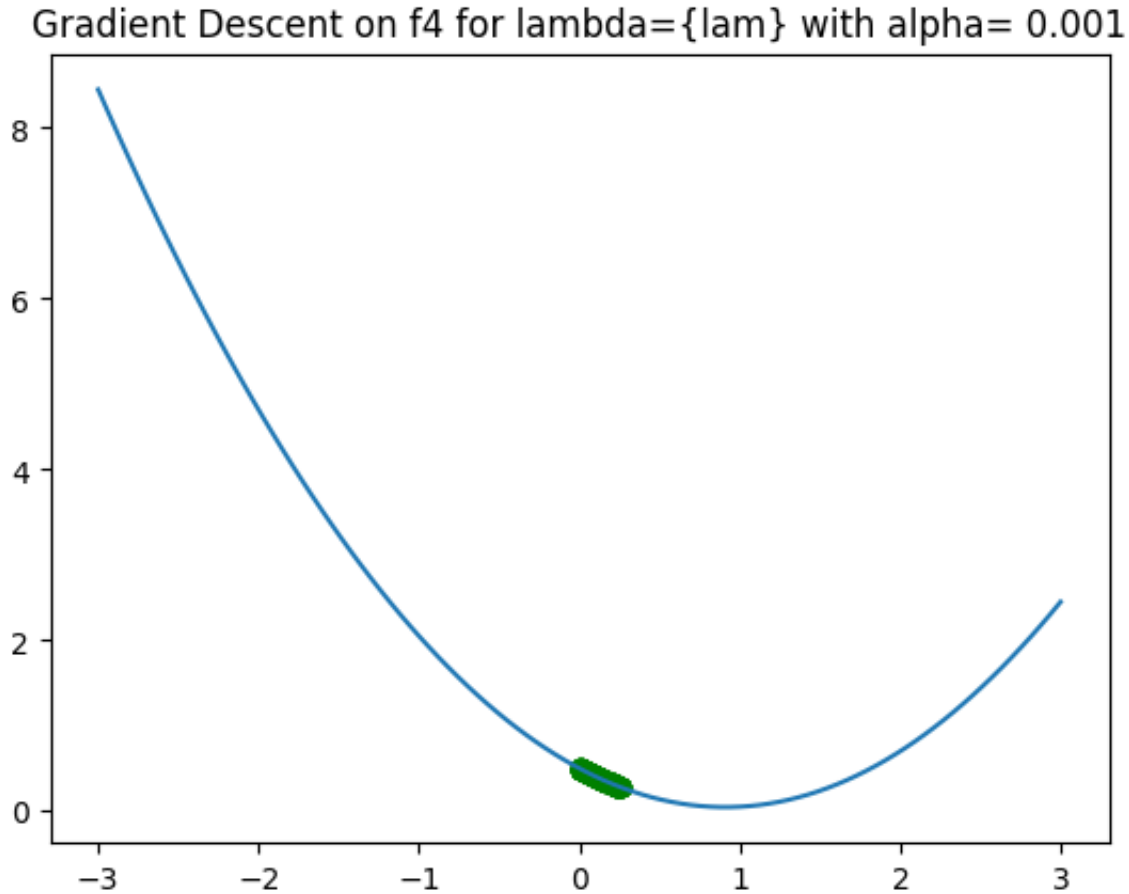
In [ ]:
```
iters = []
err_vals = []
labels = []
err_xtrue = []
lam =[0.1,0.5,0.9]

print("n = 1")
xtrue = np.ones(1).T
for i in lam:
    lam = i

    el1, el2, el3 = function_testing(f4, grad_f4, x0 = [0], title=f"Gradient

    iters.append(el1)
    err_vals.append(el2)
    labels.append(f"Alpha = 1e-1 and lambda = {lam}")
    err_xtrue.append(el3)

    el1, el2, el3 = function_testing(f4, grad_f4, x0 = [0], title="Gradient
    iters.append(el1)
    err_vals.append(el2)
    labels.append("Alpha = 1e-3")
    err_xtrue.append(el3)
```

```
    el1, el2, el3 = function_testing(f4, grad_f4, x0 = [0], title="Gradient
    iters.append(el1)
    err_vals.append(el2)
    labels.append("Backtracking")
    err_xtrue.append(el3)

plot_error(iters, err_vals, labels)
plot_error(iters, err_xtrue, labels, title="Error (distance from x_true)")
```

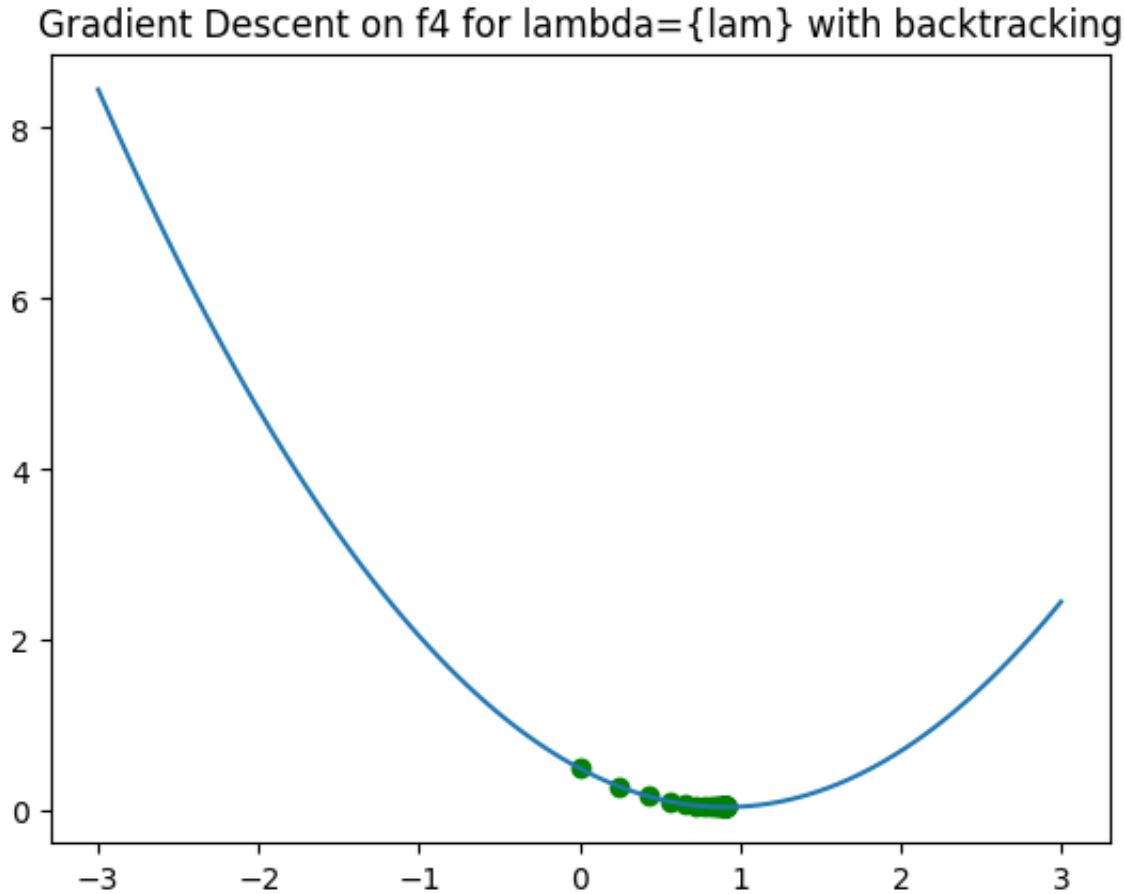
```
n = 1
```



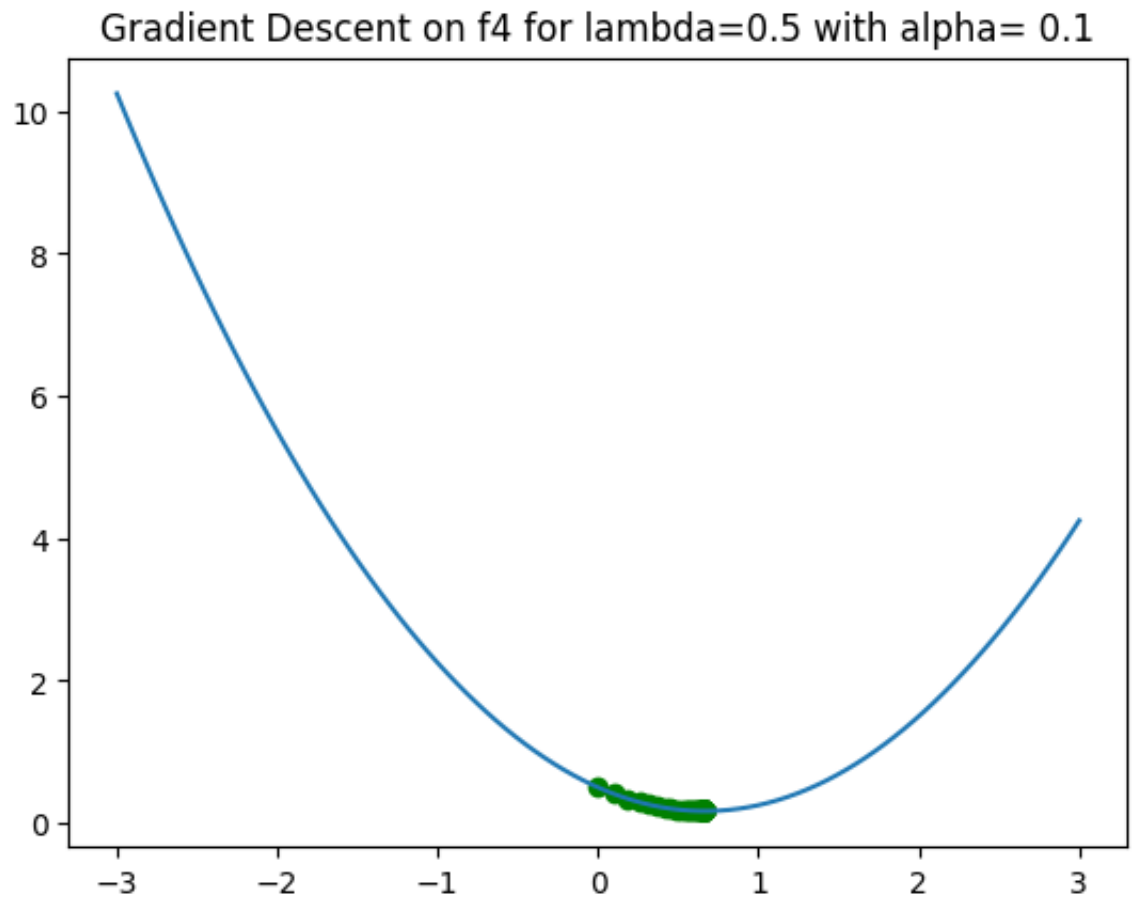

```
Minimum Found = [0.9090909] with 159 iterations
```



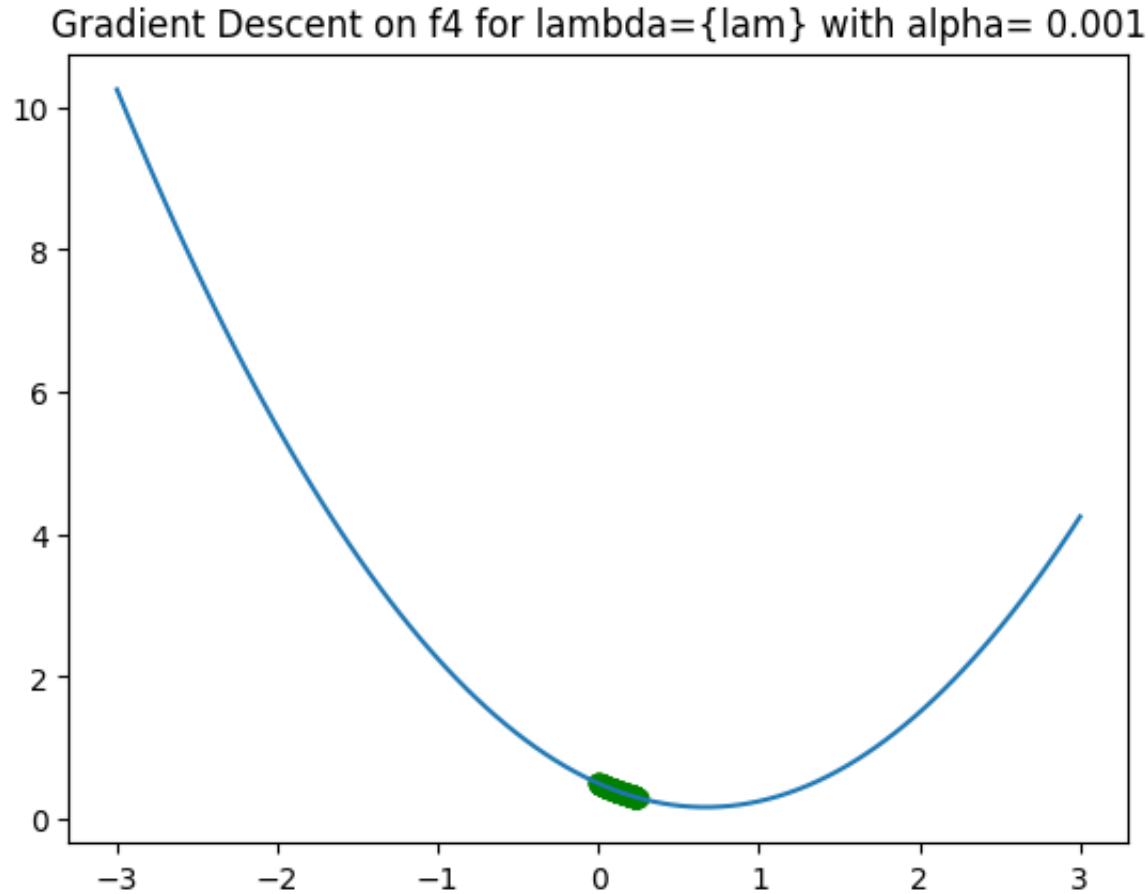

```
Minimum Found = [0.25492299] with 300 iterations
```



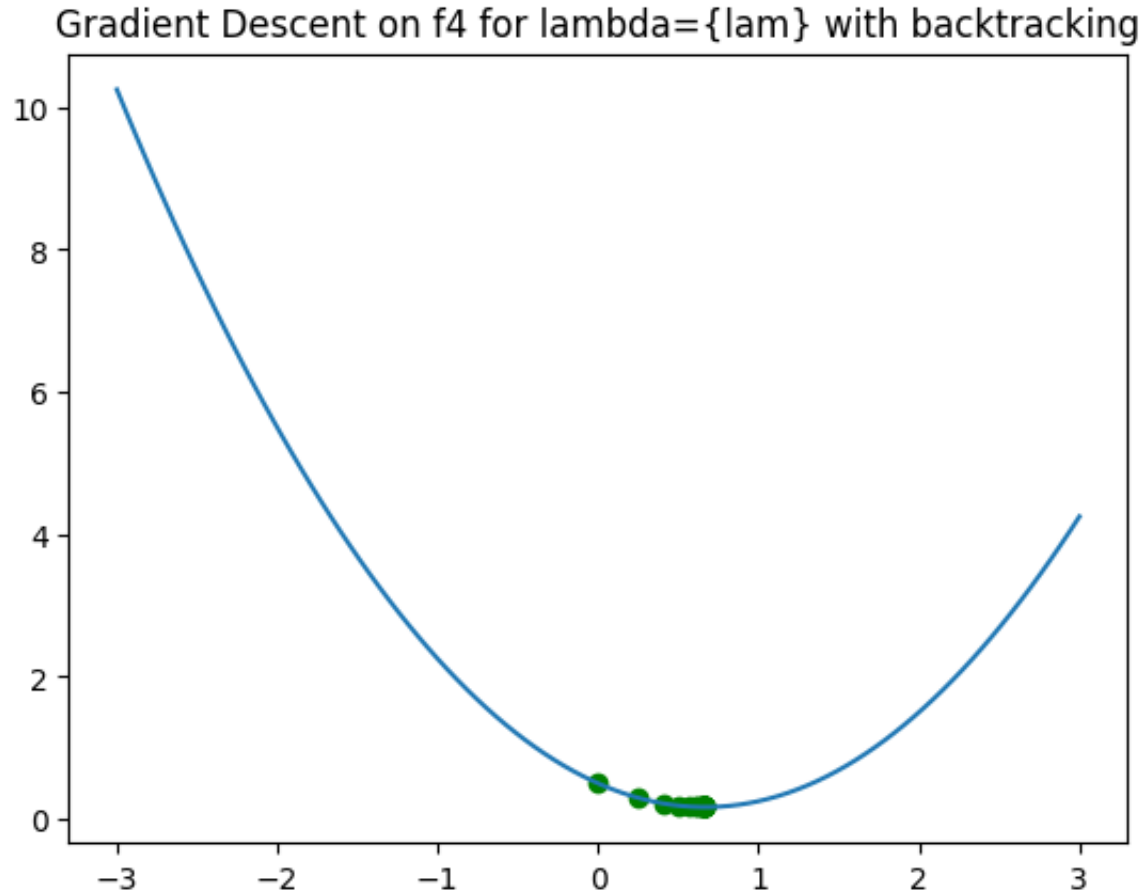

```
Minimum Found = [0.9090909] with 58 iterations
```



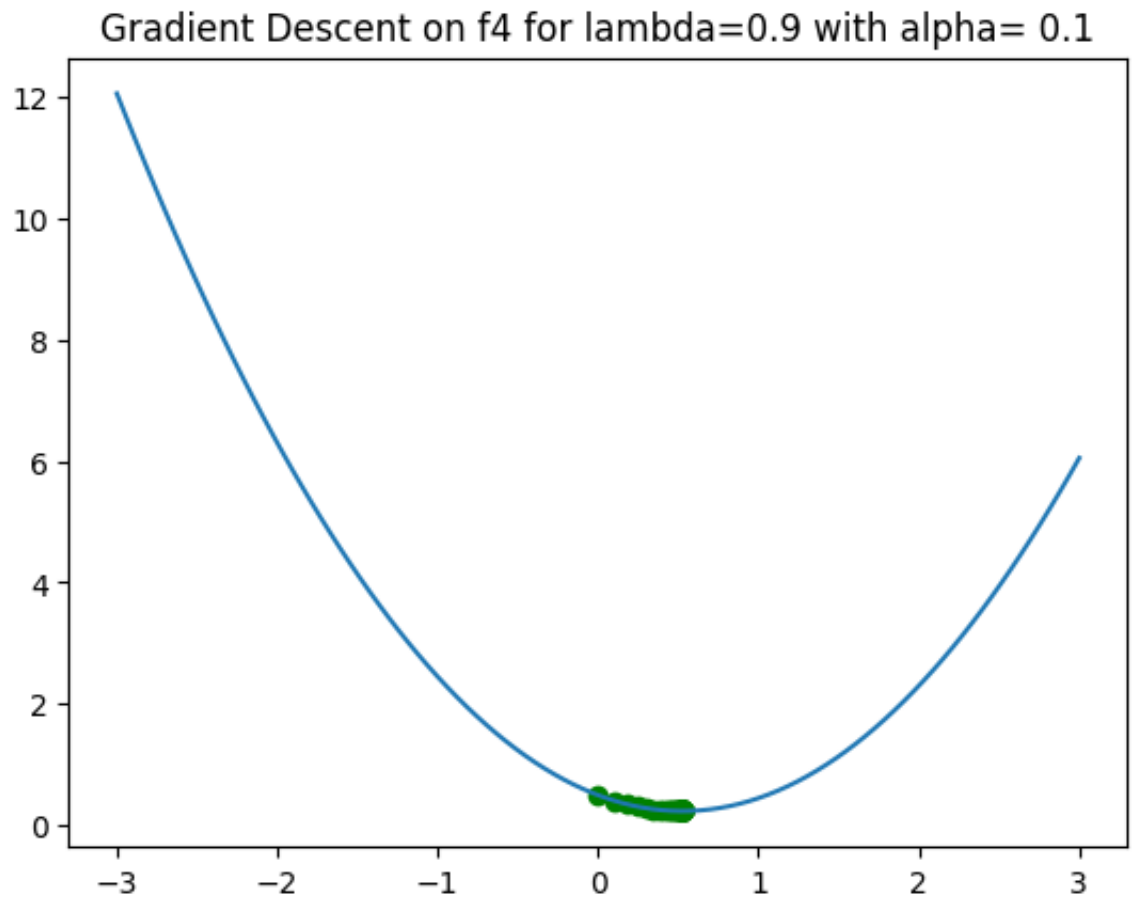

```
Minimum Found = [0.66666666] with 114 iterations
```



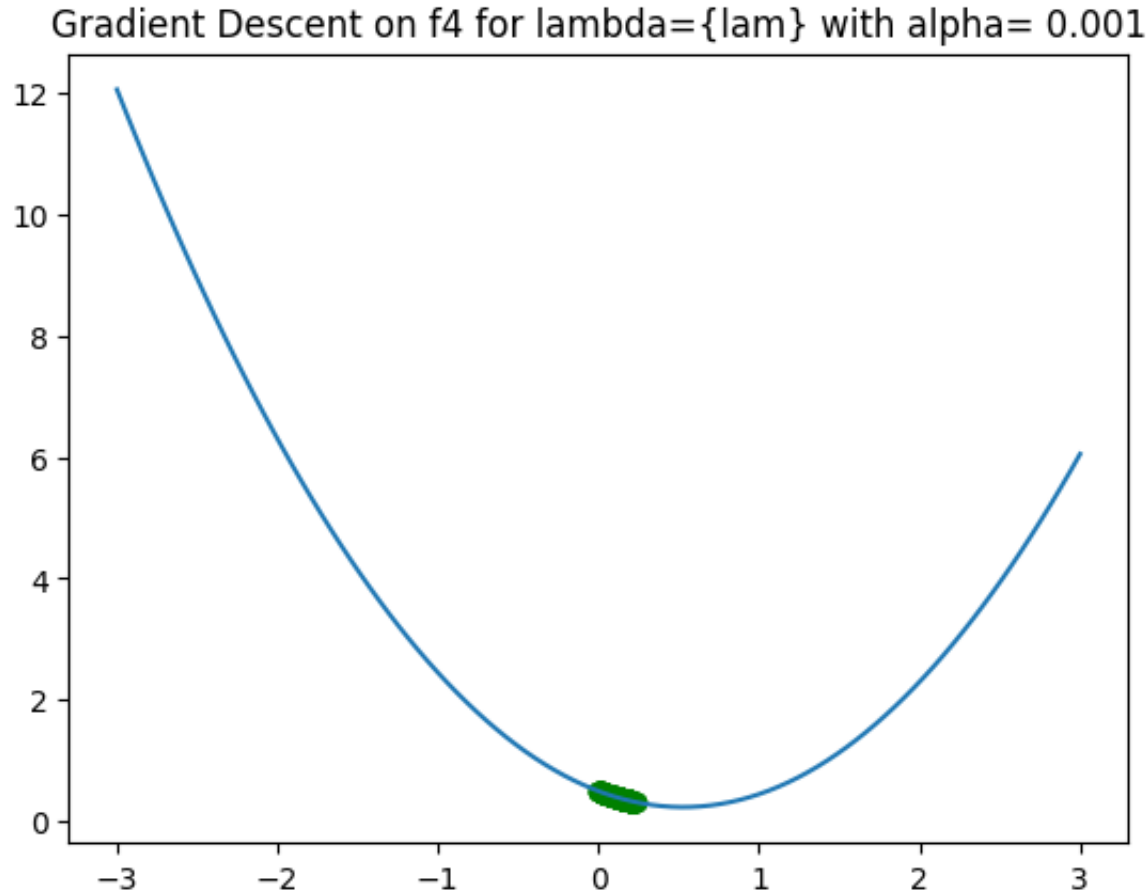

```
Minimum Found = [0.24108645] with 300 iterations
```



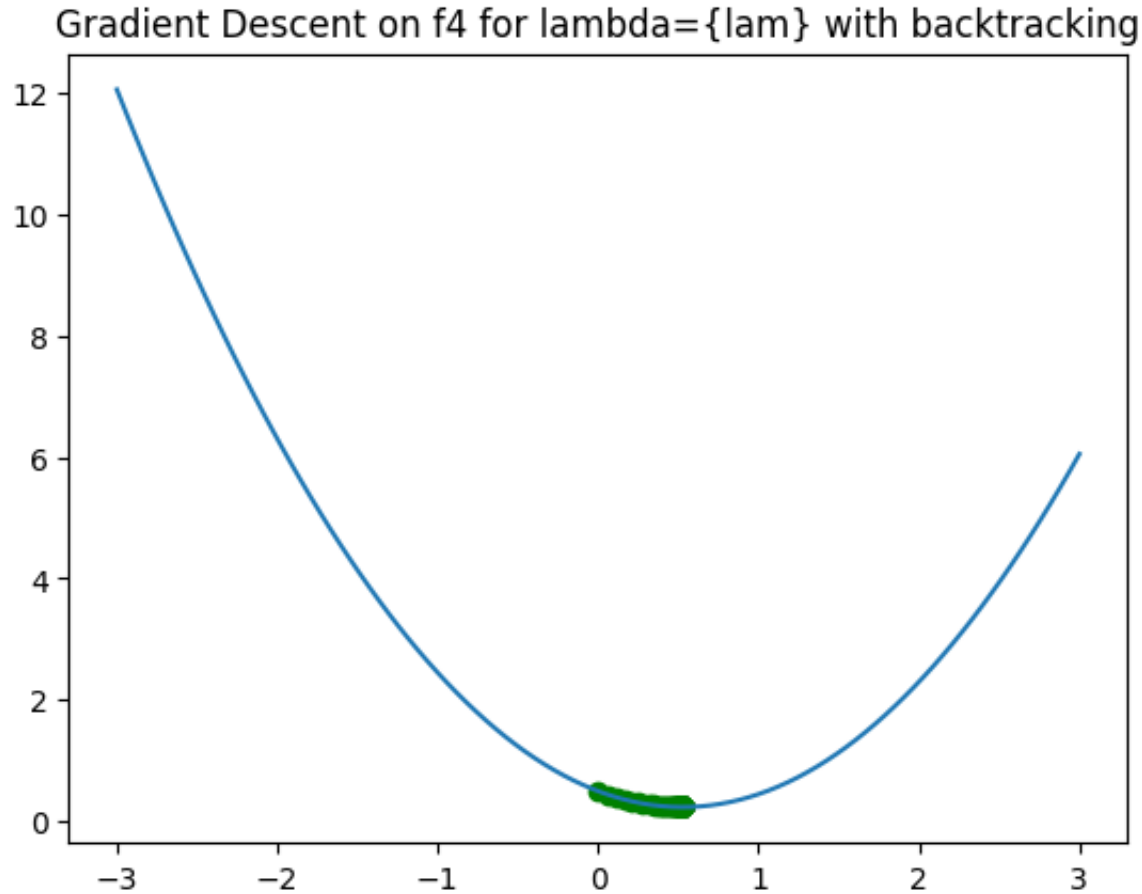

```
Minimum Found = [0.66666666] with 40 iterations
```



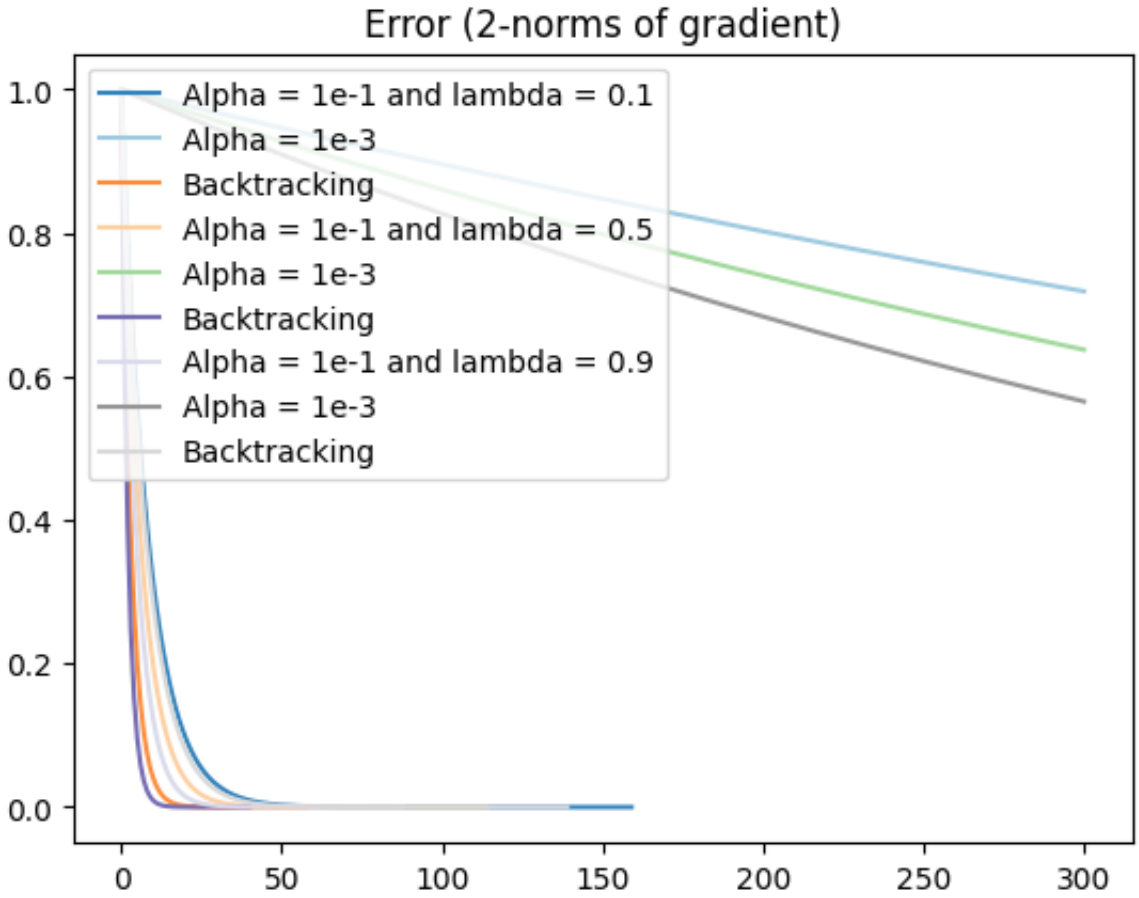

```
Minimum Found = [0.52631578] with 88 iterations
```

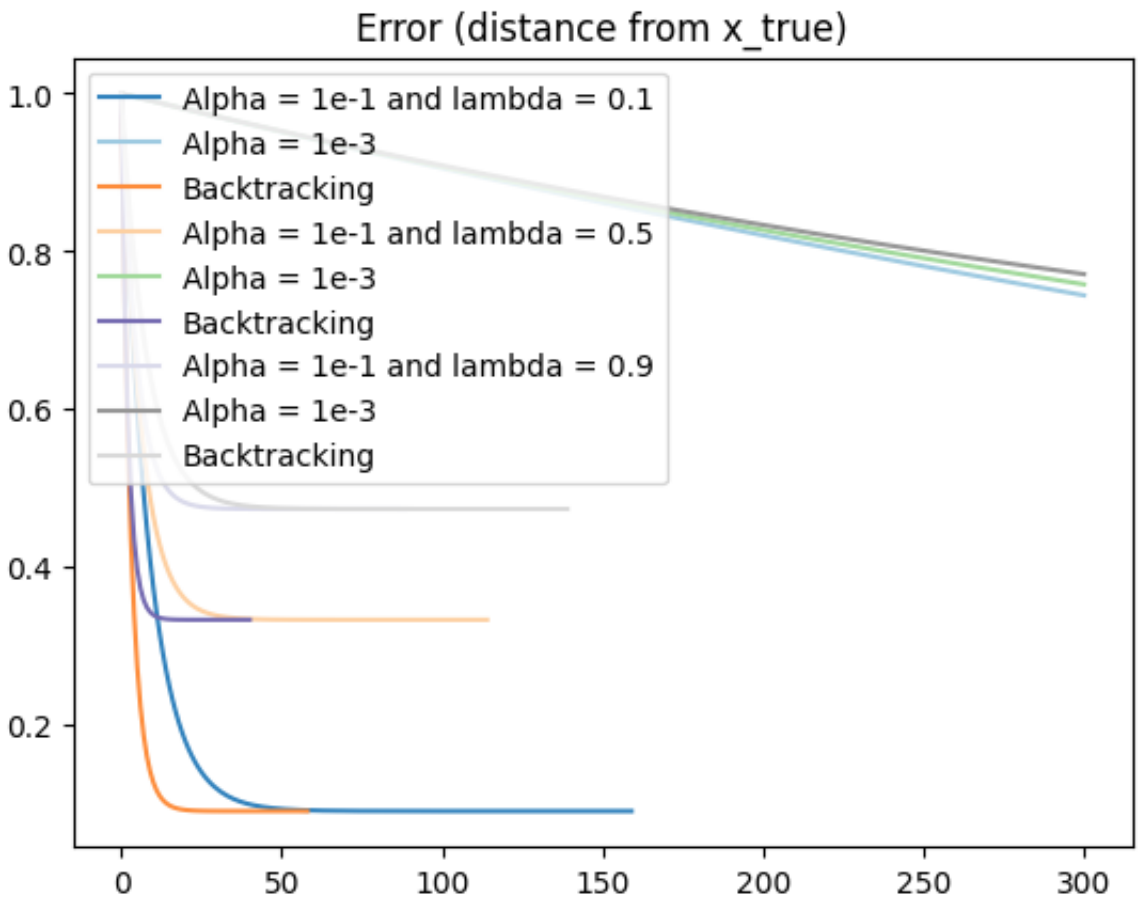



```
Minimum Found = [0.22826586] with 300 iterations
```




```
Minimum Found = [0.52631578] with 139 iterations
```

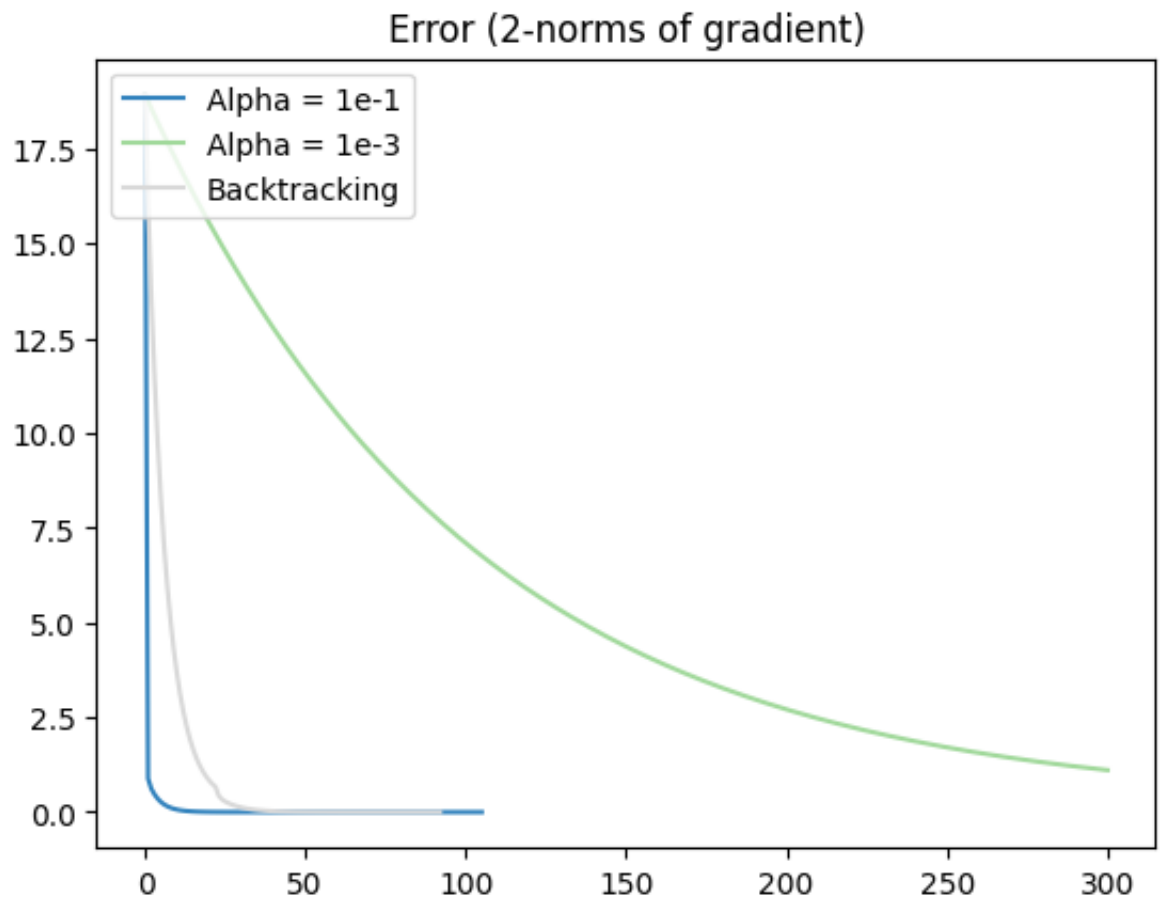

Error (2-norms of gradient)
Alpha = 1e-1 and lambda = 0.1
Alpha = 1e-3
Backtracking
Alpha = 1e-1 and lambda = 0.5
Alpha = 1e-3
Backtracking
Alpha = 1e-1 and lambda = 0.9
Alpha = 1e-3
Backtracking
1.0
0.8
0.6
0.4
0.2
0.0
0
50
100
150
200
250
300

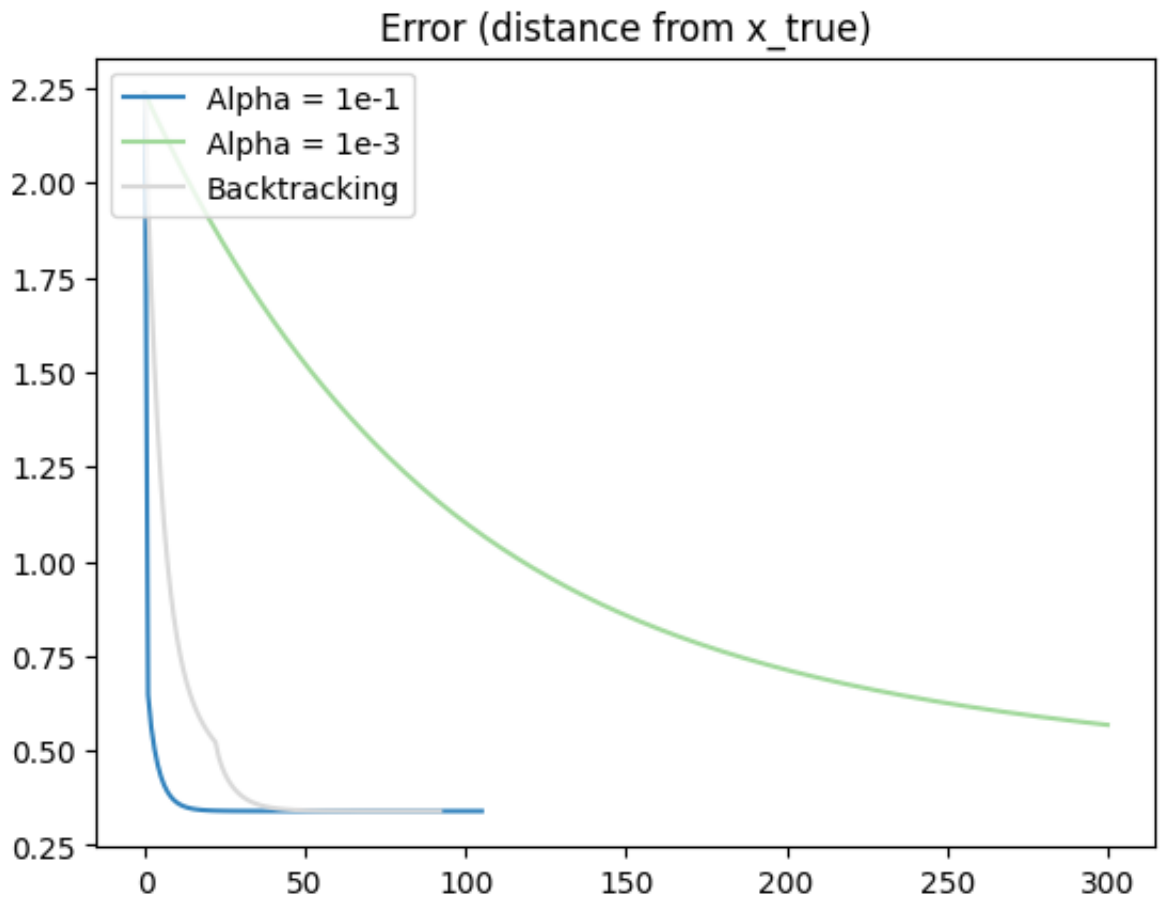


In [ ]:
```
iters = []
err_vals = []
labels = []
err_xtrue = []

print("n = 5")
xtrue = np.ones(5).T
x0 = np.zeros(5)

el1, el2, el3 = function_testing(f4, grad_f4, x0 = x0, title="Gradient Desce
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-1")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f4, grad_f4, x0 = x0, title="Gradient Desce
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-3")
err_xtrue.append(el3)

el1, el2, el3 = function_testing(f4, grad_f4, x0 = x0, title="Gradient Desce
iters.append(el1)
```

```
err_vals.append(el2)
labels.append("Backtracking")
err_xtrue.append(el3)

plot_error(iters, err_vals, labels)
plot_error(iters, err_xtrue, labels, title="Error (distance from x_true)")
```

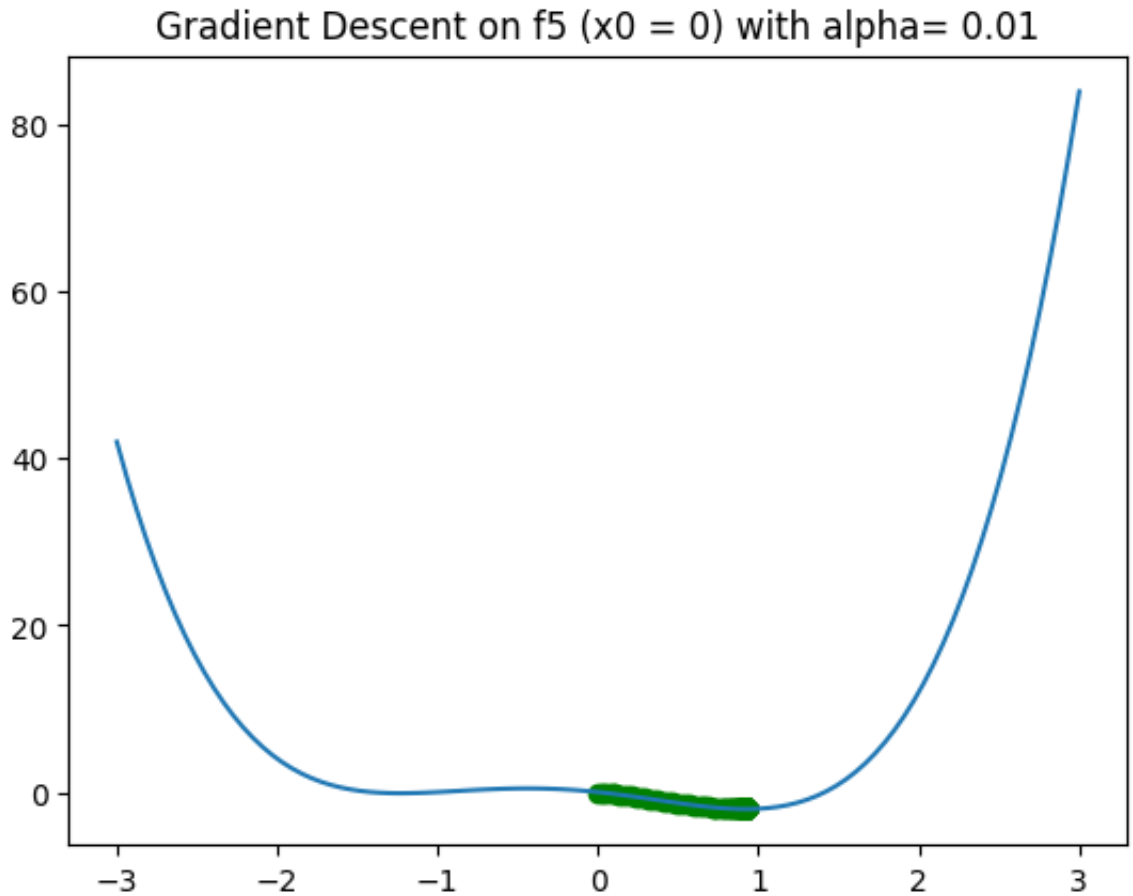
```
n = 5
Minimum Found = [0.76993005 0.80585415 0.85763082 0.93466091 1.02572651] wit
h 105 iterations
Minimum Found = [0.64299981 0.68372015 0.74670989 0.85301257 1.10266371] wit
h 300 iterations
Minimum Found = [0.76993003 0.80585414 0.85763082 0.93466094 1.0257265 ] wit
h 92 iterations
```



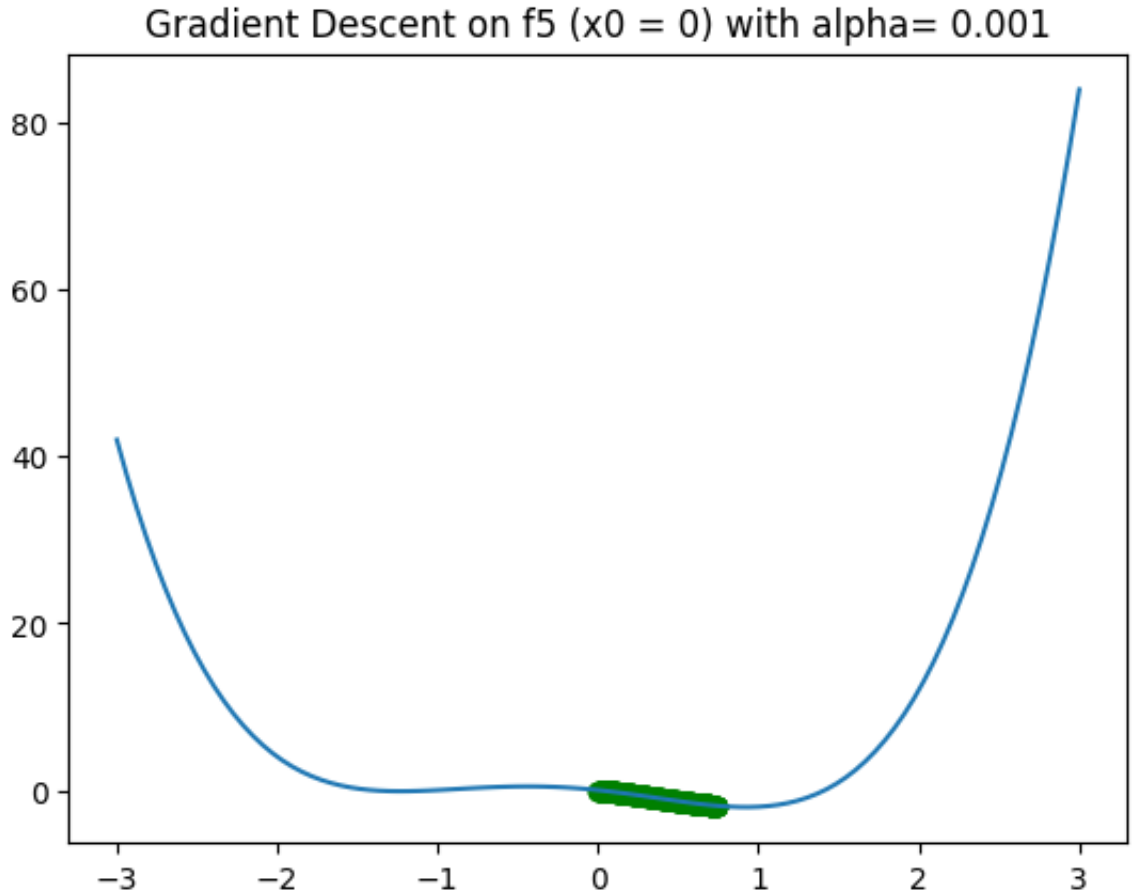


In [ ]:
```
iters = []
err_vals = []
labels = []

el1, el2 = function_testing(f5, grad_f5, x0 = [0], title="Gradient Descent c
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-1 (x0 = 0)")

el1, el2 = function_testing(f5, grad_f5, x0 = [0], title="Gradient Descent c
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-3 (x0 = 0)")

el1, el2 = function_testing(f5, grad_f5, x0 = [0], title="Gradient Descent c
iters.append(el1)
err_vals.append(el2)
labels.append("Backtracking (x0 = 0)")

plot_error(iters, err_vals, labels)
iters = []
err_vals = []
labels = []
```

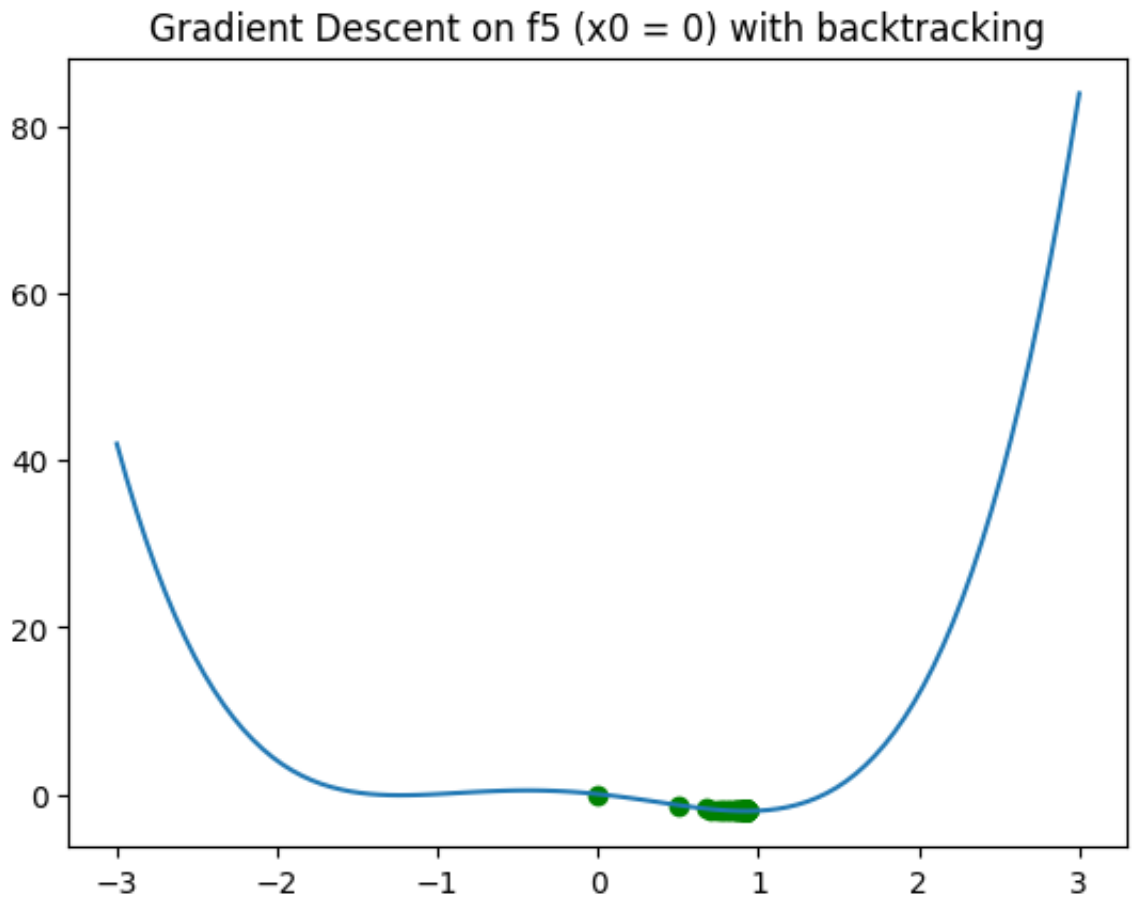
```
el1, el2 = function_testing(f5, grad_f5, x0 = [-3], title="Gradient Descent
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-1 (x0 = -3)")

el1, el2 = function_testing(f5, grad_f5, x0 = [-3], title="Gradient Descent
iters.append(el1)
err_vals.append(el2)
labels.append("Alpha = 1e-3 (x0 = -3)")

el1, el2 = function_testing(f5, grad_f5, x0 = [-3], title="Gradient Descent
iters.append(el1)
err_vals.append(el2)
labels.append("Backtracking (x0 = -3)")

plot_error(iters, err_vals, labels)
```



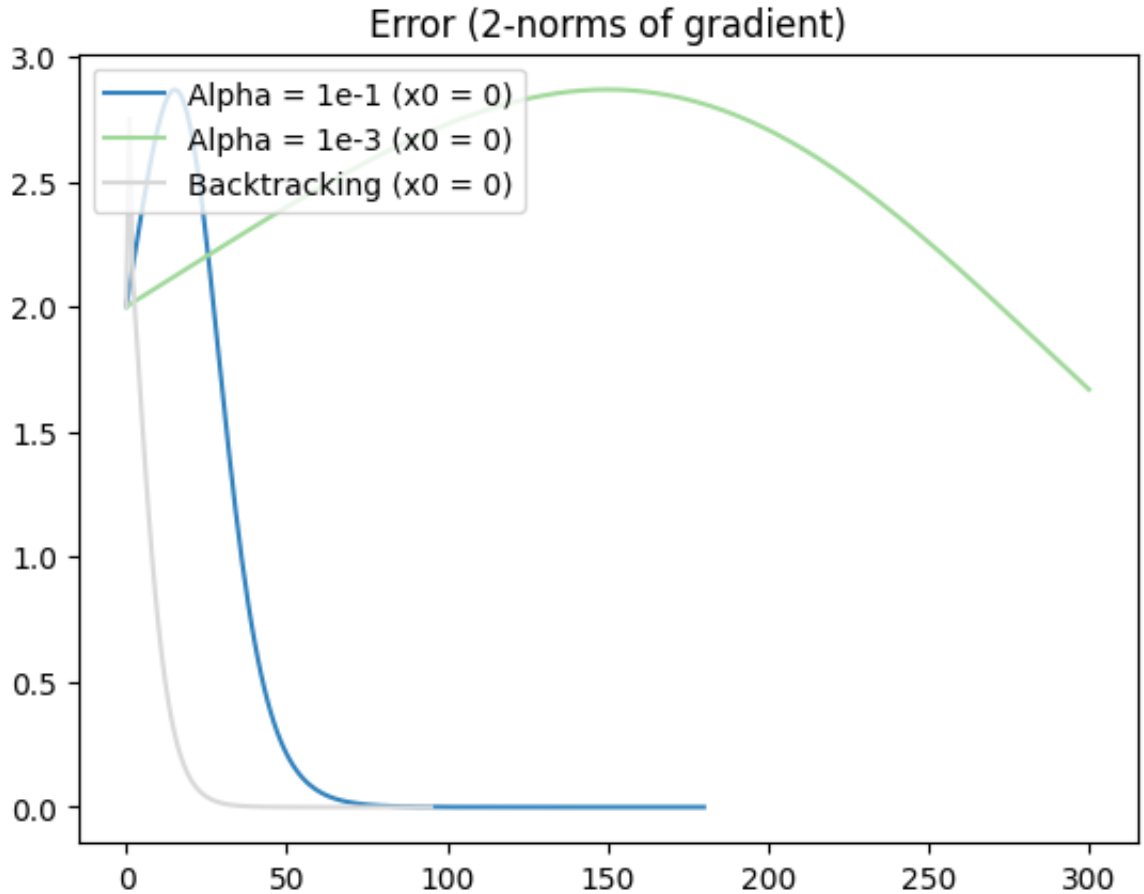

```
Minimum Found = [0.92222479] with 180 iterations
```

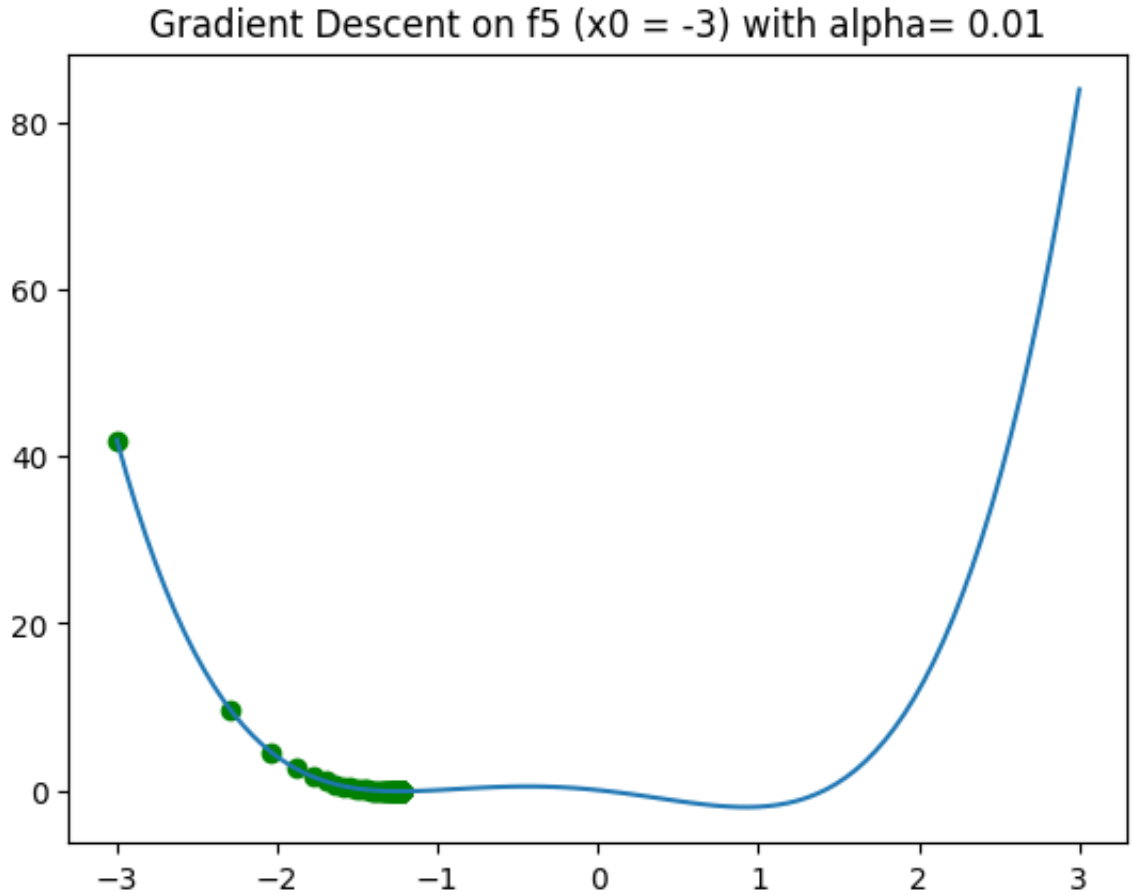



```
Minimum Found = [0.74222729] with 300 iterations
```

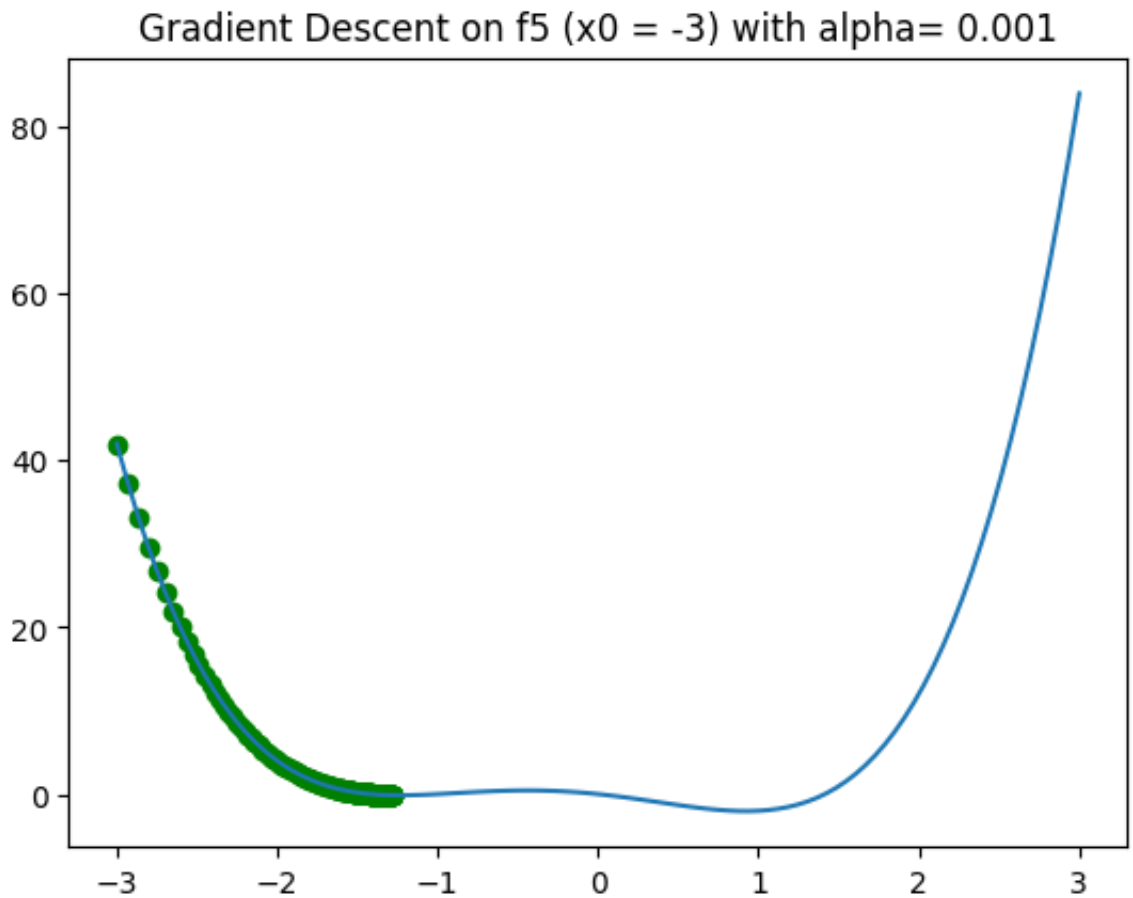



```
Minimum Found = [0.9222248] with 95 iterations
```

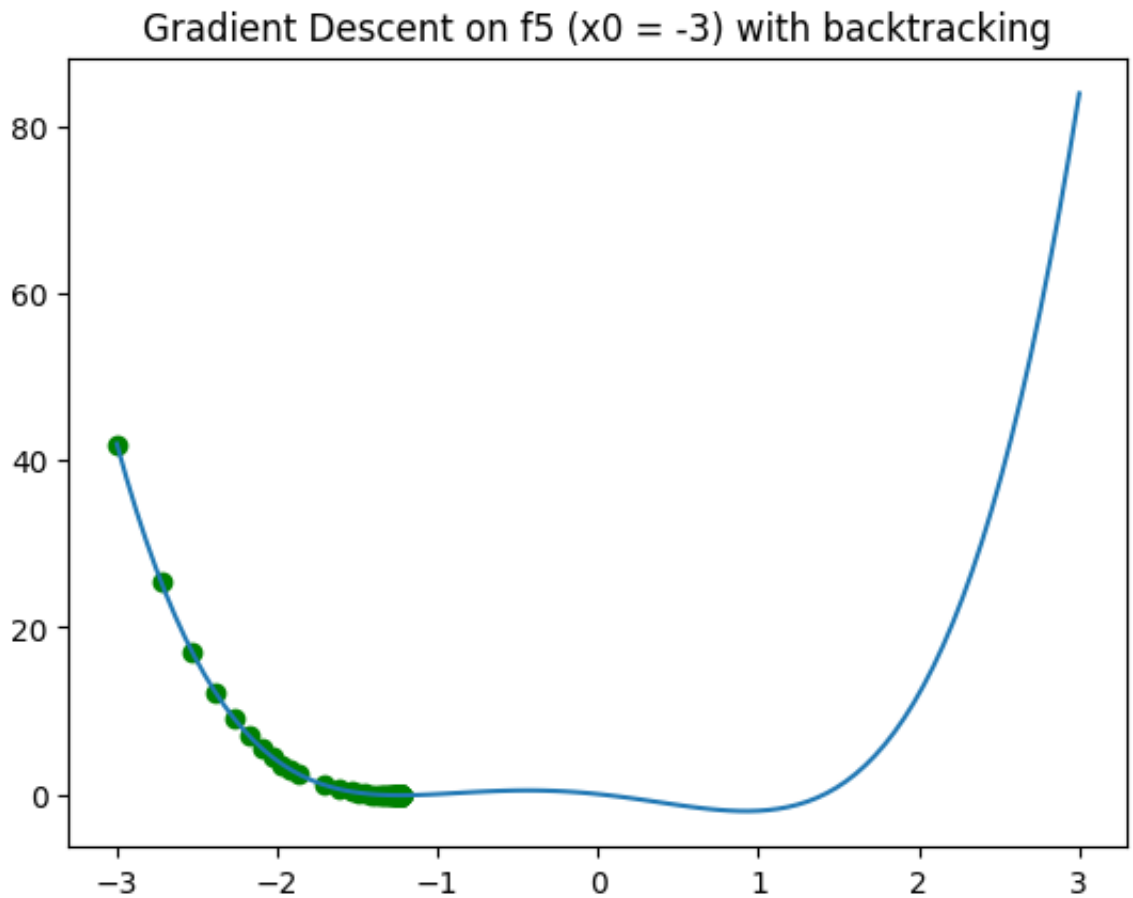

Error (2-norms of gradient)
Alpha = 1e-1 (x0 = 0)
Alpha = 1e-3 (x0 = 0)
Backtracking (x0 = 0)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
50
100
150
200
250
300



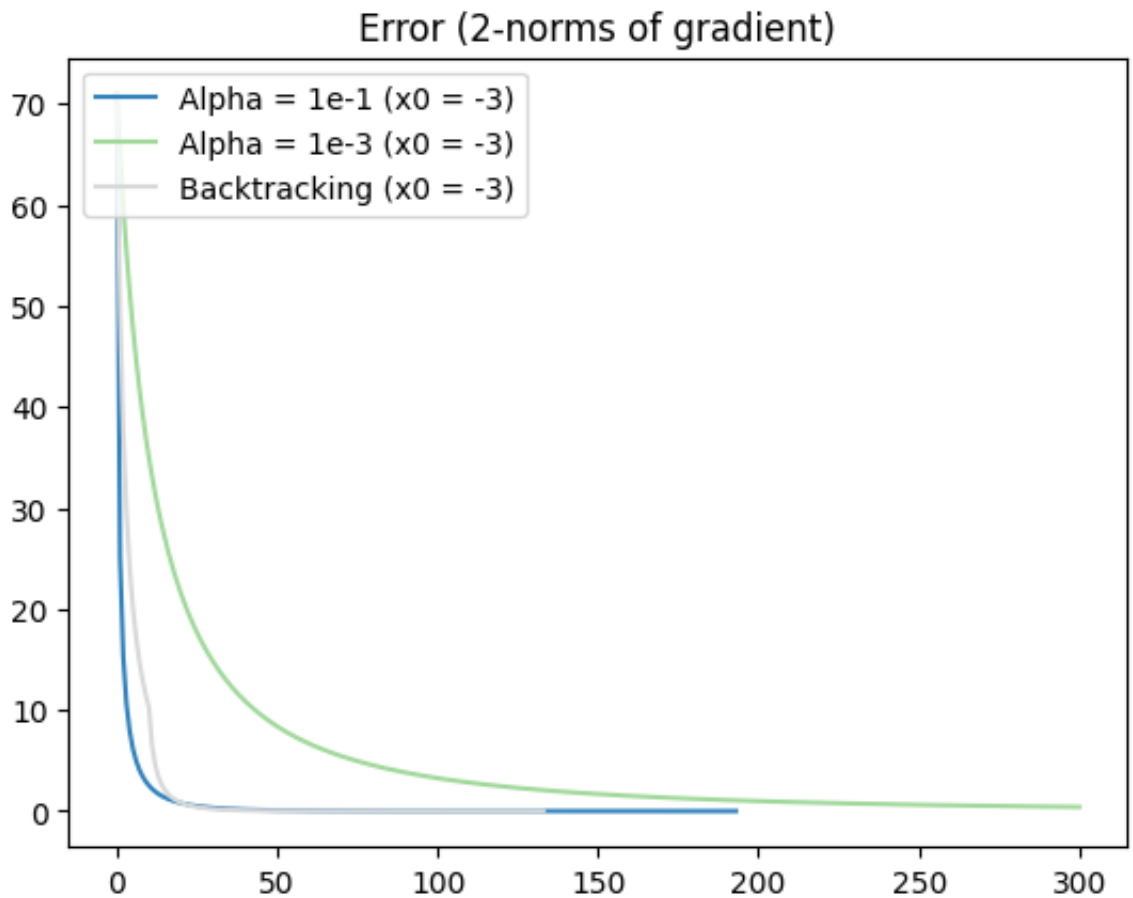

```
Minimum Found = [-1.23224005] with 193 iterations
```




```
Minimum Found = [-1.28699186] with 300 iterations
```




```
Minimum Found = [-1.2322399] with 133 iterations
```



part2 : SGD

- Without Replacement: The np.random.shuffle function ensures that the samples for each batch come from the shuffled dataset, meaning that no sample is reused within a single batch.
- Batch Size: The batches are taken in chunks of batch_size, and since shuffling happens at the start of each epoch, the dataset is fully used once per epoch without repeating any elements within the same epoch.

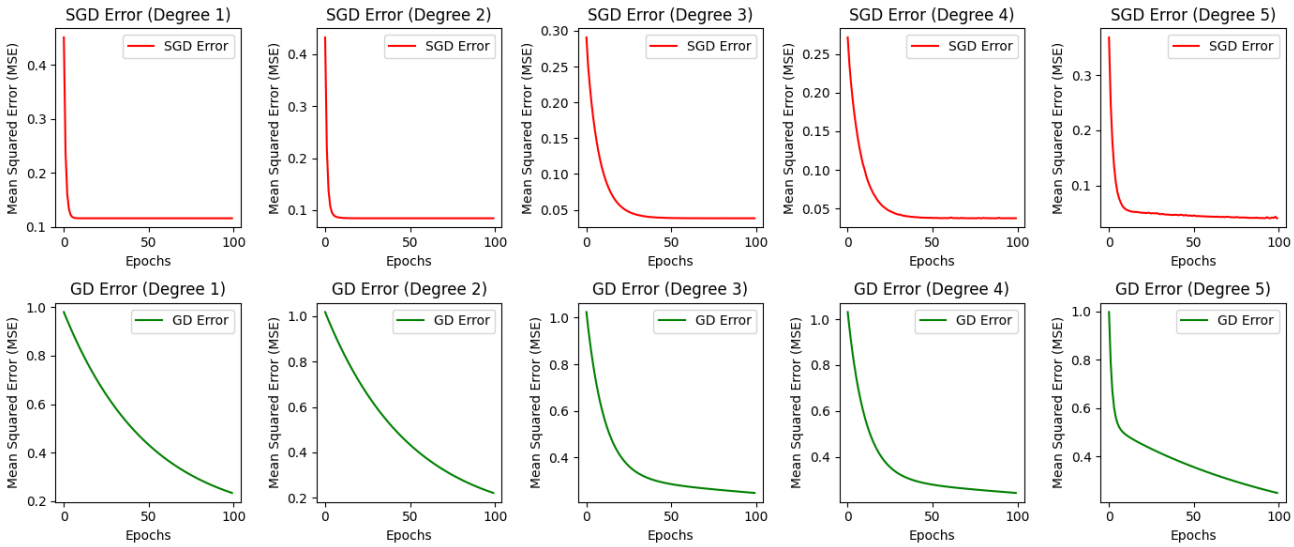
In [ ]:
```
#with predifined function PolynomialFeatures
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import train_test_split
random.seed(42)
```

```
# Load the data
data = pd.read_csv('/Users/negin/Desktop/first year/New folder (2)/static/Ho
X = data['x'].values.reshape(-1, 1)
y = data['y'].values

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, ran

# Normalize the data
scaler_X = StandardScaler()
scaler_y = StandardScaler()

X_train_scaled = scaler_X.fit_transform(X_train)
X_test_scaled = scaler_X.transform(X_test)

y_train_scaled = scaler_y.fit_transform(y_train.reshape(-1, 1)).flatten()
y_test_scaled = scaler_y.transform(y_test.reshape(-1, 1)).flatten()

# Hyperparameters
learning_rate = 0.01
batch_size = 16
epochs = 100

# Stochastic Gradient Descent
def stochastic_gradient_descent(X, y, theta, learning_rate, batch_size, epoc
    m = len(y)
    mse_list = []  # List to track MSE during training
    for epoch in range(epochs):
        # Shuffle the data
        indices = np.arange(m)
        np.random.shuffle(indices)
        X = X[indices]
        y = y[indices]

        for i in range(0, m, batch_size):
            X_batch = X[i:i+batch_size]
            y_batch = y[i:i+batch_size]

            predictions = X_batch @ theta
            errors = predictions - y_batch
            gradient = (1 / len(X_batch)) * (X_batch.T @ errors)
            theta -= learning_rate * gradient

        # Compute MSE at the end of each epoch
        predictions = X @ theta
        mse = mean_squared_error(y, predictions)
        mse_list.append(mse)
    return theta, mse_list

# Gradient Descent
```

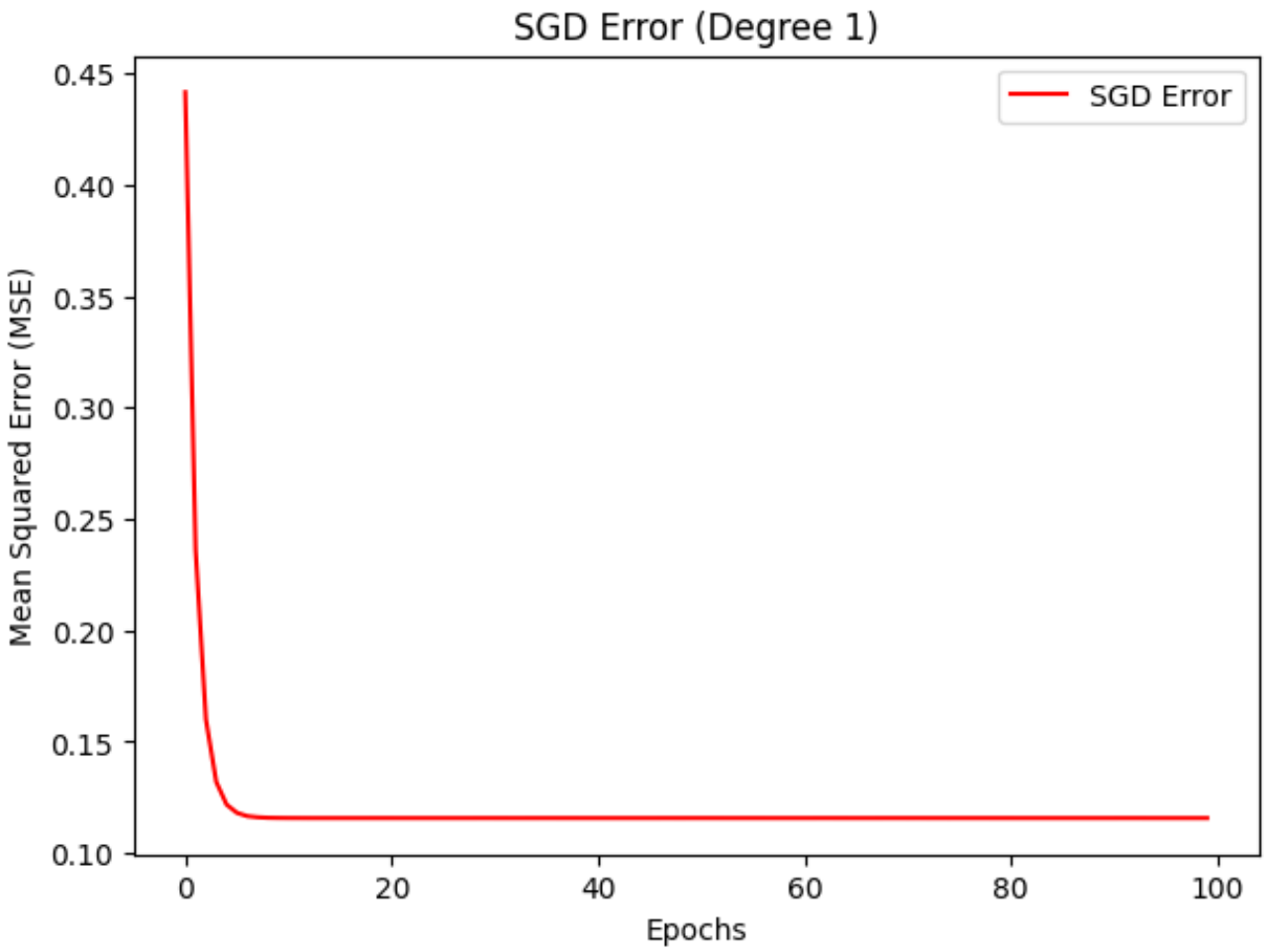
```
def gradient_descent(X, y, theta, learning_rate, epochs):
    m = len(y)
    mse_list = []  # List to track MSE during training
    for epoch in range(epochs):
        predictions = X @ theta
        errors = predictions - y
        gradient = (1 / m) * (X.T @ errors)
        theta -= learning_rate * gradient

        # Compute MSE at the end of each epoch
        mse = mean_squared_error(y, predictions)
        mse_list.append(mse)
    return theta, mse_list

# Evaluate Models with varying degree k
def evaluate_models(X_train, y_train, X_test, y_test, degree, learning_rate,
    # Initialize theta
    theta_sgd = np.random.randn(X_train.shape[1]) * 0.01
    theta_gd = np.random.randn(X_train.shape[1]) * 0.01

    # Train with SGD
    theta_sgd, mse_sgd = stochastic_gradient_descent(X_train, y_train, theta

    # Train with GD
    theta_gd, mse_gd = gradient_descent(X_train, y_train, theta_gd, learning

    # Predictions
    y_pred_sgd = X_test @ theta_sgd
    y_pred_gd = X_test @ theta_gd

    # Compute MSE for test set
    mse_sgd_test = mean_squared_error(y_test, y_pred_sgd)
    mse_gd_test = mean_squared_error(y_test, y_pred_gd)

    return theta_sgd, theta_gd, mse_sgd, mse_gd, mse_sgd_test, mse_gd_test,

# Vary the degree of the polynomial and observe the results
degrees = [1,2, 3, 4, 5]  # Test different polynomial degrees
results = {}

for degree in degrees:
    print(f"\nEvaluating with Polynomial Degree = {degree}")

    # Generate polynomial features
    poly = PolynomialFeatures(degree=degree)
    X_train_poly = poly.fit_transform(X_train_scaled)
    X_test_poly = poly.transform(X_test_scaled)

    # Run evaluation for this degree
    theta_sgd, theta_gd, mse_sgd, mse_gd, mse_sgd_test, mse_gd_test, mse_sgc
```

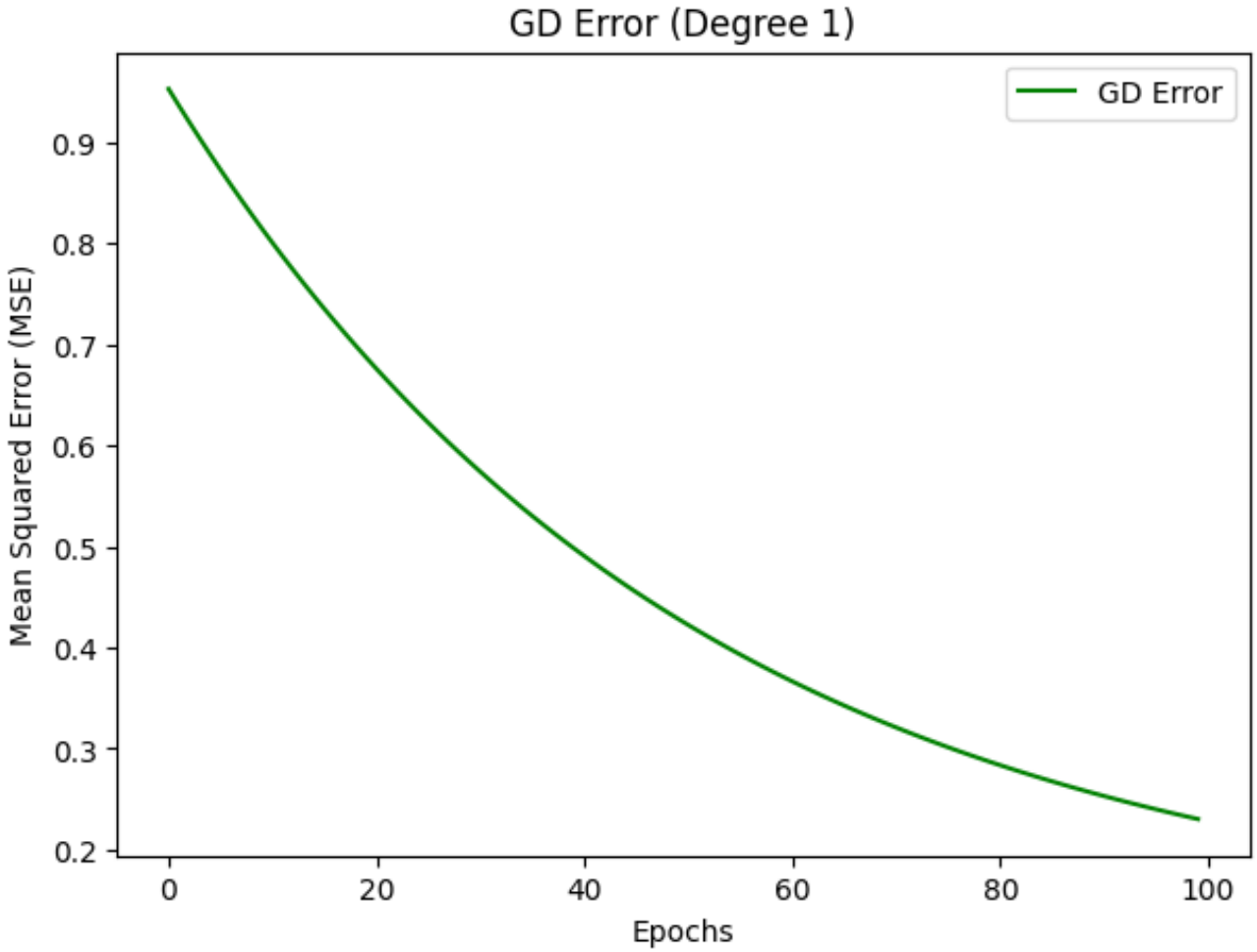
```
        X_train_poly, y_train_scaled, X_test_poly, y_test_scaled, degree, le

    # Store results
    results[degree] = {
        'theta_sgd': theta_sgd,
        'theta_gd': theta_gd,
        'mse_sgd_train': mse_sgd_train[-1],
        'mse_gd_train': mse_gd_train[-1],
        'mse_sgd_test': mse_sgd_test,
        'mse_gd_test': mse_gd_test,
        'mse_sgd': mse_sgd,
        'mse_gd': mse_gd
    }

    print(f"Parameters (SGD) for degree {degree}: {theta_sgd}")
    print(f"Parameters (GD) for degree {degree}: {theta_gd}")
    print(f"MSE (SGD - Train) for degree {degree}: {mse_sgd_train[-1]}")
    print(f"MSE (GD - Train) for degree {degree}: {mse_gd_train[-1]}")
    print(f"MSE (SGD - Test) for degree {degree}: {mse_sgd_test}")
    print(f"MSE (GD - Test) for degree {degree}: {mse_gd_test}")

# Visualize the training error change for different degrees
fig, axes = plt.subplots(2, len(degrees), figsize=(14, 6))

for i, degree in enumerate(degrees):
    axes[0, i].plot(range(epochs), results[degree]['mse_sgd'], color="red",
    axes[0, i].set_title(f"SGD Error (Degree {degree})")
    axes[0, i].set_xlabel("Epochs")
    axes[0, i].set_ylabel("Mean Squared Error (MSE)")
    axes[0, i].legend()

    axes[1, i].plot(range(epochs), results[degree]['mse_gd'], color="green",
    axes[1, i].set_title(f"GD Error (Degree {degree})")
    axes[1, i].set_xlabel("Epochs")
    axes[1, i].set_ylabel("Mean Squared Error (MSE)")
    axes[1, i].legend()

plt.tight_layout()
plt.show()
```

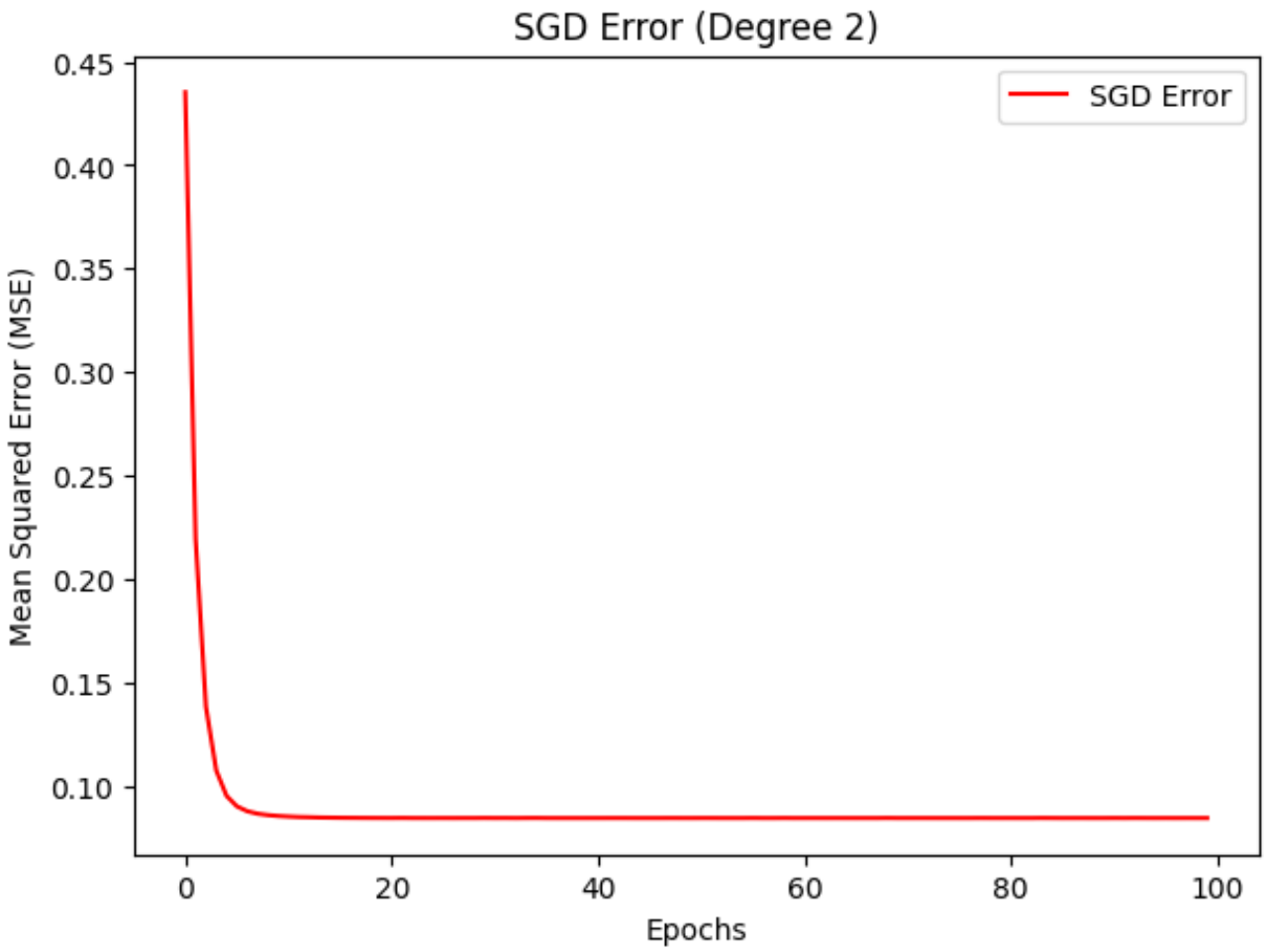
```
Evaluating with Polynomial Degree = 1
Parameters (SGD) for degree 1: [7.85363328e-04 9.41256563e-01]
Parameters (GD) for degree 1: [-0.00470494  0.60005812]
MSE (SGD - Train) for degree 1: 0.11575939459638392
MSE (GD - Train) for degree 1: 0.23392433824568953
MSE (SGD - Test) for degree 1: 0.1315382073113079
MSE (GD - Test) for degree 1: 0.2280479130120618

Evaluating with Polynomial Degree = 2
Parameters (SGD) for degree 2: [ 0.19920432  0.94137648 -0.19716102]
Parameters (GD) for degree 2: [ 0.03581835  0.59211634 -0.08399159]
MSE (SGD - Train) for degree 2: 0.08442751491106579
MSE (GD - Train) for degree 2: 0.22085352548403678
MSE (SGD - Test) for degree 2: 0.09129059867573468
MSE (GD - Test) for degree 2: 0.2154358815079619

Evaluating with Polynomial Degree = 3
Parameters (SGD) for degree 3: [ 0.19745274  1.43504274 -0.19900099 -0.27322
137]
Parameters (GD) for degree 3: [ 0.04175462  0.32183943 -0.08501437  0.260918
76]
MSE (SGD - Train) for degree 3: 0.03818489105959236
MSE (GD - Train) for degree 3: 0.24823578303804567
MSE (SGD - Test) for degree 3: 0.03776091305752591
MSE (GD - Test) for degree 3: 0.26781734438724003

Evaluating with Polynomial Degree = 4
Parameters (SGD) for degree 4: [ 0.16059129  1.43423462 -0.07620883 -0.27300
694 -0.04903972]
Parameters (GD) for degree 4: [ 0.06373185  0.30802846 -0.00572627  0.267562
92 -0.05700022]
MSE (SGD - Train) for degree 4: 0.037249961253470396
MSE (GD - Train) for degree 4: 0.24346493474011818
MSE (SGD - Test) for degree 4: 0.03664472869003756
MSE (GD - Test) for degree 4: 0.2561694813680618

Evaluating with Polynomial Degree = 5
Parameters (SGD) for degree 5: [ 0.15891463  1.17962844 -0.06934774  0.07454
13  -0.04921076 -0.09463993]
Parameters (GD) for degree 5: [ 0.05105219  0.31065736 -0.00615061  0.211779
22 -0.05403766  0.00270362]
MSE (SGD - Train) for degree 5: 0.04037816812176686
MSE (GD - Train) for degree 5: 0.25053466905363536
MSE (SGD - Test) for degree 5: 0.039752363817116526
MSE (GD - Test) for degree 5: 0.25895704473428444
```

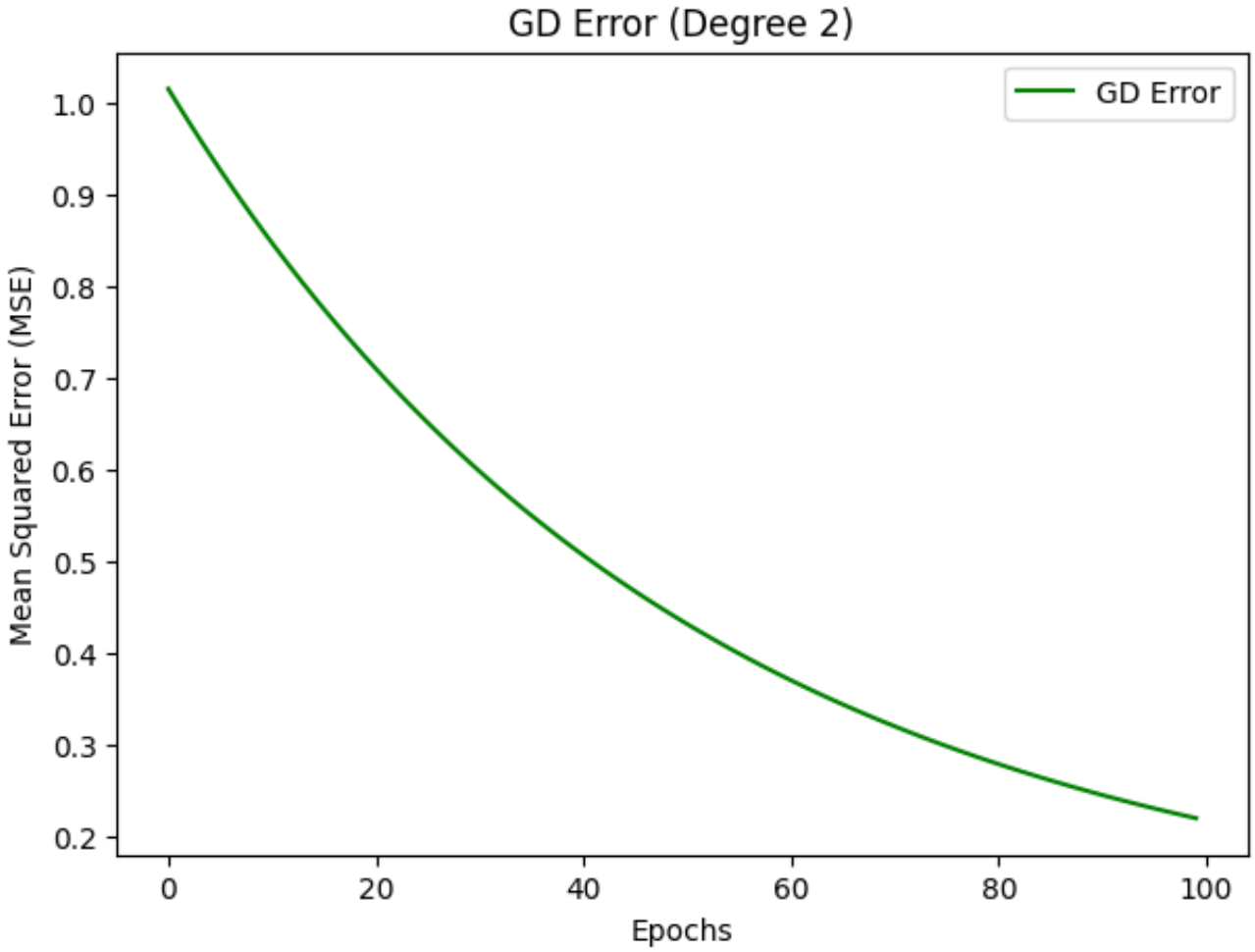


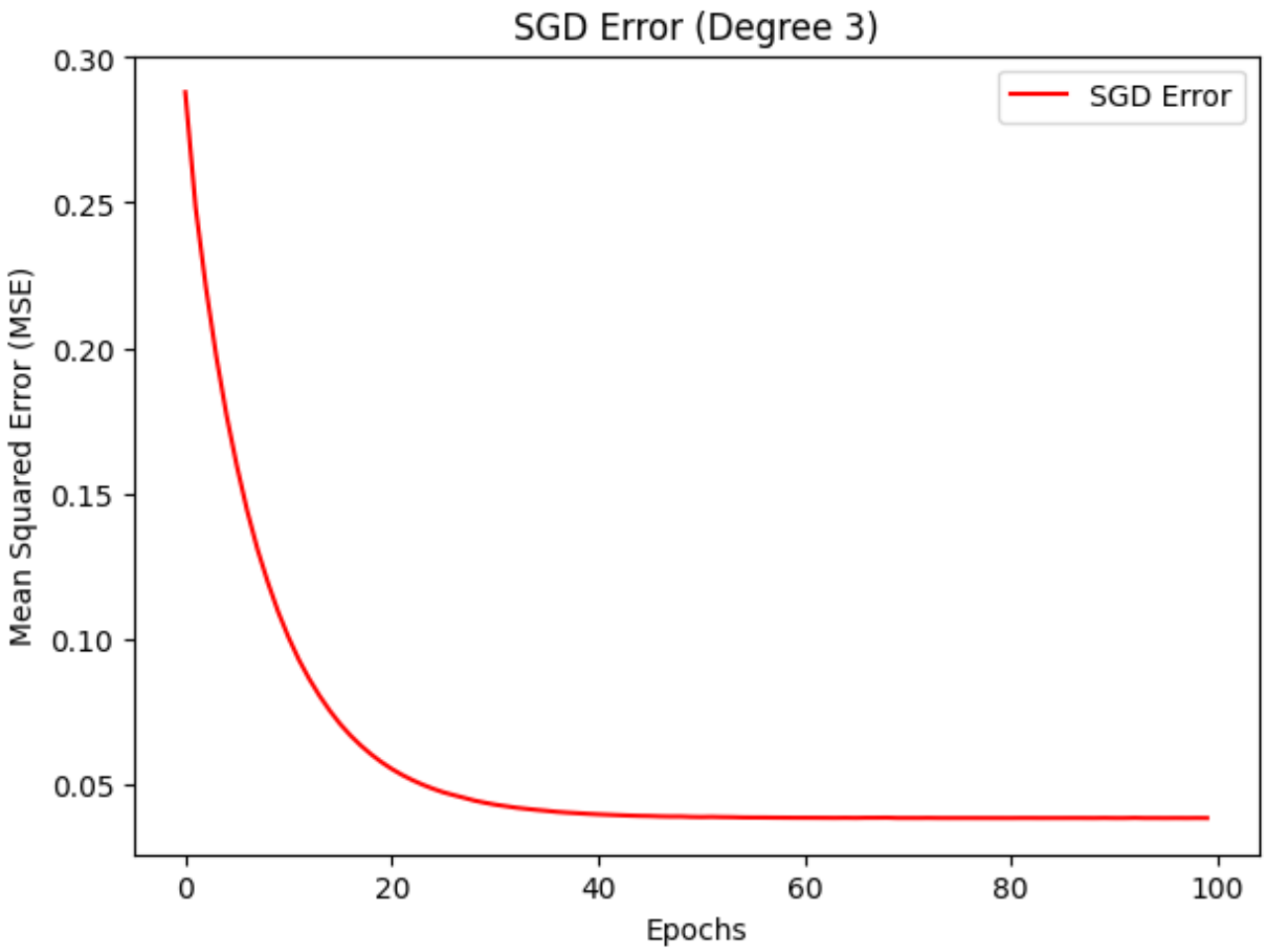
In [ ]:
```
# Visualize the training error change for different degrees separately
for i, degree in enumerate(degrees):
    # Plot SGD error
    plt.figure(figsize=(7, 5))
    plt.plot(range(epochs), results[degree]['mse_sgd'], color="red", label="
    plt.title(f"SGD Error (Degree {degree})")
    plt.xlabel("Epochs")
    plt.ylabel("Mean Squared Error (MSE)")
    plt.legend()
    plt.show()

    # Plot GD error
    plt.figure(figsize=(7, 5))
    plt.plot(range(epochs), results[degree]['mse_gd'], color="green", label=
    plt.title(f"GD Error (Degree {degree})")
    plt.xlabel("Epochs")
    plt.ylabel("Mean Squared Error (MSE)")
    plt.legend()
    plt.show()
```

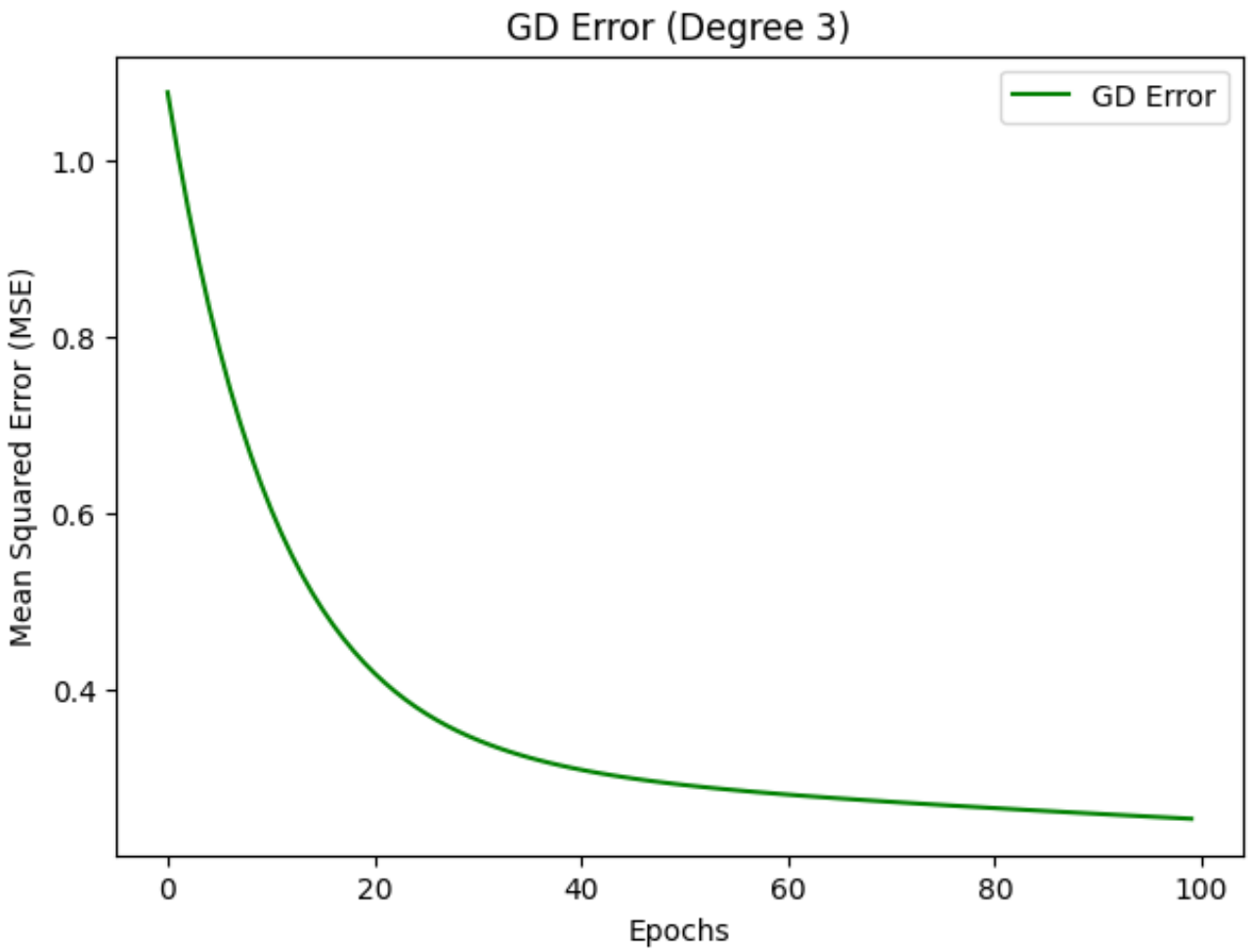

SGD Error (Degree 1)
SGD Error
Mean Squared Error (MSE)
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0
20
40
60
80
100
Epochs

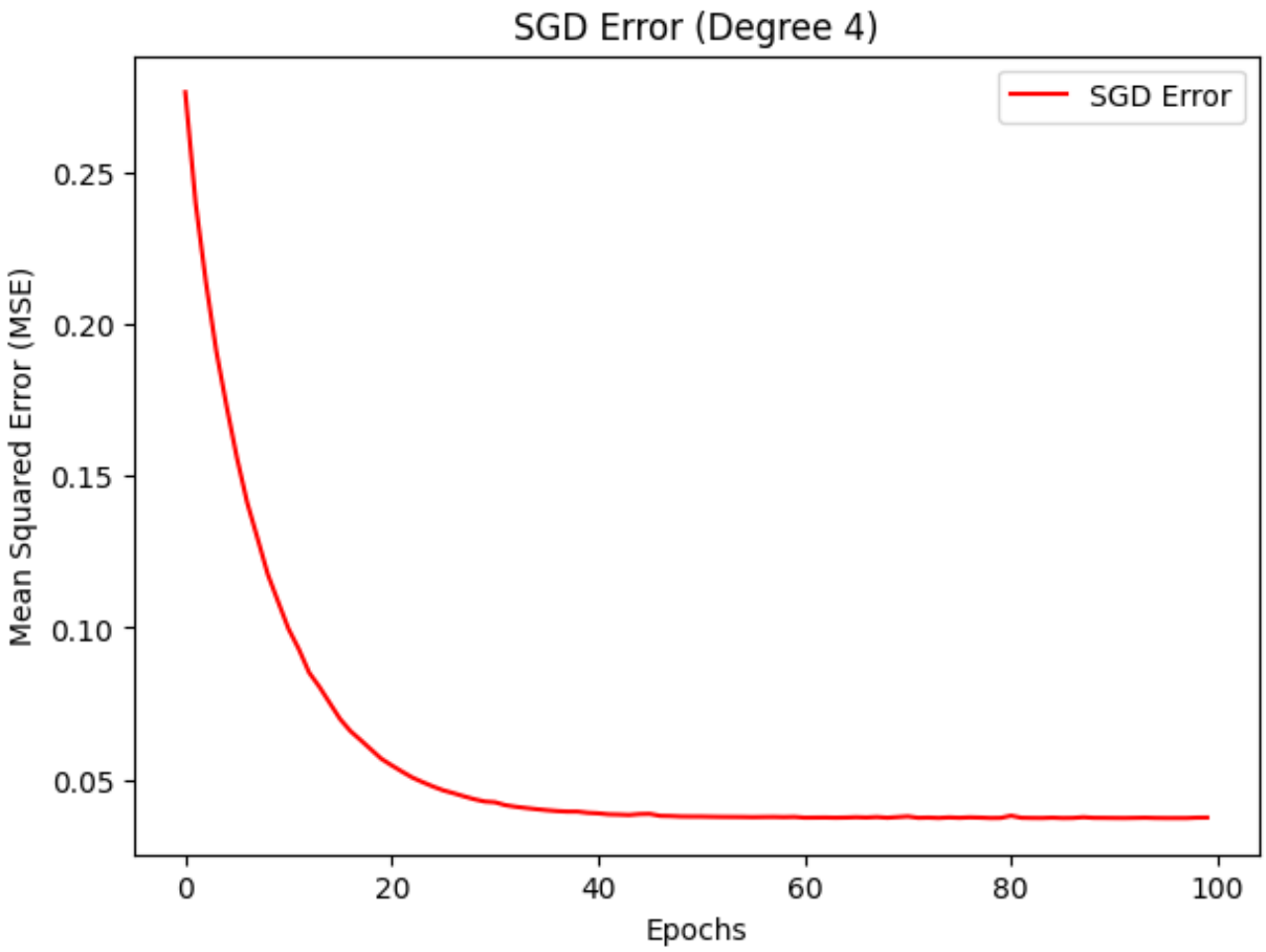

GD Error (Degree 1)
GD Error
Mean Squared Error (MSE)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
20
40
60
80
100
Epochs

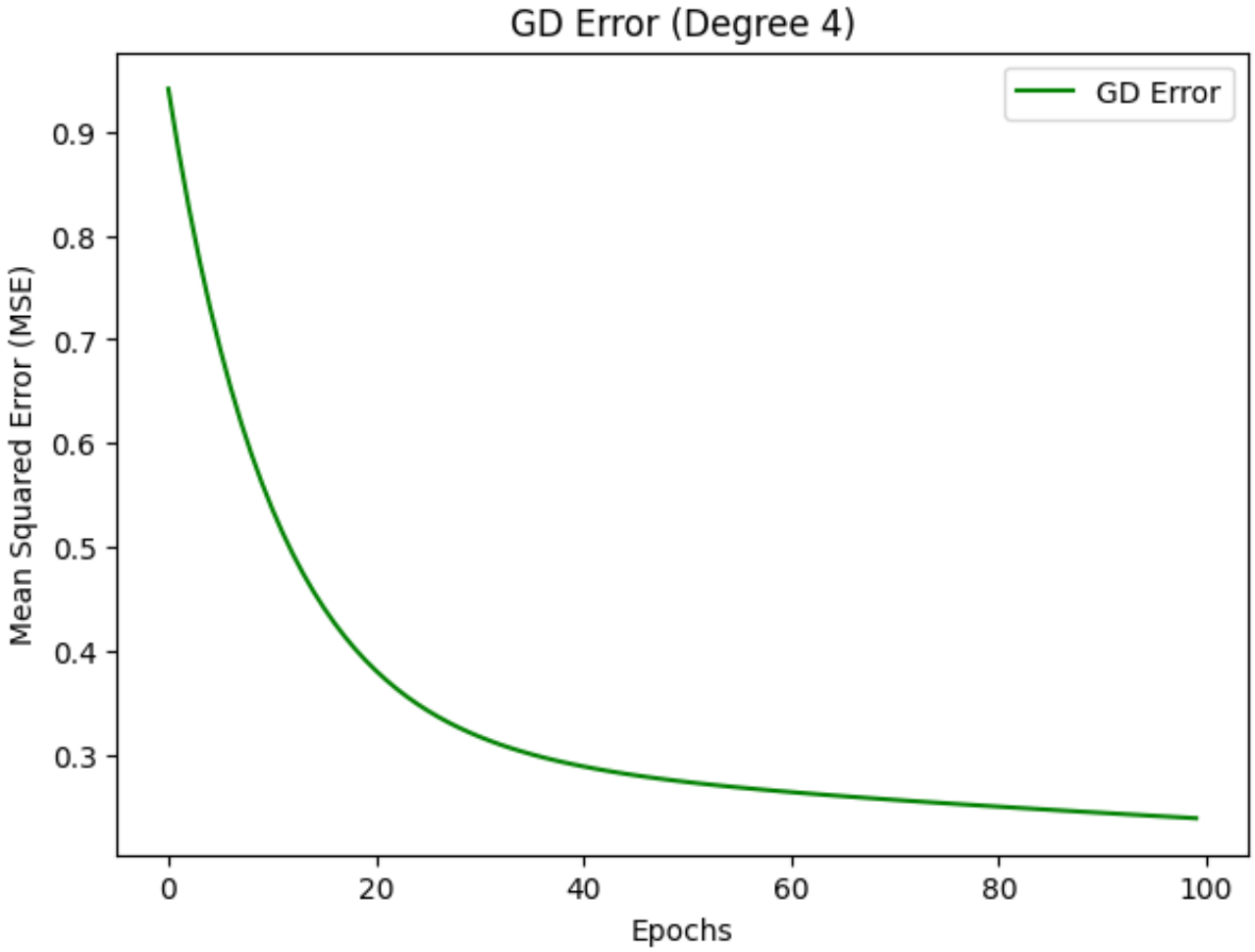

SGD Error (Degree 2)
SGD Error
Mean Squared Error (MSE)
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0
20
40
60
80
100
Epochs

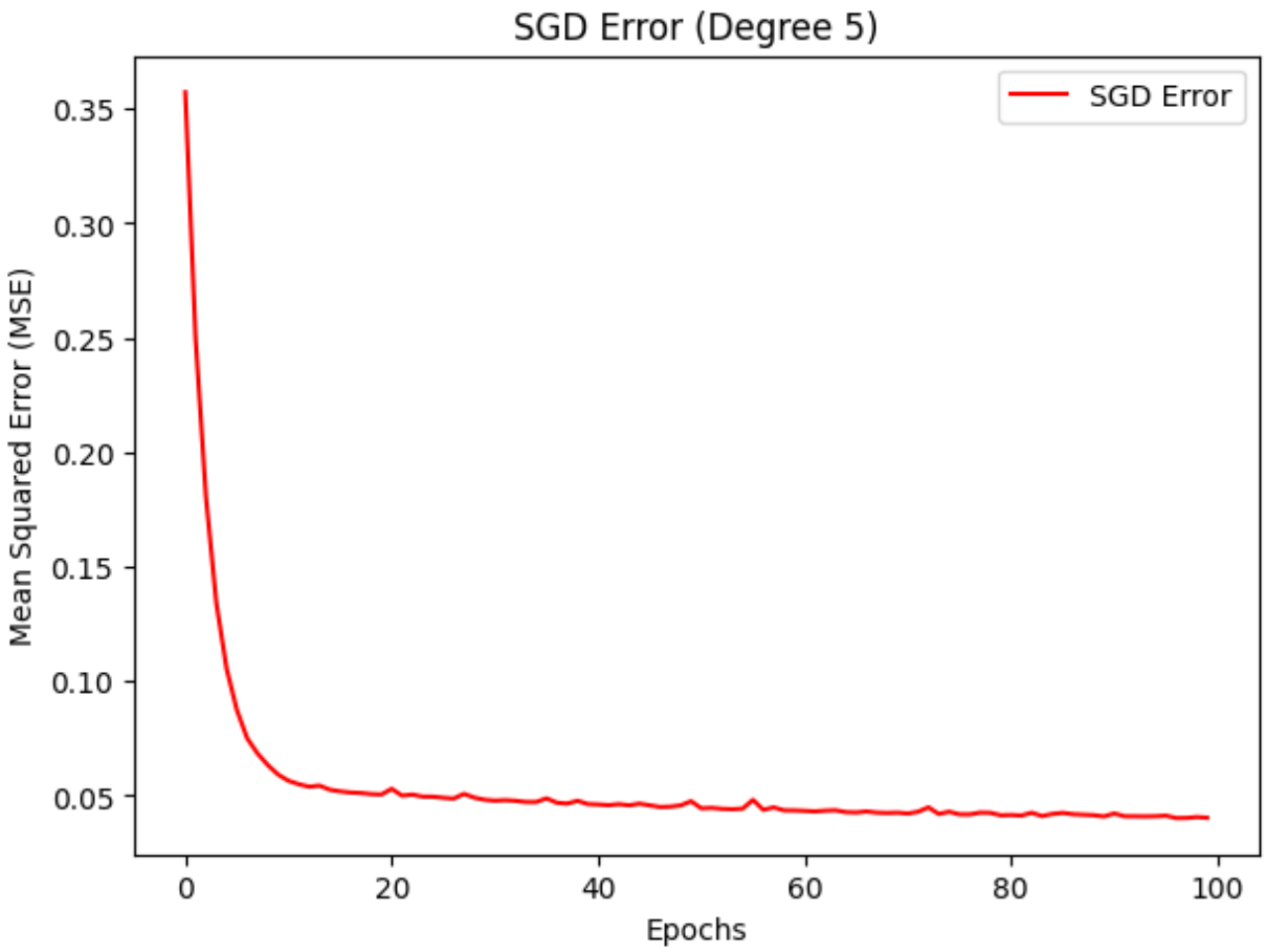

GD Error (Degree 2)
GD Error
Mean Squared Error (MSE)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
20
40
60
80
100
Epochs

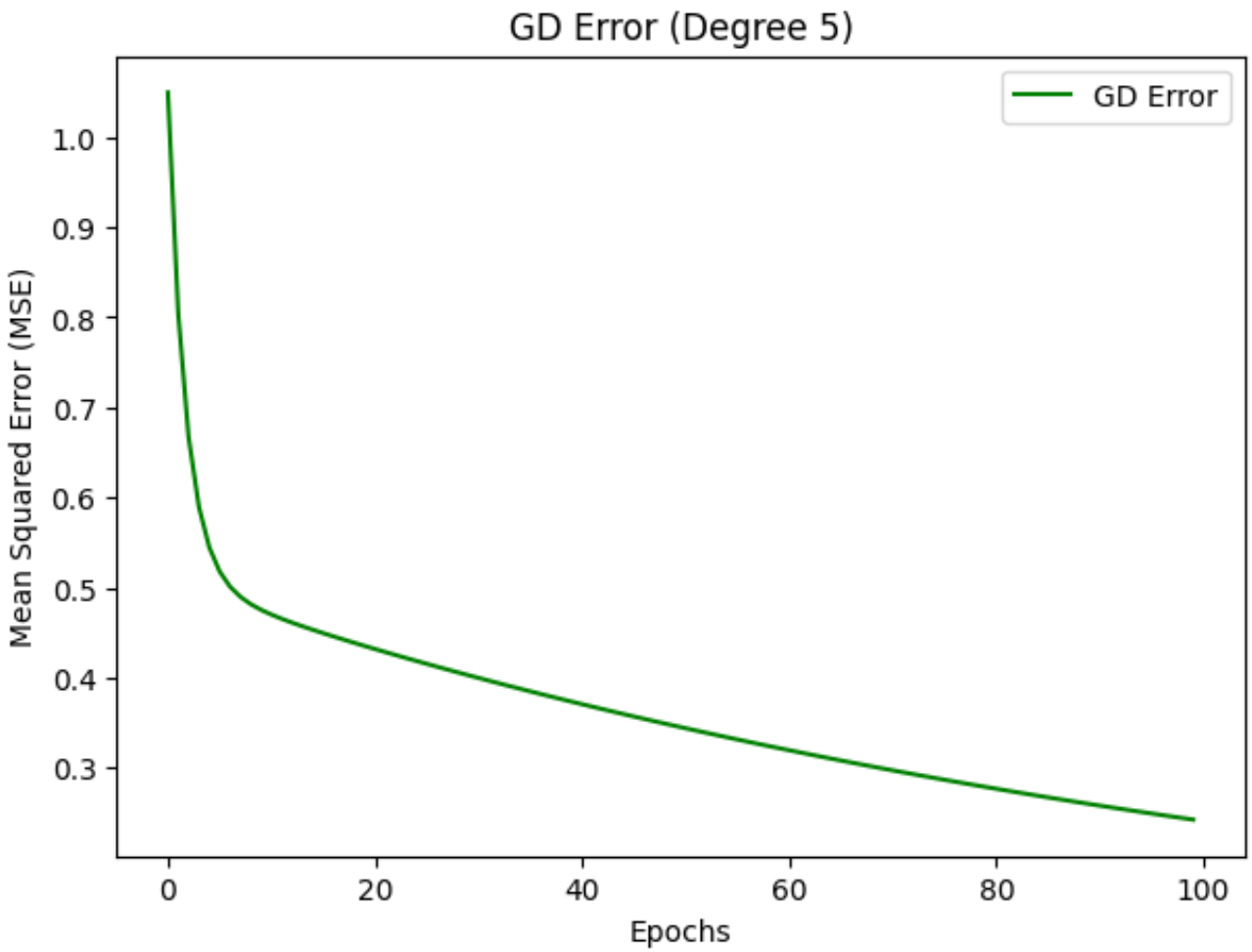

SGD Error (Degree 3)
SGD Error
Mean Squared Error (MSE)
0.30
0.25
0.20
0.15
0.10
0.05
0
20
40
60
80
100
Epochs


GD Error (Degree 3)
GD Error
Mean Squared Error (MSE)
1.0
0.8
0.6
0.4
0
20
40
60
80
100
Epochs


SGD Error (Degree 4)
SGD Error
Mean Squared Error (MSE)
0.25
0.20
0.15
0.10
0.05
0
20
40
60
80
100
Epochs


GD Error (Degree 4)
GD Error
Mean Squared Error (MSE)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
20
40
60
80
100
Epochs


SGD Error (Degree 5)
SGD Error
Mean Squared Error (MSE)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0
20
40
60
80
100
Epochs



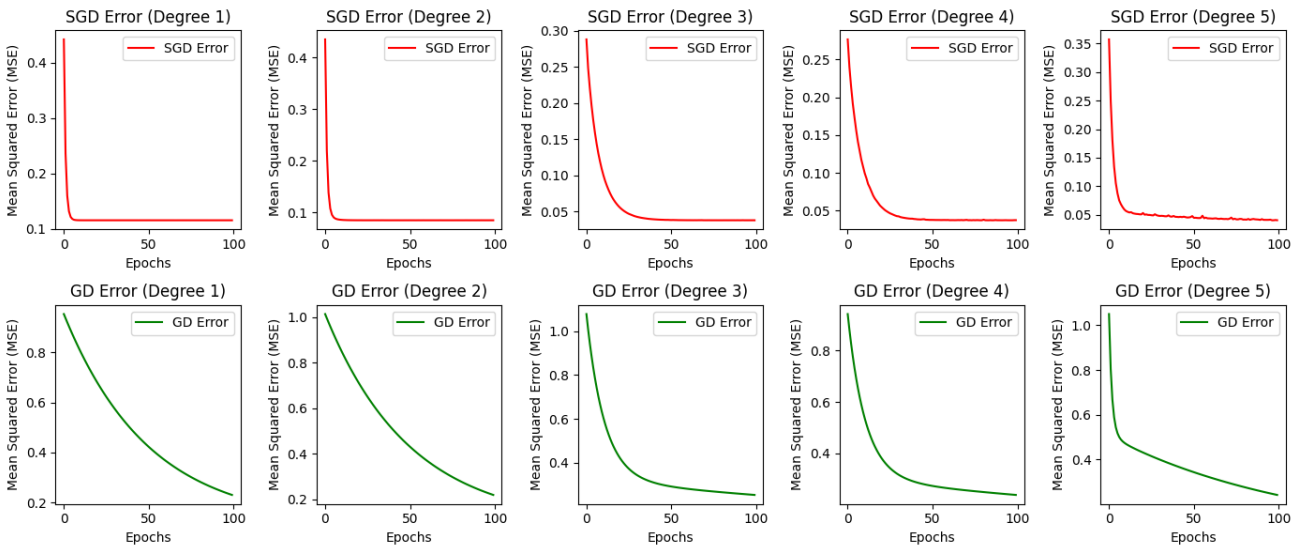
In [ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import train_test_split
import itertools

# Load the data
data = pd.read_csv('/Users/negin/Desktop/first year/New folder (2)/static/Ho
X = data['x'].values.reshape(-1, 1)
y = data['y'].values

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, ran

# Normalize the data
scaler_X = StandardScaler()
scaler_y = StandardScaler()

X_train_scaled = scaler_X.fit_transform(X_train)
```

```
X_test_scaled = scaler_X.transform(X_test)

y_train_scaled = scaler_y.fit_transform(y_train.reshape(-1, 1)).flatten()
y_test_scaled = scaler_y.transform(y_test.reshape(-1, 1)).flatten()

# Hyperparameters
learning_rate = 0.01
batch_size = 16
epochs = 100

# Stochastic Gradient Descent
def stochastic_gradient_descent(X, y, theta, learning_rate, batch_size, epoc
    m = len(y)
    mse_list = []  # List to track MSE during training
    for epoch in range(epochs):
        # Shuffle the data
        indices = np.arange(m)
        np.random.shuffle(indices)
        X = X[indices]
        y = y[indices]

        for i in range(0, m, batch_size):
            X_batch = X[i:i+batch_size]
            y_batch = y[i:i+batch_size]

            predictions = X_batch @ theta
            errors = predictions - y_batch
            gradient = (1 / len(X_batch)) * (X_batch.T @ errors)
            theta -= learning_rate * gradient

        # Compute MSE at the end of each epoch
        predictions = X @ theta
        mse = mean_squared_error(y, predictions)
        mse_list.append(mse)
    return theta, mse_list

# Gradient Descent
def gradient_descent(X, y, theta, learning_rate, epochs):
    m = len(y)
    mse_list = []  # List to track MSE during training
    for epoch in range(epochs):
        predictions = X @ theta
        errors = predictions - y
        gradient = (1 / m) * (X.T @ errors)
        theta -= learning_rate * gradient

        # Compute MSE at the end of each epoch
        mse = mean_squared_error(y, predictions)
        mse_list.append(mse)
    return theta, mse_list
```

```
# Evaluate Models with varying degree k
def evaluate_models(X_train, y_train, X_test, y_test, degree, learning_rate,
    # Initialize theta
    theta_sgd = np.random.randn(X_train.shape[1]) * 0.01
    theta_gd = np.random.randn(X_train.shape[1]) * 0.01

    # Train with SGD
    theta_sgd, mse_sgd = stochastic_gradient_descent(X_train, y_train, theta

    # Train with GD
    theta_gd, mse_gd = gradient_descent(X_train, y_train, theta_gd, learning

    # Predictions
    y_pred_sgd = X_test @ theta_sgd
    y_pred_gd = X_test @ theta_gd

    # Compute MSE for test set
    mse_sgd_test = mean_squared_error(y_test, y_pred_sgd)
    mse_gd_test = mean_squared_error(y_test, y_pred_gd)

    return theta_sgd, theta_gd, mse_sgd, mse_gd, mse_sgd_test, mse_gd_test,

# Vary the degree of the polynomial and observe the results
degrees = [1, 2, 3, 4, 5]  # Test different polynomial degrees
results = {}
def create_polynomial_features(X, k):
    """
    Create polynomial features manually (including interaction terms).
    """
    n_samples, n_features = X.shape
    features = [np.ones(n_samples)]  # Include a column of ones for the bias

    # Iterate over each degree from 1 to the specified degree
    for d in range(1, k + 1):
        for comb in itertools.combinations_with_replacement(range(n_features
            term = np.prod([X[:, i] ** (comb.count(i)) for i in range(n_feat
            features.append(term)

    return np.column_stack(features)
for degree in degrees:
    print(f"\nEvaluating with Polynomial Degree = {degree}")

    # Generate polynomial features
    X_train_poly = create_polynomial_features(X_train_scaled,k=degree)
    X_test_poly = create_polynomial_features(X_test_scaled,k=degree)

    # X_train_poly = poly.fit_transform(X_train_scaled)
    # X_test_poly = poly.transform(X_test_scaled)
```

```
    # Run evaluation for this degree
    theta_sgd, theta_gd, mse_sgd, mse_gd, mse_sgd_test, mse_gd_test, mse_sgc
        X_train_poly, y_train_scaled, X_test_poly, y_test_scaled, degree, le

    # Store results
    results[degree] = {
        'theta_sgd': theta_sgd,
        'theta_gd': theta_gd,
        'mse_sgd_train': mse_sgd_train[-1],
        'mse_gd_train': mse_gd_train[-1],
        'mse_sgd_test': mse_sgd_test,
        'mse_gd_test': mse_gd_test,
        'mse_sgd': mse_sgd,
        'mse_gd': mse_gd
    }

    print(f"Parameters (SGD) for degree {degree}: {theta_sgd}")
    print(f"Parameters (GD) for degree {degree}: {theta_gd}")
    print(f"MSE (SGD - Train) for degree {degree}: {mse_sgd_train[-1]}")
    print(f"MSE (GD - Train) for degree {degree}: {mse_gd_train[-1]}")
    print(f"MSE (SGD - Test) for degree {degree}: {mse_sgd_test}")
    print(f"MSE (GD - Test) for degree {degree}: {mse_gd_test}")

# Visualize the training error change for different degrees
fig, axes = plt.subplots(2, len(degrees), figsize=(14, 6))

for i, degree in enumerate(degrees):
    axes[0, i].plot(range(epochs), results[degree]['mse_sgd'], color="red",
    axes[0, i].set_title(f"SGD Error (Degree {degree})")
    axes[0, i].set_xlabel("Epochs")
    axes[0, i].set_ylabel("Mean Squared Error (MSE)")
    axes[0, i].legend()

    axes[1, i].plot(range(epochs), results[degree]['mse_gd'], color="green",
    axes[1, i].set_title(f"GD Error (Degree {degree})")
    axes[1, i].set_xlabel("Epochs")
    axes[1, i].set_ylabel("Mean Squared Error (MSE)")
    axes[1, i].legend()

plt.tight_layout()
plt.show()
```

```
Evaluating with Polynomial Degree = 1
Parameters (SGD) for degree 1: [4.18974488e-04 9.39889043e-01]
Parameters (GD) for degree 1: [-0.00163912  0.6054074 ]
MSE (SGD - Train) for degree 1: 0.11575832060521986
MSE (GD - Train) for degree 1: 0.2302192333758604
MSE (SGD - Test) for degree 1: 0.13146299547118792
MSE (GD - Test) for degree 1: 0.22474295224733487

Evaluating with Polynomial Degree = 2
Parameters (SGD) for degree 2: [ 0.19910355  0.93925454 -0.19692343]
Parameters (GD) for degree 2: [ 0.04251085  0.59280076 -0.08797349]
MSE (SGD - Train) for degree 2: 0.08442703419430955
MSE (GD - Train) for degree 2: 0.2193897317112452
MSE (SGD - Test) for degree 2: 0.09126926719763695
MSE (GD - Test) for degree 2: 0.21405374528458743

Evaluating with Polynomial Degree = 3
Parameters (SGD) for degree 3: [ 0.19714812  1.43410843 -0.19927228 -0.27559
439]
Parameters (GD) for degree 3: [ 0.04279649  0.3079469  -0.08723757  0.267524
33]
MSE (SGD - Train) for degree 3: 0.0381607188033305
MSE (GD - Train) for degree 3: 0.25289238993211144
MSE (SGD - Test) for degree 3: 0.03755345759893712
MSE (GD - Test) for degree 3: 0.2726032689266638

Evaluating with Polynomial Degree = 4
Parameters (SGD) for degree 4: [ 0.15901694  1.43287735 -0.07711062 -0.27594
766 -0.05109707]
Parameters (GD) for degree 4: [ 0.06116064  0.32312343 -0.00430016  0.260450
74 -0.0570514 ]
MSE (SGD - Train) for degree 4: 0.03738092625292884
MSE (GD - Train) for degree 4: 0.23831238237522076
MSE (SGD - Test) for degree 4: 0.03661441689291126
MSE (GD - Test) for degree 4: 0.25078518737420447

Evaluating with Polynomial Degree = 5
Parameters (SGD) for degree 5: [ 0.15921665  1.17594856 -0.06862833  0.07547
483 -0.05019027 -0.10199194]
Parameters (GD) for degree 5: [ 0.06387337  0.31754347 -0.00804909  0.222047
58 -0.05606276 -0.00279188]
MSE (SGD - Train) for degree 5: 0.04040755270681591
MSE (GD - Train) for degree 5: 0.2422740261074977
MSE (SGD - Test) for degree 5: 0.03868115218343485
MSE (GD - Test) for degree 5: 0.2506790618805646
```



Parameters:

- **SGD Parameters**:
  - (\theta_{\text{SGD}} = [0.16099206, 1.43450444, -0.07499406, -0.27289835, -0.04636222])
  - These values are fairly close to the true parameters (\theta^* = [0, 0, 4, 0, -3]) in terms of magnitude, but not exact. This suggests that the Stochastic Gradient Descent (SGD) approach has learned a decent approximation of the true parameters, although there is some deviation.
- **GD Parameters**:
  - (\theta_{\text{GD}} = [0.06687605, 0.30810065, 0.00749848, 0.2675689, -0.0635346])
  - These parameters are quite different from the true parameters as well. The (\theta_2) (which should be 4) is 0.308, and (\theta_4) (which should be -3) is 0.267, indicating that the Gradient Descent (GD) model might not have converged as well as the SGD model.

for small dataset

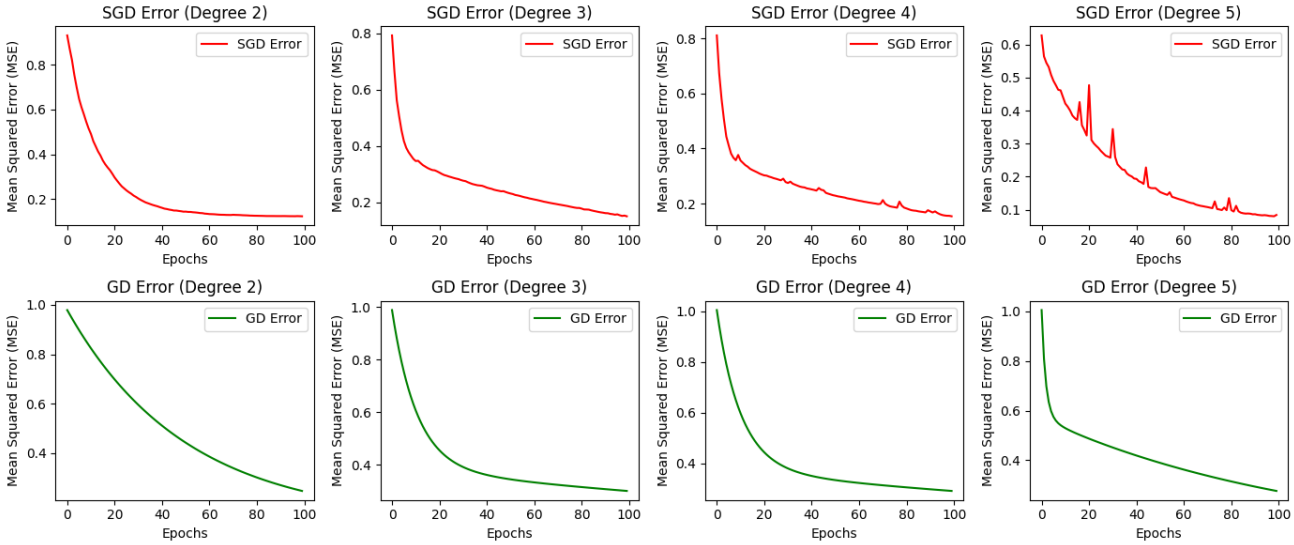
In [ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.metrics import mean_squared_error
```

```
# Load the data
data = pd.read_csv('/Users/negin/Desktop/first year/New folder (2)/static/Ho
X = data['x'].values.reshape(-1, 1)
y = data['y'].values

# Normalize the data
scaler_X = StandardScaler()
scaler_y = StandardScaler()

X_scaled = scaler_X.fit_transform(X)
y_scaled = scaler_y.fit_transform(y.reshape(-1, 1)).flatten()

# Hyperparameters
learning_rate = 0.01
batch_size = 16
epochs = 100

# Stochastic Gradient Descent
def stochastic_gradient_descent(X, y, theta, learning_rate, batch_size, epoc
    m = len(y)
    mse_list = []  # List to track MSE during training
    for epoch in range(epochs):
        # Shuffle the data
        indices = np.arange(m)
        np.random.shuffle(indices)
        X = X[indices]
        y = y[indices]

        for i in range(0, m, batch_size):
            X_batch = X[i:i+batch_size]
            y_batch = y[i:i+batch_size]

            predictions = X_batch @ theta
            errors = predictions - y_batch
            gradient = (1 / len(X_batch)) * (X_batch.T @ errors)
            theta -= learning_rate * gradient

        # Compute MSE at the end of each epoch
        predictions = X @ theta
        mse = mean_squared_error(y, predictions)
        mse_list.append(mse)
    return theta, mse_list

# Gradient Descent
def gradient_descent(X, y, theta, learning_rate, epochs):
    m = len(y)
    mse_list = []  # List to track MSE during training
    for epoch in range(epochs):
        predictions = X @ theta
```

```
        errors = predictions - y
        gradient = (1 / m) * (X.T @ errors)
        theta -= learning_rate * gradient

        # Compute MSE at the end of each epoch
        mse = mean_squared_error(y, predictions)
        mse_list.append(mse)
    return theta, mse_list

# Evaluate Model with varying degree k
def evaluate_model(X, y, degree, learning_rate, batch_size, epochs):
    # Generate polynomial features
    poly = PolynomialFeatures(degree=degree)
    X_poly = poly.fit_transform(X)

    # Initialize theta
    theta_sgd = np.random.randn(X_poly.shape[1]) * 0.01
    theta_gd = np.random.randn(X_poly.shape[1]) * 0.01

    # Train with SGD
    theta_sgd, mse_sgd = stochastic_gradient_descent(X_poly, y, theta_sgd, l

    # Train with GD
    theta_gd, mse_gd = gradient_descent(X_poly, y, theta_gd, learning_rate,

    # Predictions
    y_pred_sgd = X_poly @ theta_sgd
    y_pred_gd = X_poly @ theta_gd

    # Compute MSE for the full dataset
    mse_sgd_full = mean_squared_error(y, y_pred_sgd)
    mse_gd_full = mean_squared_error(y, y_pred_gd)

    return theta_sgd, theta_gd, mse_sgd, mse_gd, mse_sgd_full, mse_gd_full

# Vary the degree of the polynomial and observe the results
degrees = [2, 3, 4, 5]  # Test different polynomial degrees
results = {}

for degree in degrees:
    print(f"\nEvaluating with Polynomial Degree = {degree}")

    # Run evaluation for this degree
    theta_sgd, theta_gd, mse_sgd, mse_gd, mse_sgd_full, mse_gd_full = evalua
        X_scaled, y_scaled, degree, learning_rate, batch_size, epochs)

    # Store results
    results[degree] = {
        'theta_sgd': theta_sgd,
        'theta_gd': theta_gd,
```

```
        'mse_sgd_full': mse_sgd_full,
        'mse_gd_full': mse_gd_full,
        'mse_sgd': mse_sgd,
        'mse_gd': mse_gd
    }

    print(f"Parameters (SGD) for degree {degree}: {theta_sgd}")
    print(f"Parameters (GD) for degree {degree}: {theta_gd}")
    print(f"MSE (SGD - Full) for degree {degree}: {mse_sgd_full}")
    print(f"MSE (GD - Full) for degree {degree}: {mse_gd_full}")

# Visualize the training error change for different degrees
fig, axes = plt.subplots(2, len(degrees), figsize=(14, 6))

for i, degree in enumerate(degrees):
    axes[0, i].plot(range(epochs), results[degree]['mse_sgd'], color="red",
    axes[0, i].set_title(f"SGD Error (Degree {degree})")
    axes[0, i].set_xlabel("Epochs")
    axes[0, i].set_ylabel("Mean Squared Error (MSE)")
    axes[0, i].legend()

    axes[1, i].plot(range(epochs), results[degree]['mse_gd'], color="green",
    axes[1, i].set_title(f"GD Error (Degree {degree})")
    axes[1, i].set_xlabel("Epochs")
    axes[1, i].set_ylabel("Mean Squared Error (MSE)")
    axes[1, i].legend()

plt.tight_layout()
plt.show()
```

```
Evaluating with Polynomial Degree = 2
Parameters (SGD) for degree 2: [ 0.13542204  0.90962992 -0.17023411]
Parameters (GD) for degree 2: [ 0.03811733  0.58691413 -0.08779205]
MSE (SGD - Full) for degree 2: 0.12241621140317469
MSE (GD - Full) for degree 2: 0.24459944704076492

Evaluating with Polynomial Degree = 3
Parameters (SGD) for degree 3: [ 0.15106397  0.74437241 -0.19049079  0.01822
407]
Parameters (GD) for degree 3: [ 0.03984525  0.33376198 -0.08858121  0.232678
4 ]
MSE (SGD - Full) for degree 3: 0.150357292119849
MSE (GD - Full) for degree 3: 0.29902834973065984

Evaluating with Polynomial Degree = 4
Parameters (SGD) for degree 4: [ 0.13280638  0.72180091  0.02914869  0.03915
877 -0.08279973]
Parameters (GD) for degree 4: [ 0.05897055  0.32816523 -0.00305206  0.235363
66 -0.0593305 ]
MSE (SGD - Full) for degree 4: 0.1542061359252437
MSE (GD - Full) for degree 4: 0.290853588532657

Evaluating with Polynomial Degree = 5
Parameters (SGD) for degree 5: [ 0.12175596  0.76840046 -0.00539108  0.41617
755 -0.0825854  -0.18911814]
Parameters (GD) for degree 5: [ 0.05827995  0.3433401   0.00694364  0.216757
45 -0.06357131 -0.01669762]
MSE (SGD - Full) for degree 5: 0.08397321341097218
MSE (GD - Full) for degree 5: 0.2742207931140096
```



Interpretation of Results

1. **Polynomial Degree = 2**

- SGD performs better than GD with a lower MSE. The parameters from SGD suggest a better fit for the data.
- GD's MSE is higher, indicating it struggles to find an optimal set of parameters.

2. **Polynomial Degree = 3**
   - SGD continues to outperform GD in terms of MSE, showing better convergence with more complex data.
   - The parameters for SGD are more refined, whereas GD shows a less optimal fit.
3. **Polynomial Degree = 4**
   - SGD still yields a lower MSE compared to GD, suggesting better optimization in higher polynomial degrees.
   - The difference in parameters grows, reflecting the increased complexity of the model.
4. **Polynomial Degree = 5**
   - SGD shows the lowest MSE again, reinforcing its superior performance for higher-degree polynomials.
   - GD’s parameters continue to be less optimal, and its MSE is higher, confirming its limitations with more complex models.

Key Observations:

- **SGD outperforms GD** consistently across all polynomial degrees, achieving **lower MSE** values.
- As the degree increases, **SGD’s parameters become more complex**, but still yield better results.
- Higher-degree polynomials improve performance but may risk overfitting the data.